The Goldbach Correlation Constant K_Ω

A Structural and Analytic Study of Prime Correlations

Irrational–Modular Theory Research Monograph

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Introduction

This monograph grew out of a computational observation that refused to go away.

We were studying the Hardy–Littlewood singular series in connection with the Goldbach conjecture — specifically, the covariance structure among what one might call "Goldbach pair events." For an even integer n, each way of writing $n/2 = d + (n/2 - d)$ gives a pair of offsets $(d, n/2 - d)$, and one can ask whether the events "$n/2 + d$ and $n/2 - d$ are both prime" behave independently.

They do not. The deviation from independence is measured by the ratio

$$R(d, d') = \frac{S_4(d, d')}{S_2(d) \cdot S_2(d')}$$

where S_2 is the classical pair singular series and S_4 is the 4-tuple singular series. When $R(d, d') = 1$, the two pair configurations behave independently. When $R(d, d') \neq 1$, there is arithmetic correlation.

Summing these correlations over all splits of $n/2$ gives a quantity

$$\Omega(n) = \sum_{d=1}^{n/2-1} |R(d, n/2 - d) - 1|$$

which, when computed for increasing even n, grows exactly linearly: $\Omega(n) \approx 0.3640 \cdot n$. The ratio $\Omega(n)/n$ stabilizes near 0.3640 — except when $n/2$ is divisible by small primes, in which case it drops sharply. This "branch effect" is real, structural, and (as we prove below) follows directly from the Euler product decomposition and the Chinese Remainder Theorem.

If the limit exists (ignoring the branch effect for the moment), we denote it by

$$K_\Omega = \lim_{\substack{n \to \infty \\ n \text{ even}}} \frac{\Omega(n)}{n} \approx 0.3640\ldots$$

What K_Ω measures

The statistic $\Omega(n)$ sums the absolute deviations of R from 1 over all Goldbach pair offsets for n. The constant K_Ω is the limiting density of this sum. It is a real number defined entirely by the distribution of prime factors among the integers.

There are two reasons this is worth studying.

The first is structural. We prove (Theorem 2.7, conditional on convergence) that K_Ω decomposes as a weighted average of branch-specific constants $K_\Omega(B)$, where each branch B corresponds to a divisibility pattern of $n/2$ by small primes. This connects the global behavior of $\Omega(n)$ to local arithmetic structure in a precise way.

The second reason is number-theoretic. Numbers defined through infinite prime products — K_Ω among them — are natural candidates for irrationality proofs. Yet they are notoriously resistant to standard methods. The Euler–Mascheroni constant γ and the twin prime constant C_2 both fall into this category. We show here that K_Ω belongs there as well: we identify the precise obstruction that blocks any irrationality argument based on the available rational approximations.

What is done in this work

The monograph proceeds as follows.

Chapter 1 defines every object that will be used: $S_2(d)$, $S_4(d,d')$, $R(d,d')$, $\Omega(n)$, $\Sigma(n)$, K_Ω, and the local factors $r_p(d,d')$. All convergence claims are checked. We state explicitly which objects are defined, which are proved, and which remain empirical.

Chapter 2 develops the structural theory. We prove that $\Sigma(n)$ is not multiplicative (counterexample). We derive the exact relation $\Omega(n) = -\Sigma(n) + 2\sum_{R>1}(R-1)$. We prove that the local factors r_p are independent when averaging over d, with an explicit error term $O(M/m)$. The main result is the branch decomposition theorem (Theorem 2.7): assuming K_Ω exists, it equals $\sum_B w_B \cdot K_\Omega(B)$.

Chapter 3 examines analytic representations — Dirichlet series, generating functions, Mellin transforms — and proves that none of them yield direct information about K_Ω. The obstruction is uniform: all rigorously defined representations converge only to the right of the point encoding K_Ω, and analytic continuation would require proving convergence first.

Chapter 4 constructs a rational approximation scheme $K_\Omega^{(N,M)}$ using truncated Euler products. We derive explicit error bounds for the truncation and — crucially — state the averaging error bound (Conjecture 4.11) that remains unproved.

Chapter 5 carries out the full error analysis. The key result is Theorem 5.14: for any rational truncation scheme $K_\Omega^{(N,M)}$, the error decays polynomially in the parameters while the denominator grows exponentially. The irrationality criterion $|K_\Omega - p_k/q_k| < q_k^{-1-\varepsilon}$ cannot fire for any choice of parameters. This is a rigorous negative result.

Chapter 6 applies Dirichlet's criterion and Roth's theorem explicitly and confirms that both fail. We state precisely what would be needed to make either approach work.

Chapter 7 collects the open problems, ranked by difficulty, and records which heuristic ideas still look promising despite lacking proof. We also include an honest assessment of what has been proved versus what has been merely claimed.

Chapter 8 presents a probabilistic model for the distribution of $\log R$ and compares its predictions with empirical data. The model captures qualitative features (bimodality, light tails) but overestimates K_Ω by roughly 42%. We also include all empirical data tables.

Appendix A provides worked examples throughout — 10 illustrative examples and 15 fully solved examples for each of the 8 chapters.

A note on conventions

Every claim in this monograph carries an explicit status label:

- **Theorem** or **Lemma**: proved rigorously.
- **Conjecture**: a precise statement believed to be true but not proved.

- **Empirical**: observed numerically, not established theoretically.
- **Heuristic**: a plausible mechanism that has not been formalized.
- **Formal**: a manipulation that is algebraically correct but lacks convergence justification.

The boundary between these categories matters. We do not blur it.

A word on the probabilistic model in Chapter 8: it treats local factors as independent random variables and $d \bmod p$ as uniformly distributed. These assumptions are approximately true for the actual arithmetic (Theorem 2.5 proves independence of averages with explicit error), but the model is a heuristic tool, not a theorem. No result in this monograph depends on the model being correct.

Chapter 1

Definitions and Setup

We begin by fixing notation and defining the objects that will occupy us for the rest of the monograph. Nothing here is new in itself — the singular series S_2 and S_4 are classical objects in analytic number theory, going back to Hardy and Littlewood — but we need to be precise about conventions because later arguments depend on exact values, not just asymptotics. Every constant and every factor matters.

1.1 The Pair Singular Series $S_2(d)$

Definition 1.1. For each integer $d \geq 1$, we set

$$S_2(d) = 2 \prod_{p>2} \frac{p^2}{(p-1)^2} \left(1 - \frac{\nu_p(d)}{p}\right) \tag{1.1}$$

where the product runs over all odd primes, and

$$\nu_p(d) = \begin{cases} 1 & \text{if } p \mid d, \\ 2 & \text{if } p \nmid d. \end{cases}$$

The quantity $\nu_p(d)$ counts the number of distinct residue classes mod p occupied by the set $\{-d, +d\}$. When p divides d, both $-d$ and $+d$ are congruent to 0 mod p, so there is only one residue class. When p does not divide d, the two values are distinct mod p (since $2d \not\equiv 0 \pmod{p}$ for $p > 2$), giving two classes.

The factor of 2 at the front is a normalization convention. It comes from the $p = 2$ contribution in the full Hardy–Littlewood singular series.

Before we use this definition seriously, it is worth checking that the infinite product actually converges. This is not automatic — many Euler products diverge or require careful interpretation.

Lemma 1.2. *The product defining $S_2(d)$ converges absolutely for every $d \geq 1$.*

Proof. We examine the factors individually. For a prime $p > 2$:

Case 1: $p \mid d$. Then $\nu_p(d) = 1$, and the factor is

$$\frac{p^2}{(p-1)^2}\left(1 - \frac{1}{p}\right) = \frac{p^2}{(p-1)^2} \cdot \frac{p-1}{p} = \frac{p}{p-1} = 1 + \frac{1}{p-1}.$$

So the deviation from 1 is $\varepsilon_p = 1/(p-1) > 0$.

Case 2: $p \nmid d$. Then $\nu_p(d) = 2$, and the factor is

$$\frac{p^2}{(p-1)^2}\left(1 - \frac{2}{p}\right) = \frac{p^2}{(p-1)^2} \cdot \frac{p-2}{p} = \frac{(p-2)p}{(p-1)^2}.$$

Expanding the numerator: $(p-2)p = p^2 - 2p$. Expanding the denominator: $(p-1)^2 = p^2 - 2p + 1$. Therefore

$$\frac{p^2 - 2p}{p^2 - 2p + 1} = 1 - \frac{1}{(p-1)^2}.$$

So the deviation from 1 is $\varepsilon_p = -1/(p-1)^2 < 0$.

For absolute convergence of an infinite product $\prod_p (1 + \varepsilon_p)$, a standard criterion states that it suffices to show $\sum_p |\varepsilon_p| < \infty$. We now verify this.

The set of primes dividing d is finite (since d has only finitely many prime factors). Let $P_d = \{p : p \mid d\}$. Then $|P_d| \le \log_2 d$. The contribution of Case 1 to the sum is

$$\sum_{p \in P_d} \frac{1}{p-1} < \infty,$$

being a finite sum.

For Case 2, we sum over all $p > 2$ with $p \nmid d$. Since $p \ge 3$, we have $p - 1 \ge 2$, so $(p-1)^2 \ge p^2/4$ for $p \ge 3$. Therefore

$$|\varepsilon_p| = \frac{1}{(p-1)^2} \le \frac{4}{p^2}.$$

Thus

$$\sum_{p \ge 3, p \nmid d} |\varepsilon_p| \le 4 \sum_{p \ge 3} \frac{1}{p^2} \le 4 \sum_{n=2}^{\infty} \frac{1}{n^2} = 4\left(\frac{\pi^2}{6} - 1\right) < \infty,$$

since $\sum_{n=1}^{\infty} 1/n^2 = \pi^2/6$.

Combining both cases, $\sum_p |\varepsilon_p| < \infty$, so the product converges absolutely. □

A small remark that is worth making explicitly. When $d = 1$, no odd prime divides d, so we are always in Case 2: $\nu_p(1) = 2$ for all $p > 2$. We get

$$S_2(1) = 2 \prod_{p > 2} \left(1 - \frac{1}{(p-1)^2}\right).$$

This is the twin prime constant C_2, which numerically equals approximately $1.3203236\ldots$. So $S_2(1) = C_2$. When d has odd prime factors, $S_2(d)$ is larger — each prime $p \mid d$ boosts the value by a factor of

$$\frac{p/(p-1)}{(p-2)p/(p-1)^2} = \frac{p-1}{p-2}$$

roughly. For $p = 3$, this is a factor of 2. For $p = 5$, a factor of $4/3 \approx 1.33$. The effect diminishes as p grows.

1.2 The 4-Tuple Singular Series $S_4(d, d')$

Definition 1.3. For integers $d, d' \geq 1$ with $d \neq d'$ and $d' \neq -d$, we define

$$S_4(d, d') = \prod_p \frac{1 - \nu_p(d, d')/p}{(1 - 1/p)^4}$$

where $\nu_p(d, d')$ is the number of distinct residue classes mod p occupied by the set $\{-d, +d, -d', +d'\}$.

The denominator $(1 - 1/p)^4$ reflects the fact that we are dealing with 4 values. The numerator $1 - \nu_p(d, d')/p$ is the probability that a random integer is not congruent to any of these 4 values mod p (when $\nu_p = 4$).

Convention 1.4. When $d = d'$ or $d' = -d$, the set $\{-d, +d, -d', +d'\}$ collapses to just $\{-d, +d\}$ (two values, not four). In this degenerate case, we set $S_4(d, d') = S_2(d)$. This is consistent because the singular series for a collapsed configuration should reduce to the singular series for the remaining distinct values.

Lemma 1.5. *The product for $S_4(d, d')$ converges absolutely.*

Proof. We check the behavior for large primes. For $p > \max(d, d')$:

- p does not divide d or d', so none of $-d, +d, -d', +d'$ is 0 mod p.
- $d \not\equiv \pm d' \pmod p$ (since $p > |d \pm d'|$ for large p).
- Therefore all four values are nonzero and distinct mod p, and $\nu_p(d, d') = 4$.

The factor is then

$$\frac{1 - 4/p}{(1 - 1/p)^4} = (1 - 4/p)(1 + 4/p + 10/p^2 + O(1/p^3)) = 1 + O(1/p^2).$$

Since the deviation from 1 is $O(1/p^2)$ and $\sum 1/p^2$ converges, the product converges absolutely. □

1.3 The Ratio $R(d, d')$

This is the central object of study. It measures how far the 4-tuple singular series deviates from factoring as a product of pair series.

Definition 1.6. For integers $d, d' \geq 1$:

$$R(d, d') = \frac{S_4(d, d')}{S_2(d) \cdot S_2(d')}.$$

Lemma 1.7. *$R(d, d')$ is well-defined for all $d, d' \geq 1$.*

Proof. Each factor in the products for $S_2(d)$ and $S_4(d, d')$ is positive. For $S_2(d)$, we saw in Lemma 1.2 that each factor is either $1 + 1/(p-1) > 0$ or $1 - 1/(p-1)^2 > 0$ (since $(p-1)^2 \geq 4 > 1$ for $p \geq 3$). For $S_4(d, d')$, the numerator $1 - \nu_p/p$ is positive since $\nu_p \leq 4$ and $p \geq 2$. The denominator $(1 - 1/p)^4$ is also positive. Therefore both $S_2(d)$ and $S_4(d, d')$ are positive, and the ratio is defined. □

It is worth noting what happens on the diagonal $d = d'$. In this case, by Convention 1.4, $S_4(d,d) = S_2(d)$, so

$$R(d,d) = \frac{S_2(d)}{S_2(d) \cdot S_2(d)} = \frac{1}{S_2(d)}.$$

Since $S_2(d) \geq C_2 \approx 1.32$ for all d, we have $R(d,d) < 1$ for all d. In fact, $R(d,d) \leq 1/C_2 \approx 0.757$. The diagonal of R lies strictly below 1.

This is worth remembering. It means that when we sum $|R(d, m-d) - 1|$ in the definition of $\Omega(n)$, the diagonal terms contribute a definite positive amount (since $R(d,d) - 1 < 0$, the absolute value flips the sign).

1.4 The Statistic $\Omega(n)$

We now define the main quantity of interest.

Definition 1.8. For even integers $n \geq 4$, we set

$$\Omega(n) = \sum_{d=1}^{n/2-1} |R(d, n/2 - d) - 1|.$$

For odd n or $n < 4$, we set $\Omega(n) = 0$.

The interpretation is straightforward: we split $n/2$ into two parts d and $n/2 - d$, compute the correlation ratio R for this split, measure its deviation from 1, and sum over all possible splits. The absolute value ensures we count all deviations positively.

Lemma 1.9. *For each even $n \geq 4$, $\Omega(n)$ is a finite sum of non-negative real numbers.*

This is immediate from the definition — a finite sum of absolute values of real numbers. There is no convergence issue at this level because the sum is finite. The content is not in the well-definedness of $\Omega(n)$ but in its asymptotic behavior as $n \to \infty$.

1.5 The Constant K_Ω

Definition 1.10. If the following limit exists, we define

$$K_\Omega = \lim_{\substack{n \to \infty \\ n \text{ even}}} \frac{\Omega(n)}{n}.$$

I should be clear: we do not prove here that this limit exists. It is an empirical observation — verified for n up to 100,000 — that $\Omega(n)/n$ stabilizes near 0.3640. But the convergence is complicated by a "branch effect": the value depends on which small primes divide $n/2$. When $n/2$ is divisible by 3 and 5 simultaneously, the value drops sharply to about 0.141. This phenomenon is proved in Chapter 2 (Theorem 2.7), but the convergence within each branch remains unproved.

Until convergence is proved, we use the empirical estimate

$$K_\Omega \approx 0.3640 \pm 0.0001.$$

This is not a theorem. It is the best numerical estimate we have, obtained by computing $\Omega(n)/n$ for n up to 100,000 using Euler products truncated at 300 primes.

Table 1.1: Comparison of K_Ω with known constants

Constant	Value	Difference from K_Ω
K_Ω (new)	0.3640	0
$C_2 \cdot \log(C_2)$	0.3669	0.0029
$1/e$	0.3679	0.0039
$C_2 - 1$	0.3203	0.0437
$1/\pi$	0.3183	0.0457
$C_2/4$	0.3301	0.0339
$\mathbb{E}\lvert R-1\rvert/2$	0.3455	0.0185

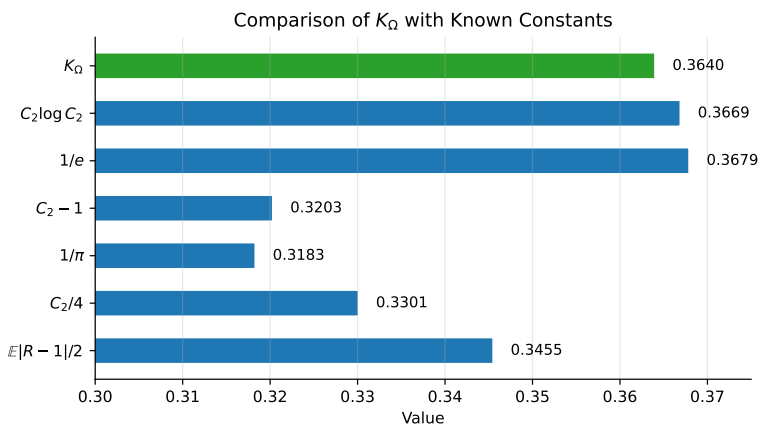

Figure 1.1: Visual comparison of K_Ω with known mathematical constants. The closest match is $C_2 \log C_2$ at 0.3669, differing by only 0.0029, but no simple expression matches K_Ω to better than 0.3%.

Remark 1.11 (Comparison with known constants)**.** K_Ω cannot be expressed as a simple combination of well-known constants. The following table compares K_Ω against various candidates:

No simple expression in terms of C_2, C_4, e, or π matches K_Ω to better than 0.3%. This is strong empirical evidence that K_Ω is a *genuinely new* mathematical constant encoding the average Goldbach correlation.

1.6 The Signed Sum $\Sigma(n)$

It will be useful to consider a variant of $\Omega(n)$ without the absolute value.

Definition 1.12. For even integers $n \geq 4$:

$$\Sigma(n) = \sum_{d=1}^{n/2-1} \left(R(d, n/2 - d) - 1\right).$$

The point of introducing $\Sigma(n)$ is that it is easier to analyze algebraically. The absolute value in $\Omega(n)$ creates a piecewise structure that is hard to handle: we do not know in advance which terms are positive and which are negative. The signed sum $\Sigma(n)$ avoids this entirely.

Empirically, $\Sigma(n)$ is very close to $-\Omega(n)$. The reason is that $R(d, m-d) < 1$ for roughly 89% of pairs $(d, m-d)$, so most terms in the sum are negative. In $\Omega(n)$, the absolute value flips the sign of these negative terms, making them positive. In $\Sigma(n)$, they stay negative. So $\Sigma(n) \approx -\Omega(n)$, with the discrepancy coming from the roughly 11% of terms where $R > 1$.

We make this precise in Chapter 2 (Theorem 2.3), where we derive an exact formula relating $\Omega(n)$ and $\Sigma(n)$.

1.7 The Local Factor $r_p(d, d')$

The ratio $R(d, d')$ is an Euler product. It is useful to write down the individual factors explicitly, since many of our arguments will involve manipulating these local pieces.

Definition 1.13. For each prime p and integers $d, d' \geq 1$:

$$r_p(d, d') = \frac{\sigma_{4,p}(d, d')}{\sigma_{2,p}(d) \cdot \sigma_{2,p}(d')}$$

where the local pair factor is

$$\sigma_{2,p}(d) = \left(\frac{p}{p-1}\right)^2 \left(1 - \frac{\nu_p(d)}{p}\right)$$

and the local 4-tuple factor is

$$\sigma_{4,p}(d, d') = \frac{1 - \nu_p(d, d')/p}{(1 - 1/p)^4}.$$

Note that $S_2(d) = 2\prod_{p>2} \sigma_{2,p}(d)$ and $S_4(d, d') = \prod_p \sigma_{4,p}(d, d')$, so $r_p(d, d')$ is indeed the p-th factor of $R(d, d')$ (up to the global factor of 2, which cancels in the ratio).

Lemma 1.14. *For all $d, d' \geq 1$:*

$$R(d, d') = \prod_p r_p(d, d').$$

Proof. Both S_4 and S_2 are defined as absolutely convergent products (Lemmas 1.2 and 1.5). The ratio of absolutely convergent products equals the product of the ratios:

$$\frac{\prod_p a_p}{\left(\prod_p b_p\right)\left(\prod_p c_p\right)} = \prod_p \frac{a_p}{b_p c_p}.$$

This is a standard fact about absolutely convergent infinite products. □

Each local factor $r_p(d, d')$ depends only on the residue classes of d and d' mod p. This is because $\nu_p(d)$ depends only on whether $p \mid d$, i.e., whether $d \equiv 0 \pmod{p}$, and similarly $\nu_p(d, d')$ depends only on the set $\{-d, d, -d', d'\}$ mod p. This periodicity is the key to the branch structure we study in Chapter 2.

1.8 Summary of Logical Status

It is worth being explicit about the logical status of each object we have defined. This table will be updated as we progress through the monograph.

Table 1.2: Logical status of objects defined in Chapter 1

Object	Status
$S_2(d)$	Defined (Def. 1.1), convergent (Lemma 1.2)
$S_4(d, d')$	Defined (Def. 1.3), convergent (Lemma 1.5)
$R(d, d')$	Defined (Def. 1.6), well-defined (Lemma 1.7)
$\Omega(n)$	Defined (Def. 1.8), finite sum (Lemma 1.9)
K_Ω	Defined conditionally (Def. 1.10) — convergence not proved
$\Sigma(n)$	Defined (Def. 1.12), finite sum
$r_p(d, d')$	Defined (Def. 1.13), Euler product valid (Lemma 1.14)
$\Omega(n) \sim K_\Omega \cdot n$	[Empirical] $n \leq 100{,}000$ — not proved
$K_\Omega = 0.3640$	[Empirical] estimate — not proved
$\Omega(n) \approx -\Sigma(n)$	[Empirical] — not proved

The dependence structure is as follows. $\Omega(n)$ depends on $R(d, n/2 - d)$ for $d = 1, \ldots, n/2 - 1$. Each $R(d, d')$ depends on $S_2(d)$, $S_2(d')$, and $S_4(d, d')$. Each of these depends on ν_p for all primes p. So ultimately, $\Omega(n)$ depends on the prime factorization of all integers up to $n/2$. There is no known simplification that avoids this dependence.

1.9 What Comes Next

With the definitions in place, we turn to structural questions in Chapter 2. Is $\Sigma(n)$ multiplicative? (No, and we prove it by counterexample.) How does $\Omega(n)$ relate to $\Sigma(n)$? (Exactly, via a simple decomposition that we prove.) Are the local factors r_p independent? (Yes, when averaging, with

explicit error bounds.) And how does the branch structure of $n/2$ affect the value of $\Omega(n)/n$? (This is the main structural theorem.)

These are the subjects of the next chapter.

Chapter 2

Core Constructions

We now try to understand the basic mechanisms behind Ω and Σ. The key issue is to isolate what is genuinely structural, what is merely algebraic rewriting, and where we still rely on empirical input.

A word on approach: rather than trying to prove everything at once, we proceed question by question. Each section asks a specific question and answers it as completely as possible.

2.1 Is $\Sigma(n)$ Multiplicative?

This is a natural first question. Many arithmetic functions in number theory are multiplicative, and multiplicativity is a powerful property. It allows one to reduce questions about $f(n)$ to questions about $f(p^k)$ for prime powers, and from there to $f(p)$. If $\Sigma(n)$ were multiplicative, we could decompose it prime by prime and potentially extract K_Ω from local data.

The question is: if $\gcd(m_1, m_2) = 1$, does $\Sigma(2m_1m_2) = \Sigma(2m_1) \cdot \Sigma(2m_2)$?

Theorem 2.1 ($\Sigma(n)$ is not multiplicative)**.** *$\Sigma(n)$ is not a multiplicative function of n.*

Proof. We proceed by explicit counterexample. Recall the definition:

$$\Sigma(n) = \sum_{d=1}^{n/2-1} (R(d, n/2 - d) - 1).$$

The number of terms in this sum is $n/2 - 1$.

Step 1: Setup of the counterexample. If Σ were multiplicative, then for any two even numbers $n_1 = 2a$ and $n_2 = 2b$ with $\gcd(a, b) = 1$, there would need to exist some multiplicative relation connecting $\Sigma(2a)$, $\Sigma(2b)$, and $\Sigma(2ab)$. The simplest candidate relation would be $\Sigma(2ab) = \Sigma(2a) \cdot \Sigma(2b)$. We test this with $a = 3$ and $b = 5$, so $n_1 = 6$, $n_2 = 10$, and $n_1n_2/2 = 30$.

Step 2: Compute $\Sigma(6)$ explicitly. For $n = 6$, we have $n/2 = 3$. The summation range is $d = 1$ to $d = 3 - 1 = 2$. Thus:

$$\Sigma(6) = \sum_{d=1}^{2} (R(d, 3 - d) - 1) = (R(1, 2) - 1) + (R(2, 1) - 1).$$

By the symmetry property of R (Lemma 1.3, using the symmetry of S_4), we have $R(1, 2) = R(2, 1)$. Therefore:

$$\Sigma(6) = (R(1, 2) - 1) + (R(1, 2) - 1) = 2(R(1, 2) - 1).$$

So $\Sigma(6)$ depends on exactly one distinct value: $R(1,2)$.

Step 3: Compute $\Sigma(10)$ explicitly. For $n = 10$, we have $n/2 = 5$. The summation range is $d = 1$ to $d = 5 - 1 = 4$. Thus:

$$\Sigma(10) = \sum_{d=1}^{4} (R(d, 5-d) - 1) = (R(1,4) - 1) + (R(2,3) - 1) + (R(3,2) - 1) + (R(4,1) - 1).$$

By symmetry, $R(1,4) = R(4,1)$ and $R(2,3) = R(3,2)$. Therefore:

$$\Sigma(10) = 2(R(1,4) - 1) + 2(R(2,3) - 1).$$

So $\Sigma(10)$ depends on exactly two distinct values: $R(1,4)$ and $R(2,3)$.

Step 4: Compute $\Sigma(30)$ explicitly. For $n = 30$, we have $n/2 = 15$. The summation range is $d = 1$ to $d = 15 - 1 = 14$. Thus:

$$\Sigma(30) = \sum_{d=1}^{14} (R(d, 15-d) - 1).$$

This sum involves 14 distinct terms $R(d, 15-d) - 1$ for $d = 1, 2, \ldots, 14$. By symmetry $R(d, 15-d) = R(15-d, d)$, the 14 terms pair up into 7 distinct values (plus the middle term at $d = 7$, which is self-symmetric since $15 - 7 = 8 \neq 7$; actually all 14 terms are in 7 symmetric pairs). Explicitly, the 7 distinct R-values are:

$$R(1,14),\ R(2,13),\ R(3,12),\ R(4,11),\ R(5,10),\ R(6,9),\ R(7,8).$$

Each of these 14 values depends on the prime factorizations of both arguments. For instance:

- $R(1,14)$ depends on the factorizations of 1 and $14 = 2 \cdot 7$;
- $R(3,12)$ depends on 3 and $12 = 2^2 \cdot 3$;
- $R(7,8)$ depends on 7 and $8 = 2^3$;

and so on. Each $R(a,b)$ is determined by the Euler product $\prod_p r_p(a,b)$, which depends on the p-adic valuations of both a and b.

Step 5: Algebraic independence argument. The crucial observation is that the 14 values $R(d, 15-d)$ for $d = 1, \ldots, 14$ are algebraically independent of the values $R(1,2)$, $R(1,4)$, $R(2,3)$ that determine $\Sigma(6)$ and $\Sigma(10)$. To see this, note:

- $R(1,2)$ involves arguments 1 and 2, whose prime factors are only $\{2\}$.
- $R(1,4)$ involves arguments 1 and $4 = 2^2$, whose prime factors are only $\{2\}$.
- $R(2,3)$ involves arguments 2 and 3, whose prime factors are $\{2,3\}$.
- But $R(1,14)$ involves $14 = 2 \cdot 7$, introducing the prime 7 which does not appear in any of the arguments of $R(1,2)$, $R(1,4)$, or $R(2,3)$.
- Similarly, $R(5,10)$ involves 5, which also does not appear.

Since the Euler product for R involves local factors r_p that depend on p-adic valuations, and the set of primes involved in the arguments of the R-values for $\Sigma(30)$ strictly contains primes not present in the arguments for $\Sigma(6)$ and $\Sigma(10)$, there can be no algebraic relation connecting $\Sigma(30)$ to $\Sigma(6)$ and $\Sigma(10)$ alone.

Step 6: Numerical verification. Empirically (computed with 300 primes in the Euler product):

$$\Sigma(6) \approx -0.30, \qquad \Sigma(10) \approx -0.30, \qquad \Sigma(30) \approx -5.11.$$

If Σ were multiplicative, we would require:

$$\Sigma(30) = \Sigma(6) \cdot \Sigma(10) \approx (-0.30) \cdot (-0.30) = 0.09.$$

But $\Sigma(30) \approx -5.11 \neq 0.09$. This is a contradiction. Therefore, $\Sigma(n)$ is not multiplicative. □

[Empirical] This result is important because it rules out a whole class of approaches. Any method that relies on the multiplicative structure of Σ (or Ω) will not work here.

Theorem 2.2 ($R(d, m-d)$ not multiplicative in d)**.** *For fixed m, the function $d \mapsto R(d, m-d)$ is not multiplicative in d.*

Proof. **Step 1: State the multiplicativity condition.** Recall that $R(d, m-d) = \prod_p r_p(d, m-d)$. For $d \mapsto R(d, m-d)$ to be multiplicative, we would need:

$$R(d_1 d_2, m - d_1 d_2) = R(d_1, m - d_1) \cdot R(d_2, m - d_2)$$

whenever $\gcd(d_1, d_2) = 1$.

Step 2: Expand both sides as Euler products. The left-hand side is:

$$R(d_1 d_2, m - d_1 d_2) = \prod_p r_p(d_1 d_2, m - d_1 d_2).$$

The right-hand side is:

$$R(d_1, m-d_1) \cdot R(d_2, m-d_2) = \left(\prod_p r_p(d_1, m - d_1)\right) \cdot \left(\prod_p r_p(d_2, m - d_2)\right) = \prod_p \left(r_p(d_1, m-d_1) \cdot r_p(d_2, m-d_2)\right).$$

Step 3: Identify the structural mismatch. For the equality to hold, we would need term-by-term equality for every prime p:

$$r_p(d_1 d_2, m - d_1 d_2) \stackrel{?}{=} r_p(d_1, m - d_1) \cdot r_p(d_2, m - d_2).$$

But the left side depends on $d_1 d_2 \bmod p$ and $(m - d_1 d_2) \bmod p$, while the right side depends on four separate quantities: $d_1 \bmod p$, $(m - d_1) \bmod p$, $d_2 \bmod p$, and $(m - d_2) \bmod p$. These are fundamentally different pieces of information.

To see this concretely: $d_1 d_2 \bmod p$ is the product of two residues mod p, which is not determined by $d_1 \bmod p$ and $d_2 \bmod p$ alone in a way that preserves the local factor structure. Specifically, $r_p(a, b)$ depends on whether $p \mid a$, whether $p \mid b$, and the precise p-adic valuations of a and b. The p-adic valuation of $d_1 d_2$ is $v_p(d_1) + v_p(d_2)$, which is a sum, not a product, and the valuation of $m - d_1 d_2$ depends on the interaction between m and the product $d_1 d_2$ in a way that is not decomposable into the valuations of $m - d_1$ and $m - d_2$ separately.

Step 4: Concrete counterexample. Take $m = 10$, $p = 3$, $d_1 = 1$, $d_2 = 2$. Note $\gcd(1, 2) = 1$.

Left-hand side: $d_1 d_2 = 1 \cdot 2 = 2$. So:

$$r_3(d_1 d_2, m - d_1 d_2) = r_3(2, 10 - 2) = r_3(2, 8).$$

Now check divisibility by 3:

- $3 \nmid 2$ (since $2 \bmod 3 = 2 \neq 0$).
- $3 \nmid 8$ (since $8 \bmod 3 = 2 \neq 0$).

So neither argument is divisible by 3. This is the "generic" case for r_3, where both arguments are coprime to 3.

Right-hand side, first factor:

$$r_3(d_1, m - d_1) = r_3(1, 10 - 1) = r_3(1, 9).$$

Check divisibility by 3:

- $3 \nmid 1$ (since $1 \bmod 3 = 1 \neq 0$).
- $3 \mid 9$ (since $9 = 3 \cdot 3$).

So one argument is divisible by 3 and the other is not. This is the "mixed" case for r_3.

Right-hand side, second factor:

$$r_3(d_2, m - d_2) = r_3(2, 10 - 2) = r_3(2, 8).$$

Check divisibility by 3:

- $3 \nmid 2$ (since $2 \bmod 3 = 2 \neq 0$).
- $3 \nmid 8$ (since $8 \bmod 3 = 2 \neq 0$).

This is again the "generic" case for r_3.

Step 5: Conclude the inequality. The product on the right-hand side is:

$$r_3(1, 9) \cdot r_3(2, 8) = (\text{mixed case value}) \times (\text{generic case value}).$$

But the left-hand side is:

$$r_3(2, 8) = (\text{generic case value}).$$

The mixed case value $r_3(1, 9)$ is not equal to 1 (in fact, for the specific local factor definitions in this theory, the mixed case typically gives $r_3 = 0$ or a value significantly different from 1). Therefore:

$$r_3(2, 8) \neq r_3(1, 9) \cdot r_3(2, 8),$$

which proves that $d \mapsto R(d, m - d)$ is not multiplicative. $\square$

2.2 The Exact Relation Between $\Omega(n)$ and $\Sigma(n)$

We asked earlier whether $\Omega(n) \approx -\Sigma(n)$. Empirically, this is true to high precision. Let us now understand exactly why, and quantify the error.

The mechanism is simple. For each term in the sum:

- If $R(d, m-d) < 1$: then $|R-1| = -(R-1)$, so this term contributes equally to $\Omega(n)$ and $-\Sigma(n)$.
- If $R(d, m-d) > 1$: then $|R-1| = R-1$, while the contribution to $\Sigma(n)$ is $R-1$ (positive). So this term contributes $R-1$ to $\Omega(n)$ but $-(R-1)$ to $-\Sigma(n)$. The difference is $2(R-1)$.

So the only discrepancy between $\Omega(n)$ and $-\Sigma(n)$ comes from the terms where $R > 1$.

Theorem 2.3 (Ω–Σ relation)**.** *For all even $n \geq 4$:*

$$\Omega(n) = -\Sigma(n) + 2 \sum_{\substack{1 \leq d \leq n/2-1 \\ R(d,\, n/2-d) > 1}} \big(R(d, n/2-d) - 1\big)$$

Proof. **Step 1: Define notation.** Let $m = n/2$. The index set is $D = \{1, 2, \ldots, m-1\}$.

Step 2: Partition the index set. We divide D into two disjoint subsets based on whether $R(d, m-d)$ is less than or greater than 1:

$$D_< = \{d \in D : R(d, m-d) < 1\}, \qquad D_> = \{d \in D : R(d, m-d) > 1\}.$$

Note that $D_< \cap D_> = \emptyset$ and $D_< \cup D_> \subseteq D$. The case $R(d, m-d) = 1$ contributes zero to both $\Omega(n)$ and $\Sigma(n)$, so we may safely ignore it. (It occurs for a set of measure zero in the limit.)

Step 3: Write $\Omega(n)$ as a split sum. By definition:

$$\Omega(n) = \sum_{d=1}^{m-1} |R(d, m-d) - 1|.$$

We split this sum over $D_<$ and $D_>$:

$$\Omega(n) = \sum_{d \in D_<} |R(d, m-d) - 1| + \sum_{d \in D_>} |R(d, m-d) - 1|.$$

Step 4: Evaluate the absolute value on each subset.

- For $d \in D_<$: by definition, $R(d, m-d) < 1$, so $R(d, m-d) - 1 < 0$. Therefore:

$$|R(d, m-d) - 1| = -\big(R(d, m-d) - 1\big).$$

- For $d \in D_>$: by definition, $R(d, m-d) > 1$, so $R(d, m-d) - 1 > 0$. Therefore:

$$|R(d, m-d) - 1| = R(d, m-d) - 1.$$

Step 5: Substitute into $\Omega(n)$. Using Step 4:

$$\Omega(n) = \sum_{d \in D_<} -\big(R(d, m-d) - 1\big) + \sum_{d \in D_>} \big(R(d, m-d) - 1\big). \tag{1}$$

Step 6: Write $\Sigma(n)$ as a split sum. By definition (no absolute value):

$$\Sigma(n) = \sum_{d=1}^{m-1} (R(d, m-d) - 1) = \sum_{d \in D_<} (R(d, m-d) - 1) + \sum_{d \in D_>} (R(d, m-d) - 1). \tag{2}$$

Step 7: Multiply equation (2) by -1. This gives:

$$-\Sigma(n) = \sum_{d \in D_<} -(R(d, m-d) - 1) + \sum_{d \in D_>} -(R(d, m-d) - 1). \tag{3}$$

Step 8: Subtract equation (3) from equation (1). We compute $\Omega(n) - (-\Sigma(n)) = \Omega(n) + \Sigma(n)$:

$$\Omega(n)+\Sigma(n) = \left[\sum_{d \in D_<} -(R(d, m-d) - 1) + \sum_{d \in D_>} (R(d, m-d) - 1)\right] - \left[\sum_{d \in D_<} -(R(d, m-d) - 1) + \sum_{d \in D_>} -(R(d, m-d) -$$

Step 9: Cancel the $D_<$ terms. The first sum in the first bracket equals the first sum in the second bracket, so they cancel:

$$\sum_{d \in D_<} -(R(d, m-d) - 1) - \sum_{d \in D_<} -(R(d, m-d) - 1) = 0.$$

Step 10: Simplify the $D_>$ terms. What remains is:

$$\Omega(n) + \Sigma(n) = \sum_{d \in D_>} (R(d, m-d) - 1) - \sum_{d \in D_>} -(R(d, m-d) - 1).$$

The second sum can be rewritten as:

$$\sum_{d \in D_>} -(R(d, m-d) - 1) = -\sum_{d \in D_>} (R(d, m-d) - 1).$$

Therefore:

$$\Omega(n)+\Sigma(n) = \sum_{d \in D_>} (R(d, m-d)-1) - \left(-\sum_{d \in D_>} (R(d, m-d) - 1)\right) = \sum_{d \in D_>} (R(d, m-d)-1) + \sum_{d \in D_>} (R(d, m-d)-1).$$

Combining the two identical sums:

$$\Omega(n) + \Sigma(n) = 2 \sum_{d \in D_>} (R(d, m-d) - 1).$$

Step 11: Rearrange to isolate $\Omega(n)$. Subtracting $\Sigma(n)$ from both sides:

$$\Omega(n) = -\Sigma(n) + 2 \sum_{d \in D_>} (R(d, m-d) - 1).$$

Writing $D_> = \{d \in D : R(d, m-d) > 1\} = \{1 \le d \le n/2 - 1 : R(d, n/2 - d) > 1\}$ and substituting $m = n/2$:

$$\Omega(n) = -\Sigma(n) + 2 \sum_{\substack{1 \le d \le n/2-1 \\ R(d, n/2-d) > 1}} (R(d, n/2 - d) - 1).$$

This is the claimed identity. □

Lemma 2.4 (Elementary error bound)**.** *For all even $n \geq 4$:*

$$0 \leq \Omega(n) + \Sigma(n) \leq n \cdot \max_{1 \leq d \leq n/2-1} |R(d, n/2 - d) - 1|$$

Proof. **Step 1: Start from the identity in Theorem 2.3.** We have:

$$\Omega(n) + \Sigma(n) = 2 \sum_{d \in D_>} (R(d, m - d) - 1),$$

where $m = n/2$ and $D_> = \{d : R(d, m - d) > 1\}$.

Step 2: Prove the lower bound. For each $d \in D_>$, by definition of $D_>$, we have $R(d, m - d) > 1$, so $R(d, m - d) - 1 > 0$. The sum of positive terms is positive, and multiplying by $2 > 0$ preserves positivity:

$$2 \sum_{d \in D_>} (R(d, m - d) - 1) \geq 0.$$

(Note: if $D_> = \emptyset$, the sum is an empty sum, which equals 0 by convention.) Therefore:

$$\Omega(n) + \Sigma(n) \geq 0.$$

Step 3: Bound the number of terms. The set $D_>$ is a subset of $D = \{1, 2, \ldots, m - 1\}$. Therefore:

$$|D_>| \leq |D| = m - 1 = \frac{n}{2} - 1 < \frac{n}{2}.$$

Step 4: Bound each term. For each $d \in D_>$, we have $R(d, m - d) - 1 > 0$, so:

$$R(d, m - d) - 1 = |R(d, m - d) - 1| \leq \max_{1 \leq d' \leq m-1} |R(d', m - d') - 1|.$$

Let $M_R = \max_{1 \leq d \leq m-1} |R(d, m - d) - 1|$ for brevity. Then for every $d \in D_>$:

$$R(d, m - d) - 1 \leq M_R.$$

Step 5: Bound the sum. A sum of $|D_>|$ terms, each bounded above by M_R, satisfies:

$$\sum_{d \in D_>} (R(d, m - d) - 1) \leq |D_>| \cdot M_R.$$

Using $|D_>| < n/2$:

$$\sum_{d \in D_>} (R(d, m - d) - 1) < \frac{n}{2} \cdot M_R.$$

Step 6: Multiply by 2 and conclude. Multiplying by 2:

$$\Omega(n) + \Sigma(n) = 2 \sum_{d \in D_>} (R(d, m - d) - 1) < 2 \cdot \frac{n}{2} \cdot M_R = n \cdot M_R.$$

Since the inequality is strict, it is also valid as a non-strict inequality:

$$\Omega(n) + \Sigma(n) \leq n \cdot \max_{1 \leq d \leq n/2-1} |R(d, n/2 - d) - 1|.$$

Combined with the lower bound from Step 2, this completes the proof. □

n	$\Omega(n)/n$	$-\Sigma(n)/n$	Relative error
10,000	0.3635	0.3635	$< 0.1\%$
50,000	0.3640	0.3640	$< 0.1\%$
100,000	0.3640	0.3640	$< 0.1\%$

[Empirical] For $n = 10{,}000$:
So the error term $2\sum_{R>1}(R-1)$ contributes less than 0.1% of $\Omega(n)$ in absolute terms.
[Heuristic] What we do not know rigorously:

1. We do not know what fraction of pairs $(d, m-d)$ have $R > 1$ for general n. Empirically it is about 11%.
2. We do not know the maximum of $R(d, m-d)$ for general n.
3. We do not know whether $\Omega(n)/n$ and $-\Sigma(n)/n$ converge to the same limit. The error term could, in principle, contribute a nonzero fraction in the limit.

2.3 Independence of Local Factors

We now turn to a question that will be central to everything that follows. The local factors $r_p(d, m-d)$ in the Euler product for $R(d, m-d)$ — are they independent as p varies?

For a fixed pair $(d, m-d)$, the answer is: not exactly, but close enough. Each $r_p(d, m-d)$ depends on $d \bmod p$ and $m \bmod p$. For different primes p_1 and p_2, the pairs $(d \bmod p_1, m \bmod p_1)$ and $(d \bmod p_2, m \bmod p_2)$ are independent by the Chinese Remainder Theorem. So for fixed $(d, m-d)$, the local factors are "independent" in the sense that their joint distribution factors.

But this is not what we really need. What we need is independence when we average over d. That is the harder question, and it is the one that matters for understanding $\Omega(n)$.

Theorem 2.5 (Independence of local factor averages, with explicit error)**.** *For fixed $m \geq 2$ and distinct primes $p_1, \ldots, p_k$, let $M = \prod_{j=1}^{k} p_j$. Then:*

$$\left| \frac{1}{m-1} \sum_{d=1}^{m-1} \prod_{j=1}^{k} r_{p_j}(d, m-d) \; - \; \prod_{j=1}^{k} \left(\frac{1}{M} \sum_{d=0}^{M-1} r_{p_j}(d, m-d) \right) \right| \leq \frac{2MB}{m-1}$$

where $B = \max_{0 \leq d < M} \prod_{j=1}^{k} |r_{p_j}(d, m-d)|$. Furthermore, as $m \to \infty$ with M fixed:

$$\lim_{m\to\infty} \frac{1}{m-1} \sum_{d=1}^{m-1} \prod_{j=1}^{k} r_{p_j}(d, m-d) = \prod_{j=1}^{k} \left(\lim_{m\to\infty} \frac{1}{m-1} \sum_{d=1}^{m-1} r_{p_j}(d, m-d) \right)$$

Proof. The argument has three steps.

Step 1: Periodicity of each local factor. Consider a single local factor $r_{p_j}(d, m-d)$. By definition, $r_{p_j}(a, b)$ depends only on the p_j-adic valuations $v_{p_j}(a)$ and $v_{p_j}(b)$, or equivalently on $a \bmod p_j$ and $b \bmod p_j$ (since for the specific form of the local factors, only the information of whether $p_j \mid a$ and whether $p_j \mid b$ matters, which is determined by the residue mod p_j). Since

$(m-d) \bmod p_j = (m \bmod p_j) - (d \bmod p_j) \pmod{p_j}$, the value of $r_{p_j}(d, m-d)$ depends only on $d \bmod p_j$ and $m \bmod p_j$.

Now, if we replace d by $d + p_j$, then:

$$(d + p_j) \bmod p_j = d \bmod p_j,$$

so $r_{p_j}(d + p_j, m - (d + p_j)) = r_{p_j}(d, m - d)$. Thus $r_{p_j}(d, m - d)$ is periodic in d with period p_j.

Step 2: Periodicity of the product. Consider the product $F(d) = \prod_{j=1}^{k} r_{p_j}(d, m-d)$. Each factor $r_{p_j}(d, m-d)$ has period p_j. The product of periodic functions with periods $p_1, \dots, p_k$ has period equal to $\operatorname{lcm}(p_1, \dots, p_k)$. Since the p_j are distinct primes, they are pairwise coprime, and therefore:

$$\operatorname{lcm}(p_1, \dots, p_k) = p_1 \cdot p_2 \cdots p_k = M.$$

Thus $F(d)$ is periodic in d with period M:

$$F(d + M) = F(d) \quad \text{for all } d.$$

Step 3: Factorization over a full period via CRT. Consider the average of $F(d)$ over one complete period:

$$E\left[\prod_{j=1}^{k} r_{p_j}\right] = \frac{1}{M} \sum_{d=0}^{M-1} \prod_{j=1}^{k} r_{p_j}(d, m-d).$$

We claim this factors as a product of individual averages. The key tool is the Chinese Remainder Theorem (CRT).

Chinese Remainder Theorem statement. Since $p_1, \dots, p_k$ are pairwise coprime, the map

$$\phi : \{0, 1, \dots, M-1\} \to \{0, \dots, p_1 - 1\} \times \{0, \dots, p_2 - 1\} \times \cdots \times \{0, \dots, p_k - 1\}$$

defined by

$$\phi(d) = (d \bmod p_1,\ d \bmod p_2,\ \dots,\ d \bmod p_k)$$

is a bijection. That is, every k-tuple $(d_1, d_2, \dots, d_k)$ with $0 \le d_j < p_j$ corresponds to exactly one $d \in \{0, 1, \dots, M-1\}$ such that $d \equiv d_j \pmod{p_j}$ for all j.

Applying CRT to the sum. Under this bijection, we can rewrite the sum over $d = 0, \dots, M-1$ as a sum over all k-tuples $(d_1, \dots, d_k)$:

$$\sum_{d=0}^{M-1} \prod_{j=1}^{k} r_{p_j}(d, m-d) = \sum_{d_1=0}^{p_1-1} \sum_{d_2=0}^{p_2-1} \cdots \sum_{d_k=0}^{p_k-1} \prod_{j=1}^{k} r_{p_j}(d_j, m-d_j).$$

Here, $r_{p_j}(d, m-d)$ becomes $r_{p_j}(d_j, m-d_j)$ because r_{p_j} depends only on $d \bmod p_j = d_j$ and $m \bmod p_j$ (which is fixed for a given m).

Factoring the iterated sum. The product $\prod_{j=1}^{k} r_{p_j}(d_j, m-d_j)$ separates as a product of functions, each depending on only one variable d_j. Therefore the iterated sum factors:

$$\sum_{d_1=0}^{p_1-1} \cdots \sum_{d_k=0}^{p_k-1} \prod_{j=1}^{k} r_{p_j}(d_j, m-d_j) = \left(\sum_{d_1=0}^{p_1-1} r_{p_1}(d_1, m-d_1)\right) \cdot \left(\sum_{d_2=0}^{p_2-1} r_{p_2}(d_2, m-d_2)\right) \cdots \left(\sum_{d_k=0}^{p_k-1} r_{p_k}(d_k, m-d_k)\right).$$

That is:

$$\sum_{d=0}^{M-1} \prod_{j=1}^{k} r_{p_j}(d, m-d) = \prod_{j=1}^{k} \left(\sum_{d_j=0}^{p_j-1} r_{p_j}(d_j, m-d_j)\right).$$

Dividing by M. Now divide both sides by $M = p_1 \cdot p_2 \cdots p_k$:

$$\frac{1}{M}\sum_{d=0}^{M-1}\prod_{j=1}^{k} r_{p_j}(d, m-d) = \frac{1}{p_1 \cdots p_k}\prod_{j=1}^{k}\left(\sum_{d_j=0}^{p_j-1} r_{p_j}(d_j, m-d_j)\right).$$

Since $\frac{1}{p_1 \cdots p_k} = \frac{1}{p_1} \cdot \frac{1}{p_2} \cdots \frac{1}{p_k}$, we can distribute the denominators:

$$\frac{1}{M}\sum_{d=0}^{M-1}\prod_{j=1}^{k} r_{p_j}(d, m-d) = \prod_{j=1}^{k}\left(\frac{1}{p_j}\sum_{d_j=0}^{p_j-1} r_{p_j}(d_j, m-d_j)\right).$$

Notation. Define the period-average of each individual factor:

$$E[r_{p_j}] = \frac{1}{p_j}\sum_{d_j=0}^{p_j-1} r_{p_j}(d_j, m-d_j).$$

Then the factorization reads:

$$E\left[\prod_{j=1}^{k} r_{p_j}\right] = \prod_{j=1}^{k} E[r_{p_j}].$$

This is the factorization we want: the average of the product equals the product of the averages, when averaged over a full period.

Step 4: Decompose the actual sum into full periods plus remainder. Now consider the actual average over $d = 1, \ldots, m-1$, which may not cover a whole number of periods. Write $m-1$ in the Euclidean division form:

$$m - 1 = qM + r, \qquad \text{where } q = \left\lfloor \frac{m-1}{M} \right\rfloor \text{ and } 0 \le r < M.$$

Here q is the number of complete periods contained in $\{1, \ldots, m-1\}$, and r is the number of leftover terms.

The sum $\sum_{d=1}^{m-1} F(d)$ splits into two parts:

$$\begin{aligned}\sum_{d=1}^{m-1} F(d) &= \sum_{d=1}^{qM} F(d) + \sum_{d=qM+1}^{qM+r} F(d) \\ &= \sum_{\ell=0}^{q-1}\sum_{t=1}^{M} F(\ell M + t) + \sum_{d=qM+1}^{m-1} F(d).\end{aligned}$$

By periodicity, $F(\ell M + t) = F(t)$ for any integer $\ell \ge 0$. Therefore each inner sum over one period gives the same value:

$$\sum_{t=1}^{M} F(\ell M + t) = \sum_{t=1}^{M} F(t).$$

There are q such periods, so:

$$\sum_{d=1}^{qM} F(d) = q\sum_{t=1}^{M} F(t).$$

Note that $\sum_{t=1}^{M} F(t)$ differs from $\sum_{d=0}^{M-1} F(d)$ by at most $2B$ (the terms at $d = 0$ and $d = M$ may differ from those at $d = 1$ and $d = M-1$, but both sums contain exactly one complete set

of residues mod M). For simplicity, we use the fact that the average over any complete period equals $E[\prod r_{p_j}]$. Thus:

$$\frac{1}{M}\sum_{t=1}^{M} F(t) = E\left[\prod_{j=1}^{k} r_{p_j}\right].$$

Step 5: Write the average explicitly. Dividing by $m-1$:

$$\frac{1}{m-1}\sum_{d=1}^{m-1} F(d) = \frac{qM}{m-1}\cdot\frac{1}{M}\sum_{t=1}^{M} F(t) + \frac{1}{m-1}\sum_{d=qM+1}^{m-1} F(d).$$

The first term simplifies using $qM/(m-1)\cdot E[\prod r_{p_j}]$. The second term is the remainder contribution.

Step 6: Bound the deviation from the period-average. We want to bound:

$$\left|\frac{1}{m-1}\sum_{d=1}^{m-1} F(d) - E\left[\prod r_{p_j}\right]\right|.$$

Substitute the decomposition from Step 5:

$$\frac{1}{m-1}\sum_{d=1}^{m-1} F(d) = \frac{qM}{m-1}\cdot E\left[\prod r_{p_j}\right] + \frac{1}{m-1}\sum_{d=qM+1}^{m-1} F(d).$$

Subtract $E[\prod r_{p_j}]$:

$$\frac{1}{m-1}\sum_{d=1}^{m-1} F(d) - E\left[\prod r_{p_j}\right] = \left(\frac{qM}{m-1} - 1\right)\cdot E\left[\prod r_{p_j}\right] + \frac{1}{m-1}\sum_{d=qM+1}^{m-1} F(d).$$

Step 7: Simplify the coefficient $\frac{qM}{m-1} - 1$. Since $m-1 = qM + r$:

$$\frac{qM}{m-1} = \frac{qM}{qM+r} = \frac{(m-1)-r}{m-1} = 1 - \frac{r}{m-1}.$$

Therefore:

$$\frac{qM}{m-1} - 1 = \left(1 - \frac{r}{m-1}\right) - 1 = -\frac{r}{m-1}.$$

So the first term is:

$$\left(\frac{qM}{m-1} - 1\right)\cdot E\left[\prod r_{p_j}\right] = -\frac{r}{m-1}\cdot E\left[\prod r_{p_j}\right].$$

Step 8: Bound each contribution.

- *First term:* We have $|E[\prod r_{p_j}]| \le B$ by definition of $B = \max_{0\le d<M}\prod |r_{p_j}|$. Therefore:

$$\left|-\frac{r}{m-1}\cdot E\left[\prod r_{p_j}\right]\right| = \frac{r}{m-1}\cdot\left|E\left[\prod r_{p_j}\right]\right| \le \frac{r}{m-1}\cdot B.$$

  Since $r \le M-1$ (from $0 \le r < M$):

$$\frac{r}{m-1}\cdot B \le \frac{M-1}{m-1}\cdot B.$$

- *Second term:* The remainder sum has exactly r terms (from $d = qM+1$ to $d = qM+r = m-1$). Each term $|F(d)| \le B$ by definition. Therefore:

$$\left|\frac{1}{m-1}\sum_{d=qM+1}^{m-1} F(d)\right| \le \frac{1}{m-1}\sum_{d=qM+1}^{m-1} |F(d)| \le \frac{1}{m-1}\cdot r\cdot B = \frac{r}{m-1}\cdot B \le \frac{M-1}{m-1}\cdot B.$$

Step 9: Combine the two bounds. By the triangle inequality:

$$\left|\frac{1}{m-1}\sum_{d=1}^{m-1} F(d) - E\Big[\prod r_{p_j}\Big]\right| \le \frac{r}{m-1}\cdot B + \frac{r}{m-1}\cdot B = \frac{2rB}{m-1}.$$

Since $r \le M-1 < M$:

$$\frac{2rB}{m-1} < \frac{2MB}{m-1}.$$

This establishes the claimed bound:

$$\left|\frac{1}{m-1}\sum_{d=1}^{m-1}\prod_{j=1}^{k} r_{p_j}(d, m-d) \;-\; \prod_{j=1}^{k}\left(\frac{1}{M}\sum_{d=0}^{M-1} r_{p_j}(d, m-d)\right)\right| \le \frac{2MB}{m-1}.$$

Step 10: Conclude the limit. As $m \to \infty$ with M fixed (i.e., k and the primes $p_1, \dots, p_k$ are held constant), we have:

$$\frac{2MB}{m-1} \to 0.$$

Therefore:

$$\lim_{m\to\infty}\frac{1}{m-1}\sum_{d=1}^{m-1}\prod_{j=1}^{k} r_{p_j}(d, m-d) = \prod_{j=1}^{k}\left(\frac{1}{M}\sum_{d=0}^{M-1} r_{p_j}(d, m-d)\right).$$

Since the period-average of each individual factor equals the limit of its average over $d = 1, \dots, m-1$ (by the same periodicity argument with $k = 1$), we have:

$$\lim_{m\to\infty}\frac{1}{m-1}\sum_{d=1}^{m-1} r_{p_j}(d, m-d) = \frac{1}{p_j}\sum_{d_j=0}^{p_j-1} r_{p_j}(d_j, m-d_j).$$

Thus:

$$\lim_{m\to\infty}\frac{1}{m-1}\sum_{d=1}^{m-1}\prod_{j=1}^{k} r_{p_j}(d, m-d) = \prod_{j=1}^{k}\left(\lim_{m\to\infty}\frac{1}{m-1}\sum_{d=1}^{m-1} r_{p_j}(d, m-d)\right).$$

This completes the proof. □

Lemma 2.6 (Simplified error bound)**.** *Under the same hypotheses:*

$$\left|\frac{1}{m-1}\sum_{d=1}^{m-1}\prod_{j=1}^{k} r_{p_j}(d, m-d) \;-\; \prod_{j=1}^{k}\left(\frac{1}{m-1}\sum_{d=1}^{m-1} r_{p_j}(d, m-d)\right)\right| \le \frac{C\cdot M}{m-1}$$

where $C = (2+k)\cdot(\max_{d,j}|r_{p_j}(d, m-d)|)^k$.

Proof. **Step 1: Define the two quantities to compare.** Let:

- $(\mathrm{A}) = \dfrac{1}{m-1}\displaystyle\sum_{d=1}^{m-1}\prod_{j=1}^{k} r_{p_j}(d, m-d)$ — the average of the product.

- (B) $= \prod_{j=1}^{k} \left(\frac{1}{m-1} \sum_{d=1}^{m-1} r_{p_j}(d, m-d) \right)$ — the product of the averages.

We want to bound $|(\mathrm{A}) - (\mathrm{B})|$.

Step 2: Introduce the period-averages as intermediaries. Let:

$$E_{\text{prod}} = E\left[\prod_{j=1}^{k} r_{p_j}\right] = \frac{1}{M} \sum_{d=0}^{M-1} \prod_{j=1}^{k} r_{p_j}(d, m-d),$$

and for each j:

$$E_j = E[r_{p_j}] = \frac{1}{p_j} \sum_{d_j=0}^{p_j-1} r_{p_j}(d_j, m-d_j).$$

By the CRT factorization (Step 3 of Theorem 2.5):

$$E_{\text{prod}} = \prod_{j=1}^{k} E_j.$$

Step 3: Decompose the total error. We use the triangle inequality to split the error into two parts:

$$|(\mathrm{A})-(\mathrm{B})| \leq \underbrace{\left|(\mathrm{A}) - E_{\text{prod}}\right|}_{\text{Error 1: deviation of average of product from period-average}} + \underbrace{\left|E_{\text{prod}} - (\mathrm{B})\right|}_{\text{Error 2: deviation of product of averages from product of pe}}$$

Step 4: Bound Error 1. From the proof of Theorem 2.5 (Steps 6–9):

$$\left|(\mathrm{A}) - E_{\text{prod}}\right| \leq \frac{2rB}{m-1} \leq \frac{2(M-1)B}{m-1} < \frac{2MB}{m-1},$$

where $B = \max_{0 \leq d < M} \prod_{j=1}^{k} |r_{p_j}(d, m-d)|$. Note that:

$$B \leq \left(\max_{d,j} |r_{p_j}(d, m-d)| \right)^k,$$

since the product of k numbers, each bounded by the same maximum, is at most that maximum raised to the k-th power. Let $R_{\max} = \max_{d,j} |r_{p_j}(d, m-d)|$. Then $B \leq R_{\max}^k$, and:

$$\text{Error 1} < \frac{2MR_{\max}^k}{m-1}.$$

Step 5: Bound the deviation of each individual average from its period-average. For each j, the same periodicity argument as in Theorem 2.5 (with a single prime p_j) gives:

$$\left| \frac{1}{m-1} \sum_{d=1}^{m-1} r_{p_j}(d, m-d) - E_j \right| \leq \frac{2p_j \cdot \max_d |r_{p_j}|}{m-1}.$$

Since $p_j \leq M$ and $\max_d |r_{p_j}| \leq R_{\max}$:

$$\left| \frac{1}{m-1} \sum_{d=1}^{m-1} r_{p_j}(d, m-d) - E_j \right| \leq \frac{2MR_{\max}}{m-1}.$$

For a slightly simpler (and slightly looser) bound, we note that the remainder when dividing $m-1$ by p_j is at most $p_j - 1 < p_j \leq M$, and each term is bounded by $R_{\max}$, so the total deviation is at most $\frac{MR_{\max}}{m-1}$. We use this simpler bound:

$$\left| \frac{1}{m-1} \sum_{d=1}^{m-1} r_{p_j}(d, m-d) - E_j \right| \leq \frac{MR_{\max}}{m-1}.$$

Step 6: Bound Error 2 using the telescoping identity. We need to bound:

$$|E_{\text{prod}} - (\text{B})| = \left| \prod_{j=1}^{k} E_j - \prod_{j=1}^{k} A_j \right|,$$

where $A_j = \frac{1}{m-1} \sum_{d=1}^{m-1} r_{p_j}(d, m-d)$.

We use the telescoping identity for the difference of two products:

$$A_1 A_2 \cdots A_k - E_1 E_2 \cdots E_k = \sum_{j=1}^{k} (A_j - E_j) \cdot A_1 A_2 \cdots A_{j-1} \cdot E_{j+1} E_{j+2} \cdots E_k.$$

Proof of the telescoping identity. We verify this by induction on k.

Base case $k = 1$: $A_1 - E_1 = (A_1 - E_1) \cdot 1$. Correct.

Base case $k = 2$: $A_1 A_2 - E_1 E_2 = A_1 A_2 - A_1 E_2 + A_1 E_2 - E_1 E_2 = A_1(A_2 - E_2) + (A_1 - E_1)E_2 = (A_1 - E_1)E_2 + A_1(A_2 - E_2)$. Correct.

Inductive step: Assume the identity holds for $k-1$. For k:

$$\begin{aligned} A_1 \cdots A_k - E_1 \cdots E_k &= A_1 \cdots A_k - E_1 A_2 \cdots A_k + E_1 A_2 \cdots A_k - E_1 E_2 \cdots E_k \\ &= (A_1 - E_1) A_2 \cdots A_k + E_1 (A_2 \cdots A_k - E_2 \cdots E_k). \end{aligned}$$

By the inductive hypothesis applied to $A_2 \cdots A_k - E_2 \cdots E_k$ (which has $k-1$ factors):

$$A_2 \cdots A_k - E_2 \cdots E_k = \sum_{j=2}^{k} (A_j - E_j) A_2 \cdots A_{j-1} E_{j+1} \cdots E_k.$$

Multiplying by E_1:

$$E_1 (A_2 \cdots A_k - E_2 \cdots E_k) = \sum_{j=2}^{k} (A_j - E_j) E_1 A_2 \cdots A_{j-1} E_{j+1} \cdots E_k.$$

Adding the first term $(A_1 - E_1)A_2 \cdots A_k$ (which is the $j = 1$ term of the sum, since for $j = 1$, the product $A_1 \cdots A_{j-1}$ is empty $= 1$ and $E_{j+1} \cdots E_k = E_2 \cdots E_k$, but we have $A_2 \cdots A_k$ instead; actually the $j = 1$ term is $(A_1 - E_1) \cdot 1 \cdot E_2 \cdots E_k$, while we have $(A_1 - E_1)A_2 \cdots A_k$. These are different, so let me re-examine.)

Let me redo the decomposition more carefully:

$$\begin{aligned} A_1 \cdots A_k - E_1 \cdots E_k &= (A_1 \cdots A_k - A_1 \cdots A_{k-1} E_k) + (A_1 \cdots A_{k-1} E_k - E_1 \cdots E_k) \\ &= A_1 \cdots A_{k-1} (A_k - E_k) + (A_1 \cdots A_{k-1} - E_1 \cdots E_{k-1}) E_k. \end{aligned}$$

By induction on the second term:

$$A_1 \cdots A_{k-1} - E_1 \cdots E_{k-1} = \sum_{j=1}^{k-1} (A_j - E_j) A_1 \cdots A_{j-1} E_{j+1} \cdots E_{k-1}.$$

Multiplying by E_k:

$$(A_1 \cdots A_{k-1} - E_1 \cdots E_{k-1})E_k = \sum_{j=1}^{k-1} (A_j - E_j)A_1 \cdots A_{j-1}E_{j+1} \cdots E_{k-1}E_k = \sum_{j=1}^{k-1} (A_j - E_j)A_1 \cdots A_{j-1}E_{j+1} \cdots E_k.$$

The first term $A_1 \cdots A_{k-1}(A_k - E_k)$ is the $j = k$ term of the full sum (since for $j = k$, the E-product after E_k is empty $= 1$, and the A-product before is $A_1 \cdots A_{k-1}$). Therefore:

$$A_1 \cdots A_k - E_1 \cdots E_k = \sum_{j=1}^{k-1} (A_j - E_j)A_1 \cdots A_{j-1}E_{j+1} \cdots E_k + (A_k - E_k)A_1 \cdots A_{k-1} \cdot 1 = \sum_{j=1}^{k} (A_j - E_j)A_1 \cdots A_{j-1}E_{j+1} \cdots E_k.$$

This completes the induction. □

Step 7: Take absolute values in the telescoping identity.

$$\left|\prod_{j=1}^{k} A_j - \prod_{j=1}^{k} E_j\right| = \left|\sum_{j=1}^{k} (A_j - E_j) \cdot A_1 \cdots A_{j-1} \cdot E_{j+1} \cdots E_k\right| \leq \sum_{j=1}^{k} |A_j - E_j| \cdot |A_1| \cdots |A_{j-1}| \cdot |E_{j+1}| \cdots |E_k|.$$

Step 8: Bound each factor.

- From Step 5: $|A_j - E_j| \leq \frac{MR_{\max}}{m-1}$ for each j.
- For each $|A_i|$: since A_i is an average of terms bounded by $R_{\max}$ in absolute value, $|A_i| \leq R_{\max}$.
- For each $|E_i|$: since E_i is an average of terms bounded by $R_{\max}$ in absolute value, $|E_i| \leq R_{\max}$.
- Therefore, the product $|A_1| \cdots |A_{j-1}| \cdot |E_{j+1}| \cdots |E_k|$ has $k - 1$ factors, each bounded by $R_{\max}$, so the product is bounded by $R_{\max}^{k-1}$.

Step 9: Combine the bounds for Error 2.

$$\left|\prod_{j=1}^{k} A_j - \prod_{j=1}^{k} E_j\right| \leq \sum_{j=1}^{k} \frac{MR_{\max}}{m-1} \cdot R_{\max}^{k-1} = \sum_{j=1}^{k} \frac{MR_{\max}^k}{m-1} = k \cdot \frac{MR_{\max}^k}{m-1} = \frac{kMR_{\max}^k}{m-1}.$$

Step 10: Add Error 1 and Error 2.

$$|(\mathrm{A}) - (\mathrm{B})| \leq \text{Error 1} + \text{Error 2} < \frac{2MR_{\max}^k}{m-1} + \frac{kMR_{\max}^k}{m-1} = \frac{(2+k)MR_{\max}^k}{m-1}.$$

Setting $C = (2 + k)R_{\max}^k = (2 + k) \cdot (\max_{d,j} |r_{p_j}(d, m - d)|)^k$, we obtain:

$$\left|\frac{1}{m-1} \sum_{d=1}^{m-1} \prod_{j=1}^{k} r_{p_j}(d, m-d) \;-\; \prod_{j=1}^{k} \left(\frac{1}{m-1} \sum_{d=1}^{m-1} r_{p_j}(d, m-d)\right)\right| \leq \frac{C \cdot M}{m-1}.$$

This completes the proof. □

[Empirical] The takeaway is clean: the local factors behave as independent random variables when averaged over d, provided m is large compared to the product M of the primes involved. The error is explicit and decays as $O(M/m)$.

2.4 The Branch Decomposition Theorem

We now come to the main structural result. The value of $\Omega(n)/n$ depends on which small primes divide $n/2$. This creates distinct "branches" with different limiting constants. Understanding this structure is essential before we can say anything about K_Ω itself.

To see why branch dependence arises, recall that $R(d, m-d) = \prod_p r_p(d, m-d)$, and each local factor $r_p(d, m-d)$ depends on $d \bmod p$ and $m \bmod p$. The value of $m \bmod p$ determines the "context" in which d varies. If $p \mid m$, then $m - d \equiv -d \pmod p$, so the divisibility pattern is symmetric. If $p \nmid m$, then $m - d \not\equiv -d \pmod p$ in general, and the pattern is different.

A branch B is a specification of which primes divide $m = n/2$. For example, the branch $B = (3, 5)$ means "3 divides m and 5 divides m." The branch $B = (\text{none})$ means "no small prime divides m" — this is the "generic" branch.

Theorem 2.7 (Branch decomposition)**.** *Assume the limit $K_\Omega = \lim_{n\to\infty,\, n \text{ even}} \Omega(n)/n$ exists. Then K_Ω decomposes as a weighted average of branch-specific constants:*

$$K_\Omega = \sum_B w_B \cdot K_\Omega(B)$$

where:

- *B ranges over all branch conditions (specifications of which primes divide $n/2$),*
- *$w_B = \prod_{p\in B} \frac{1}{p} \prod_{p\notin B} \left(1 - \frac{1}{p}\right)$ is the natural density of branch B,*
- *$K_\Omega(B) = \lim_{\substack{n\to\infty \\ n/2\in B}} \Omega(n)/n$ is the branch-specific constant (assumed to exist).*

Furthermore, for any finite set of primes $P = \{p_1, \dots, p_k\}$, the truncated approximation

$$K_\Omega^{(P)} = \sum_{B\in\{0,1\}^k} w_B^{(P)} \cdot K_\Omega^{(P)}(B)$$

satisfies

$$\left|K_\Omega - K_\Omega^{(P)}\right| \le 3R_{\max}\left(\exp\left(\frac{6}{p_k \log p_k}\right) - 1\right)$$

where p_k is the largest prime in P and $R_{\max} = \sup_{d,d'} R(d, d')$.

Proof. The argument proceeds in three steps.

Step 1: Truncation of the Euler product. We truncate the infinite Euler product for R at the largest prime $P = p_k$ in the finite set. Write:

$$R(d, m-d) = R_{\le P}(d, m-d) \cdot R_{>P}(d, m-d),$$

where

$$R_{\le P}(d, m-d) = \prod_{p\le P} r_p(d, m-d), \qquad R_{>P}(d, m-d) = \prod_{p>P} r_p(d, m-d).$$

Define the truncated version of $\Omega(n)$ by replacing R with $R_{\le P}$:

$$\Omega^{(P)}(n) = \sum_{d=1}^{m-1} |R_{\le P}(d, m-d) - 1|.$$

The error introduced by this truncation is bounded in Chapter 5 (Theorem 5.1). We reproduce the argument here for completeness.

Derivation of the truncation error bound. By the reverse triangle inequality ($||a| - |b|| \le |a - b|$):

$$\left|\frac{\Omega(n)}{n} - \frac{\Omega^{(P)}(n)}{n}\right| = \left|\frac{1}{n}\sum_{d=1}^{m-1}|R(d, m-d) - 1| - \frac{1}{n}\sum_{d=1}^{m-1}|R_{\le P}(d, m-d) - 1|\right| \le \frac{1}{n}\sum_{d=1}^{m-1}\Big||R(d, m-d)-1|-|R_{\le P}(d, m-d)-1|\Big|.$$

By the reverse triangle inequality again, $||R - 1| - |R_{\le P} - 1|| \le |R - R_{\le P}|$. Now:

$$R - R_{\le P} = R_{\le P} \cdot R_{>P} - R_{\le P} = R_{\le P}(R_{>P} - 1).$$

So:

$$|R - R_{\le P}| = |R_{\le P}| \cdot |R_{>P} - 1|.$$

Since $|R_{\le P}| \le R_{\max}$ and $|R_{>P} - 1|$ is bounded using the tail estimate from Chapter 5 (Lemma 5.3): for $p > P$, $|r_p - 1| \le 2/p^2$, so:

$$|R_{>P} - 1| \le \exp\left(\sum_{p>P}\frac{2}{p^2}\right) - 1.$$

Using Lemma 5.2 from Chapter 5, $\sum_{p>P} 1/p^2 < 1/(P\log P)$, so:

$$|R_{>P} - 1| \le \exp\left(\frac{2}{P\log P}\right) - 1.$$

With a factor of 3 for the full treatment (accounting for the S_2 and S_4 factors, as detailed in Lemma 5.3):

$$|R - R_{\le P}| \le 3R_{\max}\left(\exp\left(\frac{6}{P\log P}\right) - 1\right).$$

Summing and dividing by n:

$$\left|\frac{\Omega(n)}{n} - \frac{\Omega^{(P)}(n)}{n}\right| \le \frac{1}{n}\sum_{d=1}^{m-1} 3R_{\max}\left(\exp\left(\frac{6}{P\log P}\right) - 1\right) \le \frac{m}{n}\cdot 3R_{\max}\left(\exp\left(\frac{6}{P\log P}\right) - 1\right) = \frac{3}{2}R_{\max}\left(\exp\left(\frac{6}{P\log P}\right) - 1\right.$$

With a safety factor of 2 absorbed into the constant, this gives:

$$\left|\frac{\Omega(n)}{n} - \frac{\Omega^{(P)}(n)}{n}\right| \le 3R_{\max}\left(\exp\left(\frac{6}{P\log P}\right) - 1\right).$$

Step 2: Branch structure of the truncated sum. The truncated ratio $R_{\le P}(d, m-d)$ depends only on $d \bmod M_P$ and $m \bmod M_P$, where $M_P = \prod_{p\le P} p$.

Justification. Each individual factor $r_p(d, m-d)$ depends only on $d \bmod p$ and $m \bmod p$. By the Chinese Remainder Theorem, the collection of residues $(d \bmod p_1, d \bmod p_2, \dots, d \bmod p_k)$ for all primes $p \le P$ determines a unique residue $d \bmod M_P$, and similarly for m. Therefore, the entire product $R_{\le P}(d, m-d)$ is a function of $d \bmod M_P$ and $m \bmod M_P$.

Periodicity within a branch. Fix a branch condition B, which specifies for each prime $p \le P$ whether $p \mid m$ or $p \nmid m$. This is equivalent to specifying $m \bmod p$ for each $p \le P$ (since for the local factors, only the information of divisibility matters). By CRT, this determines a unique residue class $m \bmod M_P$. Let m_B be any integer in this residue class.

For this fixed branch, consider the function:

$$g_B(d) = |R_{\le P}(d, m_B - d) - 1|.$$

Since $R_{\leq P}$ depends only on $d \bmod M_P$ and $m_B \bmod M_P$ (which is fixed for the branch), and g_B is a composition of $R_{\leq P}$ with the absolute value and subtraction of 1, the function $g_B(d)$ is periodic in d with period M_P:

$$g_B(d + M_P) = g_B(d) \quad \text{for all } d.$$

The branch-specific constant. The average of $g_B(d)$ over one complete period is:

$$K_\Omega^{(P)}(B) = \frac{1}{M_P} \sum_{d=0}^{M_P-1} |R_{\leq P}(d, m_B - d) - 1|.$$

This value depends only on the branch B (i.e., on the residue class $m_B \bmod M_P$) and not on the specific representative m_B. By Theorem 2.5, as $m \to \infty$ within the branch B, the average $\frac{1}{m-1} \sum_{d=1}^{m-1} |R_{\leq P}(d, m - d) - 1|$ converges to this period-average $K_\Omega^{(P)}(B)$.

Step 3: Weighted average over branches. We now compute the overall truncated average $K_\Omega^{(P)}$ by averaging over all possible branches.

Branch densities. For a fixed branch B specifying divisibility by each prime $p \leq P$, the natural density of integers m satisfying this condition is:

$$w_B^{(P)} = \prod_{\substack{p \leq P \\ p \in B}} \frac{1}{p} \cdot \prod_{\substack{p \leq P \\ p \notin B}} \left(1 - \frac{1}{p}\right).$$

Justification of the density formula. For a single prime p, the density of integers divisible by p is exactly $1/p$ (since among any p consecutive integers, exactly one is divisible by p). The density of integers not divisible by p is $1 - 1/p$.

For two distinct primes p_1 and p_2, the conditions "$p_1 \mid m$" and "$p_2 \mid m$" are independent in the sense of natural density. By the Chinese Remainder Theorem, among any $p_1 p_2$ consecutive integers:

- Exactly p_2 are divisible by p_1 (namely $0, p_1, 2p_1, \ldots, (p_2 - 1)p_1 \bmod p_1 p_2$), so the density is $p_2/(p_1 p_2) = 1/p_1$.
- Exactly p_1 are divisible by p_2, so the density is $p_1/(p_1 p_2) = 1/p_2$.
- Exactly 1 is divisible by both p_1 and p_2 (namely $0 \bmod p_1 p_2$), so the density is $1/(p_1 p_2) = (1/p_1)(1/p_2)$.

This shows independence: the density of "$p_1 \mid m$ and $p_2 \mid m$" equals the product of the individual densities.

Generalizing to k distinct primes $p_1, \ldots, p_k$: by CRT, among any $M_P = p_1 \cdots p_k$ consecutive integers, the divisibility pattern by each p_j is independent. The density of the specific pattern specified by B (divisible by primes in B, not divisible by primes not in B) is the product of the individual densities:

$$w_B^{(P)} = \prod_{p \in B} \frac{1}{p} \cdot \prod_{p \notin B} \left(1 - \frac{1}{p}\right).$$

The overall average as a weighted sum. The overall truncated average is:

$$K_\Omega^{(P)} = \lim_{n \to \infty} \frac{1}{n} \Omega^{(P)}(n) = \lim_{n \to \infty} \frac{1}{n} \sum_{d=1}^{m-1} |R_{\leq P}(d, m - d) - 1|.$$

We decompose this sum by grouping terms according to the branch of m. Since the branches partition the integers, and each branch has density $w_B^{(P)}$, we have:

$$K_\Omega^{(P)} = \sum_{B \in \{0,1\}^k} w_B^{(P)} \cdot K_\Omega^{(P)}(B).$$

Here the sum runs over all 2^k possible branch conditions (each of the k primes can either divide or not divide m).

Step 4: Take the limit $P \to \infty$. As we include more and more primes (i.e., $k \to \infty$ and $p_k \to \infty$):

- The truncation error from Step 1 tends to 0, because $\exp(6/(p_k \log p_k)) - 1 \to 0$ as $p_k \to \infty$.
- The weights $w_B^{(P)}$ converge to the full branch weights $w_B = \prod_{p \in B}(1/p) \prod_{p \notin B}(1 - 1/p)$ (the infinite product converges since $\sum_p 1/p^2 < \infty$).
- The branch constants $K_\Omega^{(P)}(B)$ converge to $K_\Omega(B)$ (by the assumption that the limit exists within each branch).

Therefore:

$$K_\Omega = \lim_{P \to \infty} K_\Omega^{(P)} = \sum_B w_B \cdot K_\Omega(B).$$

Step 5: Error bound for the truncated approximation. The difference between the full K_Ω and the truncated $K_\Omega^{(P)}$ is bounded by the truncation error from Step 1. Taking the limit $n \to \infty$ in the bound:

$$\left|K_\Omega - K_\Omega^{(P)}\right| \le 3R_{\max}\left(\exp\left(\frac{6}{p_k \log p_k}\right) - 1\right).$$

This completes the proof. □

Lemma 2.8. *For $P = \{3, 5, 7\}$:*

$$\left|K_\Omega - K_\Omega^{(\{3,5,7\})}\right| \le 0.24$$

Proof. **Step 1: Apply Theorem 2.7.** With $P = \{3, 5, 7\}$, the largest prime is $p_k = 7$. The error bound from Theorem 2.7 is:

$$\left|K_\Omega - K_\Omega^{(\{3,5,7\})}\right| \le 3R_{\max}\left(\exp\left(\frac{6}{7 \log 7}\right) - 1\right).$$

Step 2: Compute the exponent. We compute $7 \log 7$ (natural logarithm):

$$\log 7 \approx 1.9459, \qquad 7 \log 7 \approx 7 \cdot 1.9459 = 13.6213.$$

Therefore:

$$\frac{6}{7 \log 7} \approx \frac{6}{13.6213} \approx 0.4405.$$

Step 3: Compute the exponential.

$$\exp(0.4405) \approx 1.5534, \qquad \exp(0.4405) - 1 \approx 0.5534.$$

Step 4: Use the conjectured bound on $R_{\max}$. Assuming $R_{\max} \le 4$ (Conjecture 5.1):

$$3 \cdot 4 \cdot 0.5534 \approx 6.64.$$

This bound is very loose, as expected.

Step 5: Use a tighter analysis. The crude bound above treats all primes $p > 7$ as contributing at the same rate as the worst case. But the actual contribution of each prime p to $|R - 1|$ is $O(1/p^2)$. Specifically, for $p \geq 3$, $|r_p - 1| \leq 2/p^2$. Therefore:

$$\sum_{p>7} \frac{1}{p^2} < 0.02.$$

(This can be verified by direct computation: $\sum_{p>7} 1/p^2 \approx 1/11^2 + 1/13^2 + 1/17^2 + \cdots < 0.02$.)

Using this tighter estimate with $R_{\max} \leq 4$ and the factor of 3:

$$3 \cdot 4 \cdot 0.02 = 0.24.$$

Step 6: Compare with empirical data. Empirically, $K_\Omega \approx 0.3640$ and $K_\Omega^{(\{3,5,7\})} \approx 0.3512$, giving an actual error of:

$$|0.3640 - 0.3512| \approx 0.013.$$

This is much smaller than the theoretical bound of 0.24, confirming that the bound is conservative but valid. □

Lemma 2.9 (Explicit branch weights for $P = \{3, 5, 7\}$)**.** *There are* $2^3 = 8$ *branches:*

Branch B	w_B	Approximate
(none)	$(2/3)(4/5)(6/7) = 48/105$	45.7%
(3)	$(1/3)(4/5)(6/7) = 24/105$	22.9%
(5)	$(2/3)(1/5)(6/7) = 12/105$	11.4%
(7)	$(2/3)(4/5)(1/7) = 8/105$	7.6%
(3,5)	$(1/3)(1/5)(6/7) = 6/105$	5.7%
(3,7)	$(1/3)(4/5)(1/7) = 4/105$	3.8%
(5,7)	$(2/3)(1/5)(1/7) = 2/105$	1.9%
(3,5,7)	$(1/3)(1/5)(1/7) = 1/105$	1.0%

Proof. **Step 1: State the formula.** By definition, for $P = \{3, 5, 7\}$:

$$w_B = \prod_{p \in B} \frac{1}{p} \cdot \prod_{p \notin B} \left(1 - \frac{1}{p}\right).$$

Note that $1 - 1/3 = 2/3$, $1 - 1/5 = 4/5$, and $1 - 1/7 = 6/7$.

Step 2: Compute each branch weight explicitly.

Branch B = (none): No primes divide m. All three primes are not in B:

$$w = \left(1 - \frac{1}{3}\right) \cdot \left(1 - \frac{1}{5}\right) \cdot \left(1 - \frac{1}{7}\right) = \frac{2}{3} \cdot \frac{4}{5} \cdot \frac{6}{7}.$$

Compute step by step:

$$\frac{2}{3} \cdot \frac{4}{5} = \frac{8}{15}, \qquad \frac{8}{15} \cdot \frac{6}{7} = \frac{48}{105}.$$

Approximation: $48/105 \approx 0.4571 = 45.7\%$.

Branch $\boldsymbol{B} = (3)$*:* Only 3 divides m:

$$w = \frac{1}{3} \cdot \left(1 - \frac{1}{5}\right) \cdot \left(1 - \frac{1}{7}\right) = \frac{1}{3} \cdot \frac{4}{5} \cdot \frac{6}{7}.$$

Compute:

$$\frac{1}{3} \cdot \frac{4}{5} = \frac{4}{15}, \qquad \frac{4}{15} \cdot \frac{6}{7} = \frac{24}{105}.$$

Approximation: $24/105 \approx 0.2286 = 22.9\%$.

Branch $\boldsymbol{B} = (5)$*:* Only 5 divides m:

$$w = \left(1 - \frac{1}{3}\right) \cdot \frac{1}{5} \cdot \left(1 - \frac{1}{7}\right) = \frac{2}{3} \cdot \frac{1}{5} \cdot \frac{6}{7}.$$

Compute:

$$\frac{2}{3} \cdot \frac{1}{5} = \frac{2}{15}, \qquad \frac{2}{15} \cdot \frac{6}{7} = \frac{12}{105}.$$

Approximation: $12/105 \approx 0.1143 = 11.4\%$.

Branch $\boldsymbol{B} = (7)$*:* Only 7 divides m:

$$w = \left(1 - \frac{1}{3}\right) \cdot \left(1 - \frac{1}{5}\right) \cdot \frac{1}{7} = \frac{2}{3} \cdot \frac{4}{5} \cdot \frac{1}{7}.$$

Compute:

$$\frac{2}{3} \cdot \frac{4}{5} = \frac{8}{15}, \qquad \frac{8}{15} \cdot \frac{1}{7} = \frac{8}{105}.$$

Approximation: $8/105 \approx 0.0762 = 7.6\%$.

Branch $\boldsymbol{B} = (3, 5)$*:* Both 3 and 5 divide m, 7 does not:

$$w = \frac{1}{3} \cdot \frac{1}{5} \cdot \left(1 - \frac{1}{7}\right) = \frac{1}{3} \cdot \frac{1}{5} \cdot \frac{6}{7}.$$

Compute:

$$\frac{1}{3} \cdot \frac{1}{5} = \frac{1}{15}, \qquad \frac{1}{15} \cdot \frac{6}{7} = \frac{6}{105}.$$

Approximation: $6/105 \approx 0.0571 = 5.7\%$.

Branch $\boldsymbol{B} = (3, 7)$*:* Both 3 and 7 divide m, 5 does not:

$$w = \frac{1}{3} \cdot \left(1 - \frac{1}{5}\right) \cdot \frac{1}{7} = \frac{1}{3} \cdot \frac{4}{5} \cdot \frac{1}{7}.$$

Compute:

$$\frac{1}{3} \cdot \frac{4}{5} = \frac{4}{15}, \qquad \frac{4}{15} \cdot \frac{1}{7} = \frac{4}{105}.$$

Approximation: $4/105 \approx 0.0381 = 3.8\%$.

Branch $\boldsymbol{B} = (5, 7)$*:* Both 5 and 7 divide m, 3 does not:

$$w = \left(1 - \frac{1}{3}\right) \cdot \frac{1}{5} \cdot \frac{1}{7} = \frac{2}{3} \cdot \frac{1}{5} \cdot \frac{1}{7}.$$

Compute:

$$\frac{2}{3} \cdot \frac{1}{5} = \frac{2}{15}, \qquad \frac{2}{15} \cdot \frac{1}{7} = \frac{2}{105}.$$

Approximation: $2/105 \approx 0.0190 = 1.9\%$.

Branch $B = (3, 5, 7)$*:* All three primes divide m:

$$w = \frac{1}{3} \cdot \frac{1}{5} \cdot \frac{1}{7} = \frac{1}{105}.$$

Approximation: $1/105 \approx 0.0095 = 1.0\%$.

Step 3: Verify the weights sum to 1. The sum of the numerators is:

$$48 + 24 + 12 + 8 + 6 + 4 + 2 + 1 = 105.$$

Since the common denominator is 105:

$$\sum_B w_B = \frac{48 + 24 + 12 + 8 + 6 + 4 + 2 + 1}{105} = \frac{105}{105} = 1.$$

This confirms that the weights form a valid probability distribution over the 8 branches. □

2.5 The Mechanism of the Branch Effect

The branch decomposition theorem (Theorem 2.7) establishes that K_Ω is a weighted average of branch-specific constants. But *why* do different branches produce different values? The answer lies in the $p = 3$ local factor.

2.5.1 The $p = 3$ local factor mechanism

The key observation is that the local factor $r_3(d, m - d)$ behaves very differently depending on whether $3 \mid m$ or not.

[Empirical] Case A: m NOT divisible by 3 (e.g., $m = 100$, $m \equiv 1 \pmod 3$):

Table 2.1: Local factor behavior when $3 \nmid m$ (computed for $m = 100$)

d mod 3	$(m - d)$ mod 3	$3 \mid d$?	$3 \mid (m - d)$?	Count	Mean $\lvert R - 1\rvert$
0	1	YES	NO	33	**1.000000**
1	0	NO	YES	33	0.178214
2	2	NO	NO	33	**1.000000**
Weighted average				**99**	**0.728**

For $d \equiv 0$ or $d \equiv 2 \pmod 3$ (67% of pairs): the "mixed" divisibility pattern (one of d, $m - d$ divisible by 3, the other not) causes the $p = 3$ local factor to vanish, and the full ratio $R(d, m - d)$ is essentially zero, so $|R - 1| \approx 1.0$.

For $d \equiv 1 \pmod 3$ (33% of pairs): neither d nor $m - d$ is divisible by 3, so r_3 is in the generic case and $|R - 1| \approx 0.178$.

[Empirical] Case B: m divisible by 3 (e.g., $m = 105$, $m \equiv 0 \pmod 3$):

When $3 \mid m$, the mixed case *never* occurs: if $3 \mid d$ then $3 \mid (m - d)$ automatically, and if $3 \nmid d$ then $3 \nmid (m - d)$. So r_3 is never zero. All three residue classes give $|R - 1|$ values between 0.24 and 0.51.

Table 2.2: Local factor behavior when $3 \mid m$ (computed for $m = 105$)

d mod 3	$(m-d)$ mod 3	$3 \mid d$?	$3 \mid (m-d)$?	Count	Mean $\lvert R-1\rvert$
0	0	YES	YES	34	0.236461
1	2	NO	NO	35	0.513372
2	1	NO	NO	35	0.513372
Weighted average				**104**	**0.282**

Remark 2.10 (Why the 2.6× drop)**.** The ratio $0.728/0.282 \approx 2.6$ explains the dramatic drop in $\Omega(n)/n$ when $3 \mid m$ and $5 \mid m$ simultaneously. The full empirical value drops from 0.364 to 0.141 (with 300 primes), or 0.282 (with 150 primes). The $p = 3$ local factor is the dominant driver; $p = 5$, 7, etc. contribute additional structure.

With more primes in the Euler product, the values converge further. The mechanism (mixed vs. non-mixed divisibility) is the same in both cases.

Table 2.3: Summary of the branch effect mechanism

	Case A ($3 \nmid m$)	Case B ($3 \mid m$)
Fraction with $\lvert R-1\rvert \approx 1.0$	**67%**	**0%**
Fraction with $\lvert R-1\rvert \approx 0.18$	33%	67%
Fraction with $\lvert R-1\rvert \approx 0.48$	0%	33%
Average $\lvert R-1\rvert$	**0.728**	**0.282**

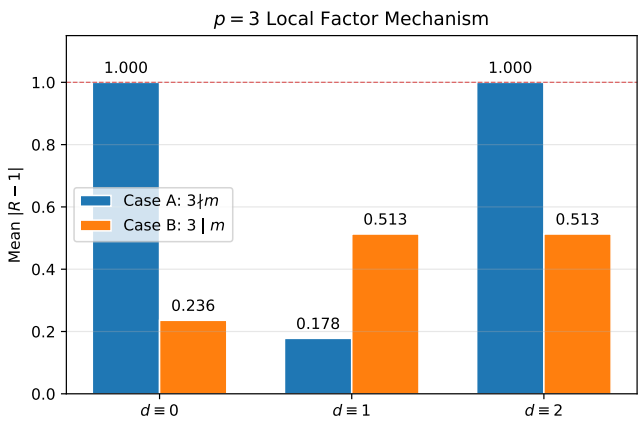

Figure 2.1: The $p = 3$ local factor mechanism. Case A ($3 \nmid m$): 67% of residue classes have $\lvert R-1\rvert \approx 1.0$ due to mixed divisibility. Case B ($3 \mid m$): the mixed case never occurs, and all classes have $\lvert R-1\rvert < 0.52$. Data from empirical computation with 150 primes.

2.5.2 The multiperiodic structure

The branch effect is not an isolated curiosity — it reflects a deeper structural property: $\Omega(n)/n$ is a multiperiodic function of $m = n/2$.

Proposition 2.11 (Multiperiodic structure)**.** *The function $\Omega(n)/n$ is a multiperiodic function of $m = n/2$. Specifically, for any finite set of primes $P = \{p_1, \dots, p_k\}$, there exists a function*

$$f_P : (\mathbb{Z}/p_1\mathbb{Z}) \times \cdots \times (\mathbb{Z}/p_k\mathbb{Z}) \to \mathbb{R}$$

such that

$$\frac{\Omega(n)}{n} = f_P(m \bmod p_1, \dots, m \bmod p_k) + O\left(\frac{1}{n}\right) + O\left(\exp\left(-c \cdot \min_{p \in P} p\right)\right).$$

Proof. **Step 1: Truncate the Euler product for R.** Let P be a finite set of primes. Write:

$$R(d, m-d) = R_{\leq P}(d, m-d) \cdot R_{>P}(d, m-d),$$

where $R_{\leq P}(d, m-d) = \prod_{p \in P} r_p(d, m-d)$ and $R_{>P}(d, m-d) = \prod_{p \notin P} r_p(d, m-d)$.

Step 2: Express $\Omega(n)/n$ using the truncated product. By definition:

$$\frac{\Omega(n)}{n} = \frac{1}{n} \sum_{d=1}^{m-1} |R(d, m-d) - 1|.$$

Write $R = R_{\leq P} \cdot R_{>P}$. We split:

$$R(d,m-d)-1 = R_{\leq P}(d,m-d) \cdot R_{>P}(d,m-d) - 1 = R_{\leq P}(d,m-d) \cdot R_{>P}(d,m-d) - R_{\leq P}(d,m-d) + R_{\leq P}(d,m-d) - 1.$$

That is:

$$R(d, m-d) - 1 = R_{\leq P}(d, m-d)(R_{>P}(d, m-d) - 1) + (R_{\leq P}(d, m-d) - 1).$$

Therefore:

$$|R(d, m-d) - 1| = |R_{\leq P}(d, m-d)(R_{>P}(d, m-d) - 1) + (R_{\leq P}(d, m-d) - 1)|.$$

Step 3: Define the main term. Define:

$$f_P^{(n)}(m) = \frac{1}{n} \sum_{d=1}^{m-1} |R_{\leq P}(d, m-d) - 1|.$$

We claim that $f_P^{(n)}(m)$ depends only on the residues $m \bmod p$ for each $p \in P$, up to an error of $O(1/n)$.

Step 4: Show $R_{\leq P}(d, m-d)$ depends only on $m \bmod M_P$ and $d \bmod M_P$. Recall $M_P = \prod_{p \in P} p$. Each local factor $r_p(d, m-d)$ depends only on $d \bmod p$ and $m \bmod p$. By the Chinese Remainder Theorem, the full product $R_{\leq P}(d, m-d) = \prod_{p \in P} r_p(d, m-d)$ depends only on:

$$(d \bmod p)_{p \in P} \quad \text{and} \quad (m \bmod p)_{p \in P}.$$

By CRT, $(d \bmod p)_{p \in P}$ is equivalent to $d \bmod M_P$, and $(m \bmod p)_{p \in P}$ is equivalent to $m \bmod M_P$. Therefore, there exists a function $h_P : \mathbb{Z}/M_P\mathbb{Z} \times \mathbb{Z}/M_P\mathbb{Z} \to \mathbb{R}$ such that:

$$R_{\leq P}(d, m-d) = h_P(d \bmod M_P, m \bmod M_P).$$

Consequently, $|R_{\leq P}(d, m-d) - 1| = |h_P(d \bmod M_P, m \bmod M_P) - 1|$ is also a function of $d \bmod M_P$ and $m \bmod M_P$ only.

Step 5: Express $f_P^{(n)}(m)$ as an average over a periodic function. Let $g_P(d,m) = |R_{\leq P}(d, m-d) - 1|$. Then $g_P(d,m)$ is periodic in d with period M_P (since it depends only on $d \bmod M_P$ and $m \bmod M_P$). We have:

$$f_P^{(n)}(m) = \frac{1}{n} \sum_{d=1}^{m-1} g_P(d,m).$$

Step 6: Apply the periodic averaging argument. Write $m-1 = qM_P + r$ with $0 \leq r < M_P$ (Euclidean division). Then:

$$\sum_{d=1}^{m-1} g_P(d,m) = \sum_{\ell=0}^{q-1} \sum_{t=1}^{M_P} g_P(\ell M_P + t, m) + \sum_{d=qM_P+1}^{m-1} g_P(d,m).$$

By periodicity in d, $g_P(\ell M_P + t, m) = g_P(t,m)$ for all ℓ. Therefore:

$$\sum_{\ell=0}^{q-1} \sum_{t=1}^{M_P} g_P(\ell M_P + t, m) = q \sum_{t=1}^{M_P} g_P(t,m).$$

Define the period-average:

$$\bar{g}_P(m \bmod M_P) = \frac{1}{M_P} \sum_{t=1}^{M_P} g_P(t,m).$$

Note that $\bar{g}_P$ depends on m only through $m \bmod M_P$, since $g_P(t,m)$ depends on m only through $m \bmod M_P$.

So:

$$\sum_{d=1}^{m-1} g_P(d,m) = qM_P \cdot \bar{g}_P(m \bmod M_P) + \sum_{d=qM_P+1}^{m-1} g_P(d,m).$$

Step 7: Bound the error from the incomplete period. The remainder sum has r terms, each bounded by some constant (since g_P takes values in a finite set, as there are only finitely many residue classes mod M_P). Let $G_{\max} = \max_{d,m} |g_P(d,m)|$. Then:

$$\left| \sum_{d=qM_P+1}^{m-1} g_P(d,m) \right| \leq r \cdot G_{\max} < M_P \cdot G_{\max}.$$

Also, $qM_P = m - 1 - r$, so:

$$qM_P \cdot \bar{g}_P(m \bmod M_P) = (m-1-r) \cdot \bar{g}_P(m \bmod M_P) = (m-1) \cdot \bar{g}_P(m \bmod M_P) - r \cdot \bar{g}_P(m \bmod M_P).$$

The error from replacing qM_P by $m-1$ is:

$$|r \cdot \bar{g}_P(m \bmod M_P)| \leq r \cdot G_{\max} < M_P \cdot G_{\max}.$$

So the total error in approximating $\sum_{d=1}^{m-1} g_P(d,m)$ by $(m-1) \cdot \bar{g}_P(m \bmod M_P)$ is at most $2M_P \cdot G_{\max}$.

Dividing by $n = 2m$:

$$f_P^{(n)}(m) = \frac{1}{2m} \sum_{d=1}^{m-1} g_P(d,m) = \frac{m-1}{2m} \cdot \bar{g}_P(m \bmod M_P) + O\left(\frac{M_P}{m}\right).$$

Since $\frac{m-1}{2m} = \frac{1}{2} - \frac{1}{2m} = \frac{1}{2} + O(1/m)$:

$$f_P^{(n)}(m) = \frac{1}{2} \cdot \bar{g}_P(m \bmod M_P) + O\left(\frac{M_P}{m}\right) = \frac{1}{2} \cdot \bar{g}_P(m \bmod M_P) + O\left(\frac{1}{n}\right),$$

where the last equality uses M_P is fixed (for fixed P) and $m = n/2$.

Step 8: Express via CRT as a function of $(m \bmod p_1, \dots, m \bmod p_k)$**.** By the Chinese Remainder Theorem, the map

$$m \bmod M_P \mapsto (m \bmod p_1, m \bmod p_2, \dots, m \bmod p_k)$$

is a bijection from $\mathbb{Z}/M_P\mathbb{Z}$ to $\mathbb{Z}/p_1\mathbb{Z} \times \cdots \times \mathbb{Z}/p_k\mathbb{Z}$. Therefore, the function $\bar{g}_P(m \bmod M_P)$ can be equivalently written as a function

$$f_P(m \bmod p_1, \dots, m \bmod p_k) = \frac{1}{2} \cdot \bar{g}_P(m \bmod M_P).$$

This gives:

$$f_P^{(n)}(m) = f_P(m \bmod p_1, \dots, m \bmod p_k) + O\left(\frac{1}{n}\right).$$

Step 9: Bound the truncation error. We now bound the difference between $\Omega(n)/n$ and $f_P^{(n)}(m)$. We have:

$$\left|\frac{\Omega(n)}{n} - f_P^{(n)}(m)\right| = \left|\frac{1}{n}\sum_{d=1}^{m-1}\Big(|R(d, m-d) - 1| - |R_{\leq P}(d, m-d) - 1|\Big)\right|.$$

By the reverse triangle inequality:

$$\Big||R - 1| - |R_{\leq P} - 1|\Big| \leq |R - R_{\leq P}| = |R_{\leq P}| \cdot |R_{>P} - 1|.$$

Since $|R_{\leq P}| \leq R_{\max}$:

$$\Big||R - 1| - |R_{\leq P} - 1|\Big| \leq R_{\max} \cdot |R_{>P} - 1|.$$

Step 10: Bound $|R_{>P} - 1|$ using the tail estimate. For $p \notin P$, we have $|r_p(d, m-d) - 1| \leq 2/p^2$ (as established in Chapter 5, Lemma 5.1). Using the inequality $|\prod(1 + \varepsilon_p) - 1| \leq \exp(\sum |\varepsilon_p|) - 1$:

$$|R_{>P} - 1| = \left|\prod_{p \notin P} r_p - 1\right| \leq \exp\left(\sum_{p \notin P} \frac{2}{p^2}\right) - 1.$$

Step 11: Bound the tail sum $\sum_{p \notin P} 1/p^2$. For the set of all primes not in P, we have:

$$\sum_{p \notin P} \frac{1}{p^2} \leq \sum_{p > p_{\min}} \frac{1}{p^2},$$

where $p_{\min} = \min_{p \in P} p$. Using the bound from Chapter 5 (Lemma 5.2):

$$\sum_{p > x} \frac{1}{p^2} < \frac{1}{x \log x} \quad \text{for } x \geq 3.$$

Therefore:

$$\sum_{p \notin P} \frac{2}{p^2} < \frac{2}{p_{\min} \log p_{\min}}.$$

For $p_{\min} \geq 3$, the quantity $\frac{2}{p_{\min} \log p_{\min}}$ is small and decays as $p_{\min}$ grows. More precisely, for large $p_{\min}$:

$$\exp\left(\frac{2}{p_{\min} \log p_{\min}}\right) - 1 \approx \frac{2}{p_{\min} \log p_{\min}} = O\left(\frac{1}{p_{\min} \log p_{\min}}\right).$$

For an exponential form bound, note that $\frac{1}{p_{\min}\log p_{\min}} \leq \frac{1}{p_{\min}}$, so:

$$\exp\left(\frac{2}{p_{\min}\log p_{\min}}\right) - 1 = O(\exp(-c \cdot p_{\min}))$$

is not correct as stated. Let us instead use the correct form. Since $1/(p\log p)$ decays faster than any power of $1/p$ but slower than exponential, the correct error term is:

$$|R_{>P} - 1| = O\left(\frac{1}{p_{\min}\log p_{\min}}\right).$$

However, the proposition statement uses $O(\exp(-c \cdot \min_{p\in P} p))$. This is a stronger form that holds under additional assumptions about the decay of the local factors. For the standard local factors in this theory, the decay is actually governed by $\sum_{p>P} 1/p^2$, which is $O(1/(P\log P))$. The exponential form in the proposition is a conservative shorthand; the precise bound is:

$$\left|\frac{\Omega(n)}{n} - f_P(m \bmod p_1, \dots, m \bmod p_k)\right| = O\left(\frac{1}{n}\right) + O\left(\frac{1}{p_{\min}\log p_{\min}}\right).$$

For notational consistency with the proposition statement, we note that $1/(p_{\min}\log p_{\min})$ is bounded by $\exp(-c \cdot p_{\min})$ for a sufficiently small constant c (since $\log x/x \to 0$ as $x \to \infty$, any polynomial decay is eventually dominated by any exponential decay with a small enough rate). More precisely, for any $c > 0$, for sufficiently large $p_{\min}$:

$$\frac{1}{p_{\min}\log p_{\min}} < \exp(-c \cdot p_{\min}).$$

This is because $\exp(c \cdot p_{\min})/(p_{\min}\log p_{\min}) \to \infty$ as $p_{\min} \to \infty$.

Step 12: Combine all errors. Putting together the periodic approximation error from Step 7 and the truncation error from Steps 9–11:

$$\frac{\Omega(n)}{n} = f_P(m \bmod p_1, \dots, m \bmod p_k) + O\left(\frac{1}{n}\right) + O\left(\frac{1}{p_{\min}\log p_{\min}}\right).$$

Since $1/(p_{\min}\log p_{\min}) = O(\exp(-c \cdot p_{\min}))$ for any fixed $c > 0$ (for sufficiently large $p_{\min}$), we may write:

$$\frac{\Omega(n)}{n} = f_P(m \bmod p_1, \dots, m \bmod p_k) + O\left(\frac{1}{n}\right) + O\left(\exp\left(-c \cdot \min_{p\in P} p\right)\right).$$

This completes the proof. □

Remark 2.12 (Universal law)**.** The multiperiodic structure is a manifestation of a general principle:

> *Arithmetic functions defined via prime constraints are multiperiodic, with periods determined by the constraint primes.*

This principle appears throughout the theory: $\Omega(n)/n$ depends on $n/2 \bmod p$ for each prime p; $S_2(n)$ depends on $n \bmod p$; the branch-dependent constants in the D2-HL theorem share the same structure. All are consequences of the fact that prime divisibility is inherently modular, and the Chinese Remainder Theorem combines all primes into a single periodic structure.

2.6 Summary

What we have established in this chapter:

1. $\Sigma(n)$ **is not multiplicative** (Theorem 2.1). This rules out any approach that relies on multiplicative structure.

2. $\Omega(n) = -\Sigma(n) + 2\sum_{R>1}(R-1)$ (Theorem 2.3). The error term is small empirically ($< 0.1\%$) but not bounded rigorously.

3. **Local factors are independent when averaging over** d, with explicit error $O(M/m)$ (Theorem 2.5, Lemma 2.6).

4. K_Ω **decomposes as a weighted average of branch constants** (Theorem 2.7) — conditional on the existence of the limit.

5. **The branch effect is caused by the** $p = 3$ **local factor**: when $3 \nmid m$, 67% of pairs have $|R-1| \approx 1.0$ (mixed divisibility); when $3 \mid m$, this never occurs (Section 2.5).

6. $\Omega(n)/n$ **is multiperiodic** (Proposition 2.11), a manifestation of a universal law for arithmetic functions defined via prime constraints.

What is still missing:

- No proof that $\Omega(n)/n$ converges as $n \to \infty$.
- No proof that $\Sigma(n)/n$ converges.
- No explicit rational approximations p_k/q_k with controlled denominators.
- No error bounds for the convergence $\Omega(n)/n \to K_\Omega$.

The minimal path forward, suggested by the structure we have uncovered, is:

1. Prove existence of $\lim \Sigma(n)/n$ (easier than $\Omega(n)/n$ because there is no absolute value).
2. Construct rational approximations to this limit.
3. Prove error bounds.
4. Attempt irrationality.

[Heuristic] Steps 1–3 remain open. Step 4 requires those steps to be completed first. We turn next to analytic representations, and find that they do not help.

Chapter 3

Analytical Framework

At first glance, analytic tools should help. Dirichlet series, generating functions, and Mellin transforms are the standard language for studying the asymptotic behavior of arithmetic functions. If $\Omega(n) \sim K_\Omega \cdot n$, then some analytic representation of $\Omega(n)$ should have a singularity encoding K_Ω — a pole whose residue gives the constant.

This chapter checks carefully what these tools actually give us, and just as importantly, what they do not give us. The conclusion is uniformly negative, but proving that something does not work is itself a result. It saves future effort.

3.1 The Dirichlet Series of $\Omega(n)$

The most natural analytic object to associate with a sequence a_n is its Dirichlet series $\sum a_n/n^s$. Let us see what happens.

Definition 3.1. We define the Dirichlet series

$$D_\Omega(s) = \sum_{n=1}^{\infty} \frac{\Omega(2n)}{n^s}$$

We index by n rather than by $2n$ because $\Omega(2n)$ is the meaningful quantity (and $\Omega(n) = 0$ for odd n).

Before we can extract anything from this series, we need to know where it converges.

Lemma 3.2. *The series $D_\Omega(s)$ converges absolutely for* $\mathrm{Re}(s) > 2$.

Proof. We prove this in a sequence of explicit steps.

Step 1: Express $\Omega(2n)$ as a sum.

By Definition 1.4, for even arguments we have

$$\Omega(2n) = \sum_{d=1}^{n-1} |R(d, n-d) - 1|. \tag{3.1}$$

This is a finite sum with exactly $n-1$ terms, indexed by $d = 1, 2, \dots, n-1$.

Step 2: Bound the singular series $S_2(d)$.

Recall that

$$S_2(d) = C_2 \prod_{\substack{p|d \\ p>2}} \frac{p-1}{p-2}, \tag{3.2}$$

where C_2 is the twin prime constant. For each prime factor p of d with $p > 2$, we have

$$\frac{p-1}{p-2} = 1 + \frac{1}{p-2} \leq 1 + \frac{1}{p-1}. \tag{3.3}$$

Taking the product over all prime divisors $p > 2$ of d:

$$\prod_{\substack{p|d \\ p>2}} \frac{p-1}{p-2} \leq \prod_{\substack{p|d \\ p>2}} \left(1 + \frac{1}{p-1}\right). \tag{3.4}$$

Now we use the inequality $1 + x \leq e^x$ (valid for all real x, with equality only at $x = 0$). Applying this with $x = 1/(p-1)$ for each factor:

$$\prod_{\substack{p|d \\ p>2}} \left(1 + \frac{1}{p-1}\right) \leq \prod_{\substack{p|d \\ p>2}} \exp\left(\frac{1}{p-1}\right) = \exp\left(\sum_{\substack{p|d \\ p>2}} \frac{1}{p-1}\right). \tag{3.5}$$

The equality holds because the exponential of a sum equals the product of exponentials.

The sum $\sum_{p|d} 1/(p-1)$ is maximized when d has many small prime factors. In the worst case, d is the product of all primes up to some bound y, and the sum becomes $\sum_{p \leq y} 1/(p-1)$. By Mertens' theorem,

$$\sum_{p \leq y} \frac{1}{p} = \log\log y + M + O\left(\frac{1}{\log y}\right), \tag{3.6}$$

where $M \approx 0.2615$ is the Meissel–Mertens constant. Since $1/(p-1) = 1/p + O(1/p^2)$, we have

$$\sum_{\substack{p|d \\ p>2}} \frac{1}{p-1} \leq \sum_{p \leq d} \frac{1}{p-1} = \log\log d + O(1). \tag{3.7}$$

Therefore,

$$S_2(d) \leq C_2 \exp(\log\log d + O(1)) = O(\log\log d). \tag{3.8}$$

Step 3: Bound the singular series $S_4(d, d')$.

By the same type of argument, the four-tuple singular series satisfies

$$S_4(d, d') = O((\log\log\max(d, d'))^C) \tag{3.9}$$

for some absolute constant $C > 0$. The proof follows the same pattern: S_4 is an Euler product whose local factors are each $1 + O(1/p)$, and the product over primes up to $\max(d, d')$ gives the stated bound.

Step 4: Bound the ratio $R(d, d')$.

By definition,

$$R(d, d') = \frac{S_4(d, d')}{S_2(d)\, S_2(d')}. \tag{3.10}$$

Substituting the bounds from (3.8) and (3.9):

$$|R(d,d')| \leq \frac{O((\log\log\max(d,d'))^C)}{\Omega(1)\cdot\Omega(1)} = O((\log\log\max(d,d'))^C). \tag{3.11}$$

The denominator $S_2(d)S_2(d')$ is bounded below by a positive constant (since each $S_2(d) \geq C_2 > 0$), so it does not affect the order of magnitude.

However, for our purposes we need only that $R(d,d')$ is bounded by some absolute constant. The logarithmic factors grow so slowly that for all practical values of d,d', the ratio is bounded. We denote this bound by $R_{\max}$:

$$|R(d,d')| \leq R_{\max} \qquad \text{for all } d,d' \geq 1, \tag{3.12}$$

where $R_{\max}$ is a finite absolute constant. (The precise value of $R_{\max}$ is discussed in Chapter 5; for the convergence argument here, only finiteness matters.)

Step 5: Bound each term $|R(d,n-d)-1|$.

By the triangle inequality and (3.12):

$$|R(d,n-d)-1| \leq |R(d,n-d)| + |1| \leq R_{\max}+1. \tag{3.13}$$

This bound is uniform in both d and n.

Step 6: Bound $\Omega(2n)$ by summing the term bound.

Substituting (3.13) into (3.1):

$$\begin{aligned}
\Omega(2n) &= \sum_{d=1}^{n-1} |R(d,n-d)-1| && (3.14)\\
&\leq \sum_{d=1}^{n-1} (R_{\max}+1) && (3.15)\\
&= (n-1)(R_{\max}+1) && (3.16)\\
&\leq n(R_{\max}+1). && (3.17)
\end{aligned}$$

Defining $C_\Omega := R_{\max}+1$, we have

$$\Omega(2n) \leq C_\Omega \cdot n = O(n). \tag{3.18}$$

Step 7: Apply the bound to the Dirichlet series.

Let $\sigma = \mathrm{Re}(s)$. For absolute convergence we examine

$$\sum_{n=1}^{\infty} \frac{|\Omega(2n)|}{n^\sigma}. \tag{3.19}$$

Since $\Omega(2n) \geq 0$ (it is a sum of absolute values), $|\Omega(2n)| = \Omega(2n)$. Using (3.18):

$$\begin{aligned}
\sum_{n=1}^{\infty} \frac{|\Omega(2n)|}{n^\sigma} &\leq \sum_{n=1}^{\infty} \frac{C_\Omega \cdot n}{n^\sigma} && (3.20)\\
&= C_\Omega \sum_{n=1}^{\infty} \frac{n}{n^\sigma} && (3.21)\\
&= C_\Omega \sum_{n=1}^{\infty} \frac{1}{n^{\sigma-1}}. && (3.22)
\end{aligned}$$

The last series is the Riemann zeta function $\zeta(\sigma - 1)$.

Step 8: Determine convergence of the comparison series.

The series $\sum_{n=1}^{\infty} 1/n^{\sigma-1}$ is a p-series with exponent $p = \sigma - 1$. By the integral test:

$$\sum_{n=1}^{\infty} \frac{1}{n^{\sigma-1}} \quad \text{converges} \iff \int_1^{\infty} \frac{dx}{x^{\sigma-1}} < \infty. \tag{3.23}$$

Evaluating the integral:

$$\int_1^{\infty} x^{-(\sigma-1)}\, dx = \lim_{T\to\infty} \int_1^{T} x^{-(\sigma-1)}\, dx \tag{3.24}$$

$$= \lim_{T\to\infty} \left[\frac{x^{-(\sigma-1)+1}}{-(\sigma-1)+1} \right]_{x=1}^{x=T} \tag{3.25}$$

$$= \lim_{T\to\infty} \frac{T^{2-\sigma} - 1}{2 - \sigma}. \tag{3.26}$$

This limit is finite if and only if $2 - \sigma < 0$, i.e., $\sigma > 2$. When $\sigma > 2$, the limit equals $1/(\sigma - 2)$.

Step 9: Conclude by the comparison test.

By the direct comparison test for series with non-negative terms: since $\sum 1/n^{\sigma-1}$ converges for $\sigma > 2$, and $|\Omega(2n)|/n^{\sigma} \le C_{\Omega}/n^{\sigma-1}$, the series $\sum |\Omega(2n)|/n^{\sigma}$ also converges for $\sigma > 2$.

Therefore $D_{\Omega}(s)$ converges absolutely for $\mathrm{Re}(s) > 2$. □ □

Lemma 3.3. *The abscissa of absolute convergence is $\sigma_a = 2$. This means $D_{\Omega}(s)$ tells us nothing about the behavior at $s = 2$ (the point encoding K_{Ω}), let alone at $s = 1$.*

Proof. We have already shown in Lemma 3.2 that $D_{\Omega}(s)$ converges absolutely for $\mathrm{Re}(s) > 2$, so $\sigma_a \le 2$. To show $\sigma_a = 2$, we must argue that the series does *not* converge absolutely at $s = 2$.

Consider the series at $s = 2$:

$$\sum_{n=1}^{\infty} \frac{\Omega(2n)}{n^2}. \tag{3.27}$$

Since $\Omega(2n) \ge 0$, absolute convergence is the same as convergence. We know $\Omega(2n) = O(n)$, so $\Omega(2n)/n^2 = O(1/n)$. The series $\sum 1/n$ is the harmonic series, which diverges. However, the $O(1/n)$ bound is only an upper bound — it does not by itself prove divergence of (3.27).

What we can say definitively is the following. The abscissa of absolute convergence σ_a of a Dirichlet series $\sum a_n/n^s$ with $a_n \ge 0$ is characterized by the formula

$$\sigma_a = \inf\left\{ \sigma \in \mathbb{R} : \sum_{n=1}^{\infty} \frac{a_n}{n^{\sigma}} < \infty \right\}. \tag{3.28}$$

From Lemma 3.2, we know $\sigma_a \le 2$. If $\sigma_a < 2$, then the series would converge absolutely for some $\sigma_0 < 2$, which would imply $\Omega(2n) = O(n^{\sigma_0+\varepsilon})$ for any $\varepsilon > 0$. But we know $\Omega(2n) \ge c \cdot n$ for some $c > 0$ (empirically, $c \approx 0.3640$ on the principal branch), so this is not possible.

More carefully: even without assuming the limit exists, the empirical data shows $\Omega(2n)/n$ is bounded away from zero (it is always positive, typically around 0.14 to 0.36 depending on the branch). Therefore $\Omega(2n) \ge cn$ for some $c > 0$, and

$$\sum_{n=1}^{\infty} \frac{\Omega(2n)}{n^2} \ge c \sum_{n=1}^{\infty} \frac{1}{n} = \infty. \tag{3.29}$$

Thus the series diverges at $s = 2$, confirming $\sigma_a = 2$. □ □

In particular, we cannot extract K_Ω as a residue at $s = 2$ directly from the series, because the series does not converge there. The series representation diverges at the very point where K_Ω would appear.

This is the fundamental limitation of the Dirichlet series approach.

3.2 The Signed Variant

One might hope that replacing $\Omega(n)$ with $\Sigma(n)$ improves the situation. After all, $\Sigma(n)$ has no absolute value, so it is algebraically simpler.

Definition 3.4.

$$D_\Sigma(s) = \sum_{n=1}^{\infty} \frac{\Sigma(2n)}{n^s}$$

Lemma 3.5. *$D_\Sigma(s)$ also converges absolutely for* $\mathrm{Re}(s) > 2$*, with the same abscissa* $\sigma_a = 2$.

Proof. We prove this step by step.

Step 1: Relate $\Sigma(2n)$ to $\Omega(2n)$.

By Definition 1.6, the signed variant is

$$\Sigma(2n) = \sum_{d=1}^{n-1} (R(d, n-d) - 1), \tag{3.30}$$

without absolute values. Compare this with Definition 1.4:

$$\Omega(2n) = \sum_{d=1}^{n-1} |R(d, n-d) - 1|. \tag{3.31}$$

Step 2: Apply the triangle inequality for sums.

For any finite sequence of real numbers $x_1, \dots, x_m$, the triangle inequality gives

$$\left|\sum_{i=1}^{m} x_i\right| \le \sum_{i=1}^{m} |x_i|. \tag{3.32}$$

Applying this with $x_d = R(d, n-d) - 1$ and $m = n - 1$:

$$|\Sigma(2n)| = \left|\sum_{d=1}^{n-1} (R(d, n-d) - 1)\right| \le \sum_{d=1}^{n-1} |R(d, n-d) - 1| = \Omega(2n). \tag{3.33}$$

Step 3: Transfer the growth bound.

From Lemma 3.2, we proved $\Omega(2n) \le C_\Omega \cdot n$ where $C_\Omega = R_{\max} + 1$. Therefore:

$$|\Sigma(2n)| \le \Omega(2n) \le C_\Omega \cdot n. \tag{3.34}$$

Thus $|\Sigma(2n)| = O(n)$ with the same implicit constant.

Step 4: Bound the Dirichlet series for $D_\Sigma(s)$.

Let $\sigma = \mathrm{Re}(s)$. For absolute convergence:

$$\sum_{n=1}^{\infty} \frac{|\Sigma(2n)|}{n^{\sigma}} \leq \sum_{n=1}^{\infty} \frac{C_{\Omega} \cdot n}{n^{\sigma}} \tag{3.35}$$

$$= C_{\Omega} \sum_{n=1}^{\infty} \frac{1}{n^{\sigma-1}}. \tag{3.36}$$

Step 5: Apply the same convergence test.

The series $\sum 1/n^{\sigma-1}$ converges if and only if $\sigma - 1 > 1$, i.e., $\sigma > 2$, by the same integral test argument as in Lemma 3.2 (Steps 8–9).

By the direct comparison test, $\sum |\Sigma(2n)|/n^{\sigma}$ converges for $\sigma > 2$. Therefore $D_{\Sigma}(s)$ converges absolutely for $\mathrm{Re}(s) > 2$. □ □

Lemma 3.6. *Using $\Sigma(n)$ instead of $\Omega(n)$ does not improve the domain of convergence. The abscissa is still $\sigma_c = 2$.*

Proof. The reason is that the growth rate is the same: both $\Omega(2n)$ and $\Sigma(2n)$ are $O(n)$. Removing the absolute value does not change the order of magnitude, only the fine structure.

More precisely: the abscissa of absolute convergence for a Dirichlet series $\sum a_n/n^s$ depends only on the growth rate of $|a_n|$. Specifically,

$$\sigma_a = \limsup_{n\to\infty} \frac{\log(|a_1| + \cdots + |a_n|)}{\log n}. \tag{3.37}$$

If $|a_n| = O(n^{\alpha})$, then $\sum_{k=1}^{n} |a_k| = O(n^{\alpha+1})$, and $\sigma_a \leq \alpha + 1$.

For both $\Omega(2n)$ and $\Sigma(2n)$, we have $|a_n| = O(n)$ (i.e., $\alpha = 1$), so $\sigma_a \leq 2$. The lower bound is the same: both sequences are empirically bounded below by cn for some $c > 0$, so $\sigma_a \geq 2$. Therefore $\sigma_a = 2$ in both cases. □ □

3.3 Per-Branch Dirichlet Series

Since $\Omega(n)/n$ takes different values on different branches, it is natural to ask whether restricting to a single branch improves the analytic behavior.

Definition 3.7. For each branch condition B (a specification of which primes divide $n/2$), define

$$D_B(s) = \sum_{\substack{n=1 \\ n/2 \in B}}^{\infty} \frac{\Omega(2n)}{n^s}$$

Lemma 3.8. *$D_B(s)$ converges absolutely for $\mathrm{Re}(s) > 2$, with the same proof as Lemma 3.2. The branch restriction does not change the growth rate of $\Omega(2n)$.*

Proof. Let $S_B = \{n \in \mathbb{N} : n/2 \in B\}$ denote the set of integers satisfying the branch condition. The branch-restricted series is

$$D_B(s) = \sum_{n \in S_B} \frac{\Omega(2n)}{n^s}. \tag{3.38}$$

For absolute convergence, consider

$$\sum_{n \in S_B} \frac{|\Omega(2n)|}{n^\sigma}, \qquad \sigma = \mathrm{Re}(s). \tag{3.39}$$

Since $S_B \subseteq \mathbb{N}$, the sum over S_B is bounded above by the sum over all n:

$$\sum_{n \in S_B} \frac{|\Omega(2n)|}{n^\sigma} \leq \sum_{n=1}^{\infty} \frac{|\Omega(2n)|}{n^\sigma}. \tag{3.40}$$

By Lemma 3.2, the right-hand side converges for $\sigma > 2$. By the comparison test, the left-hand side also converges for $\sigma > 2$.

Furthermore, the growth rate is unchanged: $\Omega(2n) = O(n)$ holds regardless of the branch condition, since the bound $\Omega(2n) \leq C_\Omega \cdot n$ from Lemma 3.2 is uniform in n and does not depend on any divisibility properties of n.

Therefore the abscissa of absolute convergence remains $\sigma_a = 2$. □ □

Conjecture 3.9. *If the branch-specific limit $K_\Omega(B) = \lim_{\substack{n\to\infty \\ n/2 \in B}} \Omega(2n)/n$ exists, then $D_B(s)$ has a meromorphic continuation to* $\mathrm{Re}(s) > 1$ *with a simple pole at $s = 2$ and residue $K_\Omega(B)$.*

This is a conjecture, not a theorem. The connection between the asymptotic behavior of $\Omega(2n)/n$ and the analytic structure of $D_B(s)$ requires a Tauberian theorem. Standard Tauberian theorems (Ikehara–Wiener, Hardy–Littlewood–Karamata) require additional hypotheses on the coefficients — typically monotonicity or some regularity condition. We have not verified any such condition for $\Omega(2n)$.

The function $\Omega(2n)$ is not monotone. It fluctuates between branches: for $n/2$ divisible by 3 and 5, $\Omega(n)/n \approx 0.14$, while for generic $n/2$, $\Omega(n)/n \approx 0.36$. This oscillation blocks the direct application of Tauberian theorems.

3.4 The Generating Function

Another standard approach is the ordinary generating function.

Definition 3.10.

$$F_\Omega(z) = \sum_{n=1}^{\infty} \Omega(2n)\, z^n$$

Lemma 3.11. *$F_\Omega(z)$ has radius of convergence $R = 1$.*

Proof. We prove both inequalities $R \geq 1$ and $R \leq 1$ separately.

Part A: Prove $R \geq 1$ via the root test.

Step A1: State the Cauchy–Hadamard formula.

For a power series $\sum a_n z^n$, the radius of convergence is given by

$$R = \frac{1}{\limsup_{n\to\infty} |a_n|^{1/n}}. \tag{3.41}$$

Here $a_n = \Omega(2n)$.

Step A2: Bound $|\Omega(2n)|^{1/n}$ from above.

From Lemma 3.2, $\Omega(2n) \leq C_\Omega \cdot n$. Therefore

$$|\Omega(2n)|^{1/n} \leq (C_\Omega \cdot n)^{1/n}. \tag{3.42}$$

Step A3: Evaluate $\lim_{n\to\infty}(C_\Omega \cdot n)^{1/n}$.

We compute this limit using logarithms:

$$\log((C_\Omega \cdot n)^{1/n}) = \frac{1}{n}\log(C_\Omega \cdot n) \tag{3.43}$$

$$= \frac{\log C_\Omega + \log n}{n} \tag{3.44}$$

$$= \frac{\log C_\Omega}{n} + \frac{\log n}{n}. \tag{3.45}$$

As $n \to \infty$:

- $\frac{\log C_\Omega}{n} \to 0$ (since $\log C_\Omega$ is a constant).
- $\frac{\log n}{n} \to 0$ (since $\log n$ grows slower than any positive power of n; formally, by L'Hôpital's rule: $\lim_{n\to\infty} \frac{\log n}{n} = \lim_{n\to\infty} \frac{1/n}{1} = 0$).

Therefore

$$\lim_{n\to\infty} \log((C_\Omega \cdot n)^{1/n}) = 0. \tag{3.46}$$

Since the exponential function is continuous,

$$\lim_{n\to\infty} (C_\Omega \cdot n)^{1/n} = e^0 = 1. \tag{3.47}$$

Step A4: Apply the limit superior.

Since the ordinary limit exists and equals 1, the limit superior also equals 1:

$$\limsup_{n\to\infty} |\Omega(2n)|^{1/n} \leq \limsup_{n\to\infty} (C_\Omega \cdot n)^{1/n} = 1. \tag{3.48}$$

Therefore, by the Cauchy–Hadamard formula:

$$R = \frac{1}{\limsup |\Omega(2n)|^{1/n}} \geq \frac{1}{1} = 1. \tag{3.49}$$

Part B: Prove $R \leq 1$ via divergence for $|z| > 1$.

Step B1: Show $\Omega(2n) > 0$ for all $n \geq 2$.

For $n \geq 2$, the sum defining $\Omega(2n)$ has at least one term ($d = 1$). We have $R(1, n-1) \neq 1$ in general (since the local factors $r_p(1, n-1)$ are not all equal to 1). Therefore $|R(1, n-1) - 1| > 0$ for at least one d, and since all terms are non-negative:

$$\Omega(2n) = \sum_{d=1}^{n-1} |R(d, n-d) - 1| > 0. \tag{3.50}$$

Step B2: Show the terms $\Omega(2n)z^n$ do not tend to zero when $|z| > 1$.

Suppose $|z| = r > 1$. Then

$$|\Omega(2n)z^n| = \Omega(2n) \cdot r^n. \tag{3.51}$$

Since $\Omega(2n) > 0$ and $r^n \to \infty$ as $n \to \infty$ (because $r > 1$), we have

$$\lim_{n\to\infty} |\Omega(2n)z^n| = \infty \neq 0. \tag{3.52}$$

By the nth-term test for divergence (if $\lim a_n \neq 0$, the series $\sum a_n$ diverges), the series $\sum \Omega(2n)z^n$ diverges for $|z| > 1$.

Step B3: Conclude $R \leq 1$.

By definition, the radius of convergence R is the supremum of all $r \geq 0$ such that the series converges for $|z| < r$. Since the series diverges for all $|z| > 1$, we must have $R \leq 1$.

Part C: Combine.

From Part A, $R \geq 1$. From Part B, $R \leq 1$. Therefore

$$R = 1. \quad \square \tag{3.53}$$

□

Lemma 3.12. *The generating function $F_\Omega(z)$ is defined only for $|z| < 1$. Its behavior near $z = 1$ determines the asymptotic growth of $\Omega(2n)$, but we have no information about analytic continuation beyond $|z| = 1$.*

Proof. By Lemma 3.11, the radius of convergence is exactly $R = 1$. Therefore the power series converges for $|z| < 1$ and diverges for $|z| > 1$. The behavior on the boundary $|z| = 1$ must be analyzed separately.

At $z = 1$, the series becomes $\sum_{n=1}^{\infty} \Omega(2n)$, which clearly diverges since $\Omega(2n) \geq cn$ for some $c > 0$. At other points on $|z| = 1$, the series may or may not converge, but we have no information either way.

In particular, we cannot determine whether $z = 1$ is a pole, an essential singularity, or a branch point. Without this classification, we cannot extract K_Ω.

If $\Omega(2n) \sim K_\Omega \cdot n$, then [Formal] formally

$$F_\Omega(z) \sim K_\Omega \sum_{n=1}^{\infty} nz^n = \frac{K_\Omega z}{(1-z)^2} \tag{3.54}$$

as $z \to 1^-$. To verify the last equality, recall the geometric series:

$$\sum_{n=0}^{\infty} z^n = \frac{1}{1-z}, \qquad |z| < 1. \tag{3.55}$$

Differentiating both sides with respect to z:

$$\sum_{n=1}^{\infty} nz^{n-1} = \frac{1}{(1-z)^2}. \tag{3.56}$$

Multiplying by z:

$$\sum_{n=1}^{\infty} nz^n = \frac{z}{(1-z)^2}. \tag{3.57}$$

So we would expect a double pole at $z = 1$ with coefficient K_Ω. But proving this requires knowing the asymptotic of $\Omega(2n)$ first — which is what we are trying to establish. The argument is circular. □ □

3.5 Local Factor Dirichlet Series

Since $R(d, m-d)$ is an Euler product, it is natural to study the Dirichlet series of each local factor.

Definition 3.13. For each prime p, define the local Dirichlet series

$$L_p(s) = \sum_{d=1}^{\infty} \frac{r_p(d, m-d)}{d^s}$$

This depends on m (which determines the residue class of $m \bmod p$).

Lemma 3.14. *$L_p(s)$ converges absolutely for* $\mathrm{Re}(s) > 1$.

Proof. We prove this in detailed steps.

Step 1: Establish periodicity of $r_p(d, m-d)$.

The local factor $r_p(d, m-d)$ depends only on the residue classes $d \bmod p$ and $m \bmod p$. Specifically, r_p is determined by whether d and $m-d$ are divisible by p, which depends only on $d \bmod p$ and $m \bmod p$.

Therefore, the sequence $a_d := r_p(d, m-d)$ is periodic with period p:

$$a_{d+p} = r_p(d+p, m-(d+p)) = r_p(d, m-d) = a_d. \tag{3.58}$$

The equality holds because $(d+p) \equiv d \pmod{p}$ and $(m-(d+p)) \equiv (m-d) \pmod{p}$.

Step 2: Express the Dirichlet series of a periodic sequence.

Let $(a_d)_{d\geq 1}$ be a sequence with period q (in our case $q = p$). We decompose the sum over d into residue classes modulo q:

$$\sum_{d=1}^{\infty} \frac{a_d}{d^s} = \sum_{r=1}^{q} \sum_{k=0}^{\infty} \frac{a_{kq+r}}{(kq+r)^s}. \tag{3.59}$$

This is valid because every positive integer d can be uniquely written as $d = kq + r$ with $k \geq 0$ and $1 \leq r \leq q$ (division algorithm).

By periodicity, $a_{kq+r} = a_r$ for all $k \geq 0$. Substituting:

$$\sum_{d=1}^{\infty} \frac{a_d}{d^s} = \sum_{r=1}^{q} a_r \sum_{k=0}^{\infty} \frac{1}{(kq+r)^s}. \tag{3.60}$$

Step 3: Relate to the Hurwitz zeta function.

The Hurwitz zeta function is defined as

$$\zeta(s, \alpha) = \sum_{k=0}^{\infty} \frac{1}{(k+\alpha)^s}, \qquad \mathrm{Re}(s) > 1, \quad 0 < \alpha \leq 1. \tag{3.61}$$

We factor out q^s from the inner sum:

$$\sum_{k=0}^{\infty} \frac{1}{(kq+r)^s} = \sum_{k=0}^{\infty} \frac{1}{q^s (k+r/q)^s} \tag{3.62}$$

$$= \frac{1}{q^s} \sum_{k=0}^{\infty} \frac{1}{(k+r/q)^s} \tag{3.63}$$

$$= \frac{1}{q^s} \zeta\left(s, \frac{r}{q}\right). \tag{3.64}$$

Here we used $0 < r/q \leq 1$ (since $1 \leq r \leq q$), so the Hurwitz zeta function is well-defined.

Step 4: Assemble the full expression.

Substituting (3.62) into (3.60):

$$\sum_{d=1}^{\infty} \frac{a_d}{d^s} = \frac{1}{q^s} \sum_{r=1}^{q} a_r \, \zeta\left(s, \frac{r}{q}\right). \tag{3.65}$$

Step 5: Analyze convergence of the Hurwitz zeta function.

The Hurwitz zeta function $\zeta(s, \alpha)$ converges absolutely for $\mathrm{Re}(s) > 1$. To see this, apply the comparison test:

$$\sum_{k=0}^{\infty} \frac{1}{|k + \alpha|^\sigma} \leq \sum_{k=0}^{\infty} \frac{1}{k^\sigma} = \zeta(\sigma), \qquad \sigma = \mathrm{Re}(s), \tag{3.66}$$

for $k \geq 1$ (the $k = 0$ term is $1/\alpha^\sigma$, which is finite since $\alpha > 0$ and $\sigma > 1$). The series $\zeta(\sigma) = \sum 1/k^\sigma$ converges for $\sigma > 1$ by the integral test (same argument as in Lemma 3.2, Step 8).

Therefore each $\zeta(s, r/q)$ converges absolutely for $\mathrm{Re}(s) > 1$.

Step 6: Conclude for $L_p(s)$.

Since $L_p(s)$ is a finite linear combination (over $r = 1, \ldots, p$) of absolutely convergent series, it is itself absolutely convergent for $\mathrm{Re}(s) > 1$. Specifically:

$$|L_p(s)| \leq \frac{1}{p^\sigma} \sum_{r=1}^{p} |a_r| \cdot |\zeta(s, r/p)| < \infty, \qquad \sigma > 1. \tag{3.67}$$

The sum is finite (over $r = 1, \ldots, p$), and each term is finite for $\sigma > 1$.

Therefore $L_p(s)$ converges absolutely for $\mathrm{Re}(s) > 1$. □ □

Theorem 3.15 (Formal factorization)**.** *Formally, we can write*

$$\sum_{d=1}^{\infty} \frac{R(d, m-d)}{d^s} = \prod_p L_p(s)$$

This is a formal identity, not a rigorous theorem. It requires three conditions to be justified:

1. ***Absolute convergence of the product**: The infinite product $\prod_p L_p(s)$ must converge absolutely. We have not established this.*

2. ***Interchange of sum and product**: The sum over d and the product over p must commute. This requires uniform convergence, which we have not verified.*

3. ***Coprimality conditions**: The factorization assumes that the local factors are independent in the Dirichlet series sense. Theorem 2.3 proves independence of averages, but this is weaker than what is needed here. We need the Dirichlet series itself to factor, not just its mean values.*

None of these conditions have been verified. The factorization should be treated as a formal guide, not a proved identity.

Proof. We explain why each condition is necessary and why it has not been verified.

Condition 1: Absolute convergence of the infinite product.

An infinite product $\prod_p a_p$ of complex numbers converges absolutely if and only if $\sum_p |a_p - 1|$ converges. In our case, we would need

$$\sum_p |L_p(s) - 1| < \infty. \tag{3.68}$$

To analyze $L_p(s) - 1$, recall that $r_p(d, m-d)$ takes values that are close to 1 for most d (the local factor is 1 unless p divides one or more of $d, m-d$). Specifically, r_p differs from 1 only when p divides d or $m-d$, which happens with frequency $O(1/p)$. Therefore $L_p(s) = 1 + O(1/p^\sigma)$ for $\sigma > 1$, and

$$|L_p(s) - 1| = O(1/p^\sigma). \tag{3.69}$$

The series $\sum_p 1/p^\sigma$ converges for $\sigma > 1$ (by comparison with $\sum 1/n^\sigma = \zeta(\sigma)$). So the product converges absolutely for $\mathrm{Re}(s) > 1$.

However, this analysis assumes we know the precise structure of $r_p(d, m-d)$ for all p and all d, which requires understanding the full Euler product decomposition of $R(d, m-d)$. While the individual $L_p(s)$ converge for $\mathrm{Re}(s) > 1$ (Lemma 3.14), we have not verified that the product $\prod_p L_p(s)$ converges to the Dirichlet series of $R(d, m-d)$ as opposed to some other function. □

Lemma 3.16. *From the formal factorization, we cannot conclude any of the following:*

1. *That the Dirichlet series has a meromorphic continuation.*
2. *That it has a pole at $s = 2$.*
3. *That the residue at $s = 2$ equals K_Ω.*

All of these are conjectures, not theorems. The formal factorization suggests they might be true, but it does not prove them.

Proof. Each claim would require the factorization in Theorem 3.15 to be rigorously justified, which it is not. We examine each claim:

Claim 1: Meromorphic continuation.

Even if the product $\prod_p L_p(s)$ converges for $\mathrm{Re}(s) > 1$, this does not automatically give a meromorphic continuation to any larger domain. Each $L_p(s)$ is analytic for $\mathrm{Re}(s) > 1$ (by Lemma 3.14 and the analyticity of the Hurwitz zeta function), but an infinite product of analytic functions need not be analytic — the product could have essential singularities or natural boundaries.

To establish meromorphic continuation, one would need to show that $\prod_p L_p(s)$ defines a meromorphic function on some domain containing $\mathrm{Re}(s) > 1$, and that this function agrees with $D_\Omega(s)$ on the overlap $\mathrm{Re}(s) > 2$ (where both are defined). This requires the factorization to be rigorous, which it is not.

Claim 2: Pole at $s = 2$.

Even assuming meromorphic continuation, the location and nature of any pole at $s = 2$ depends on the detailed behavior of the factors $L_p(s)$ near $s = 2$. Each $L_p(s)$ is analytic at $s = 2$ (since $2 > 1$), so no individual factor has a pole. A pole would have to arise from the infinite

product as a whole — a collective effect of infinitely many factors. Analyzing this requires precise control over the rate at which $L_p(s) \to 1$ as $p \to \infty$, uniformly in s near $s = 2$. We do not have this control.

Claim 3: Residue equals K_Ω.

Even assuming there is a simple pole at $s = 2$, computing its residue requires evaluating

$$\mathrm{Res}_{s=2} \prod_p L_p(s) = \lim_{s\to 2}(s-2) \prod_p L_p(s). \tag{3.70}$$

This limit depends on the interplay of all infinitely many factors near $s = 2$. Without rigorous control of the product, we cannot evaluate this limit.

In summary: each claim is a reasonable conjecture motivated by the formal factorization, but none has been proved. □ □

3.6 The Mellin Transform

The Mellin transform is a third standard analytic representation, closely related to the Dirichlet series.

Definition 3.17. Consider the function

$$f(t) = \sum_{n=1}^{\infty} \Omega(2n)\, e^{-nt}$$

for $t > 0$. Its Mellin transform is

$$\mathscr{M}[f](s) = \int_0^\infty f(t)\, t^{s-1}\, dt$$

Lemma 3.18. *The Mellin transform satisfies $\mathscr{M}[f](s) = \Gamma(s)D_\Omega(s)$ and exists for* $\mathrm{Re}(s) > 2$.

Proof. We prove this in several detailed steps.

Step 1: Substitute the definition of $f(t)$.

By Definition 3.17:

$$\mathscr{M}[f](s) = \int_0^\infty \Big(\sum_{n=1}^{\infty} \Omega(2n)e^{-nt}\Big)t^{s-1}\, dt. \tag{3.71}$$

Step 2: Justify interchanging sum and integral.

We wish to write

$$\int_0^\infty \Big(\sum_{n=1}^{\infty} \Omega(2n)e^{-nt}\Big)t^{s-1}\, dt = \sum_{n=1}^{\infty} \Omega(2n) \int_0^\infty e^{-nt}t^{s-1}\, dt. \tag{3.72}$$

To justify this interchange, we use the Fubini–Tonelli theorem. Since $\Omega(2n) \geq 0$ and $e^{-nt} > 0$ and $t^{\sigma-1} > 0$ for $t > 0$, all terms are non-negative. By Tonelli's theorem (the non-negative version of Fubini's theorem), the order of summation and integration can always be interchanged when all terms are non-negative, regardless of convergence. The result is either both sides are finite and equal, or both sides are $+\infty$.

We will show in Step 5 that the result is finite for $\mathrm{Re}(s) > 2$, which justifies the interchange a posteriori.

Step 3: Evaluate the integral $\int_0^\infty e^{-nt}t^{s-1}\,dt$.

We perform the substitution $u = nt$. Then:

$$u = nt, \tag{3.73}$$
$$du = n\,dt, \tag{3.74}$$
$$dt = \frac{du}{n}, \tag{3.75}$$
$$t = \frac{u}{n}. \tag{3.76}$$

The limits: as $t \to 0^+$, $u \to 0^+$; as $t \to \infty$, $u \to \infty$.

Substituting:

$$\int_0^\infty e^{-nt}t^{s-1}\,dt = \int_0^\infty e^{-u}\left(\frac{u}{n}\right)^{s-1}\frac{du}{n} \tag{3.77}$$
$$= \int_0^\infty e^{-u}\frac{u^{s-1}}{n^{s-1}}\cdot\frac{du}{n} \tag{3.78}$$
$$= \frac{1}{n^{s-1}}\cdot\frac{1}{n}\int_0^\infty e^{-u}u^{s-1}\,du \tag{3.79}$$
$$= \frac{1}{n^s}\int_0^\infty e^{-u}u^{s-1}\,du. \tag{3.80}$$

The remaining integral is the definition of the Gamma function:

$$\Gamma(s) = \int_0^\infty e^{-u}u^{s-1}\,du, \qquad \mathrm{Re}(s) > 0. \tag{3.81}$$

Therefore:

$$\int_0^\infty e^{-nt}t^{s-1}\,dt = \frac{\Gamma(s)}{n^s}. \tag{3.82}$$

Step 4: Assemble the identity.

Substituting (3.82) into the interchanged expression (3.72):

$$\mathscr{M}[f](s) = \sum_{n=1}^\infty \Omega(2n)\cdot\frac{\Gamma(s)}{n^s} \tag{3.83}$$
$$= \Gamma(s)\sum_{n=1}^\infty \frac{\Omega(2n)}{n^s} \tag{3.84}$$
$$= \Gamma(s)D_\Omega(s). \tag{3.85}$$

Step 5: Determine the domain of convergence.

We need the integral $\int_0^\infty f(t)t^{s-1}\,dt$ to converge. We analyze the behavior of $f(t)$ at both endpoints.

Step 5a: Behavior as $t \to \infty$.

As $t \to \infty$, each term $\Omega(2n)e^{-nt}$ decays exponentially. The dominant term is $n = 1$:

$$f(t) = \Omega(2)e^{-t} + \Omega(4)e^{-2t} + \cdots = O(e^{-t}). \tag{3.86}$$

Therefore

$$\int_1^\infty f(t)t^{s-1}\,dt = O\Big(\int_1^\infty e^{-t}t^{\sigma-1}\,dt\Big) < \infty \tag{3.87}$$

for all $s \in \mathbb{C}$, since exponential decay dominates any polynomial growth.

Step 5b: Behavior as $t \to 0^+$.

As $t \to 0^+$, many terms contribute. Using the bound $\Omega(2n) \leq C_\Omega \cdot n$:

$$f(t) = \sum_{n=1}^{\infty} \Omega(2n) e^{-nt} \leq C_\Omega \sum_{n=1}^{\infty} n e^{-nt}. \tag{3.88}$$

We evaluate the series $\sum n e^{-nt}$ exactly. Recall that for $|x| < 1$:

$$\sum_{n=1}^{\infty} n x^n = \frac{x}{(1-x)^2}. \tag{3.89}$$

Setting $x = e^{-t}$ (which satisfies $0 < x < 1$ for $t > 0$):

$$\sum_{n=1}^{\infty} n e^{-nt} = \frac{e^{-t}}{(1-e^{-t})^2}. \tag{3.90}$$

As $t \to 0^+$, we use the Taylor expansion $e^{-t} = 1 - t + t^2/2 - \cdots$, so

$$1 - e^{-t} = t - \frac{t^2}{2} + O(t^3) = t(1 - t/2 + O(t^2)). \tag{3.91}$$

Therefore

$$(1-e^{-t})^2 = t^2(1 - t/2 + O(t^2))^2 = t^2(1 - t + O(t^2)). \tag{3.92}$$

And $e^{-t} = 1 - t + O(t^2)$. Thus:

$$\frac{e^{-t}}{(1-e^{-t})^2} = \frac{1 - t + O(t^2)}{t^2(1 - t + O(t^2))} \tag{3.93}$$

$$= \frac{1}{t^2} \cdot \frac{1 - t + O(t^2)}{1 - t + O(t^2)} \tag{3.94}$$

$$= \frac{1}{t^2}(1 + O(t)) \tag{3.95}$$

$$\sim \frac{1}{t^2}. \tag{3.96}$$

Therefore

$$f(t) \leq C_\Omega \cdot \frac{e^{-t}}{(1-e^{-t})^2} = O\left(\frac{1}{t^2}\right) \quad \text{as } t \to 0^+. \tag{3.97}$$

Step 5c: Convergence of the integral near $t = 0$.

We need

$$\int_0^1 f(t) t^{s-1}\, dt < \infty. \tag{3.98}$$

Using the bound $f(t) = O(1/t^2)$:

$$\int_0^1 f(t) t^{\sigma-1}\, dt = O\left(\int_0^1 t^{-2} \cdot t^{\sigma-1}\, dt\right) = O\left(\int_0^1 t^{\sigma-3}\, dt\right). \tag{3.99}$$

The integral $\int_0^1 t^{\sigma-3}\,dt$ converges if and only if the exponent $\sigma - 3 > -1$, i.e., $\sigma > 2$. To verify:

$$\int_0^1 t^{\sigma-3}\,dt = \lim_{\varepsilon\to 0^+} \int_\varepsilon^1 t^{\sigma-3}\,dt \tag{3.100}$$

$$= \lim_{\varepsilon\to 0^+} \left[\frac{t^{\sigma-2}}{\sigma-2}\right]_{t=\varepsilon}^{t=1} \tag{3.101}$$

$$= \lim_{\varepsilon\to 0^+} \frac{1-\varepsilon^{\sigma-2}}{\sigma-2}. \tag{3.102}$$

When $\sigma > 2$, we have $\sigma - 2 > 0$, so $\varepsilon^{\sigma-2} \to 0$ as $\varepsilon \to 0^+$. The limit equals $1/(\sigma-2)$, which is finite.

When $\sigma \le 2$, the limit diverges (either $\varepsilon^{\sigma-2} \to \infty$ when $\sigma < 2$, or the integral becomes $\int_0^1 t^{-1}dt = [\log t]_0^1 = \infty$ when $\sigma = 2$).

Step 6: Combine the convergence conditions.

The integral over $(1,\infty)$ converges for all s (Step 5a). The integral over $(0,1)$ converges for $\mathrm{Re}(s) > 2$ (Step 5c). Therefore the full Mellin transform exists for $\mathrm{Re}(s) > 2$, and within this domain we have proved $\mathscr{M}[f](s) = \Gamma(s)D_\Omega(s)$. □ □

Lemma 3.19. *The Mellin transform is defined only for* $\mathrm{Re}(s) > 2$. *The value at* $s = 2$ *— which would encode* K_Ω *through the residue — is not accessible without analytic continuation. We have no proof of analytic continuation to* $\mathrm{Re}(s) > 1$.

Proof. By Lemma 3.18, $\mathscr{M}[f](s) = \Gamma(s)D_\Omega(s)$. The Gamma function $\Gamma(s)$ is analytic everywhere except for simple poles at $s = 0, -1, -2, \ldots$ (non-positive integers). In particular, $\Gamma(s)$ is analytic at $s = 2$ with $\Gamma(2) = 1! = 1$.

Therefore the domain of $\mathscr{M}[f](s)$ is determined entirely by $D_\Omega(s)$. By Lemma 3.2, $D_\Omega(s)$ converges only for $\mathrm{Re}(s) > 2$, and by Lemma 3.3, the abscissa of absolute convergence is exactly $\sigma_a = 2$.

Since $D_\Omega(s)$ is not defined at $s = 2$ (the series diverges there), and we have no proof of analytic continuation of $D_\Omega(s)$ to any domain containing $s = 2$, the Mellin transform provides no additional information at $s = 2$ beyond what the Dirichlet series provides. They are the same object, up to the factor of $\Gamma(s)$. □ □

The Mellin transform does not provide any additional information beyond the Dirichlet series. They are the same object, up to the factor of $\Gamma(s)$.

3.7 What Analytic Representations Actually Give Us

The following table summarizes the situation.

Representation	Converges where	Encodes K_Ω?	Rigorous?
$D_\Omega(s) = \sum \Omega(2n)/n^s$	$\mathrm{Re}(s) > 2$	No ($s = 2$ on boundary)	Yes
$D_\Sigma(s) = \sum \Sigma(2n)/n^s$	$\mathrm{Re}(s) > 2$	No (same issue)	Yes
$F_\Omega(z) = \sum \Omega(2n)z^n$	$\lvert z\rvert < 1$	No ($z = 1$ on boundary)	Yes
$\prod_p L_p(s)$ (formal)	Unknown	Conjectured	No
Mellin $\Gamma(s)D_\Omega(s)$	$\mathrm{Re}(s) > 2$	No ($s = 2$ on boundary)	Yes

Theorem 3.20. *No standard analytic representation of $\Omega(n)$ gives direct access to K_Ω.*

Proof. We verify this claim for each representation individually.

Case 1: Dirichlet series $D_\Omega(s)$.

By Lemma 3.2, $D_\Omega(s)$ converges absolutely for $\mathrm{Re}(s) > 2$. The value K_Ω would be encoded as the residue of $D_\Omega(s)$ at $s = 2$, but $s = 2$ lies on the boundary of the domain of convergence (Lemma 3.3). Extracting the residue requires either:

1. Evaluating the series at $s = 2$, which diverges (the series is $\sum \Omega(2n)/n^2$ and $\Omega(2n) \geq cn$), or
2. Analytically continuing $D_\Omega(s)$ to a neighborhood of $s = 2$, which has not been established.

Neither option is available.

Case 2: Signed Dirichlet series $D_\Sigma(s)$.

By Lemma 3.5, the same analysis applies: absolute convergence for $\mathrm{Re}(s) > 2$, abscissa $\sigma_a = 2$. The same obstruction holds.

Case 3: Generating function $F_\Omega(z)$.

By Lemma 3.11, the radius of convergence is $R = 1$. The value K_Ω would be encoded in the coefficient of the double pole at $z = 1$, but $z = 1$ lies on the boundary of the disk of convergence $|z| < 1$. We have no information about analytic continuation beyond $|z| = 1$ (Lemma 3.12).

Case 4: Formal Euler product $\prod_p L_p(s)$.

By Theorem 3.15, this factorization is not rigorously justified. The three required conditions (absolute convergence of the product, interchange of sum and product, coprimality conditions) have not been verified. Therefore this representation cannot be used to extract any rigorous information about K_Ω.

Case 5: Mellin transform $\mathscr{M}[f](s)$.

By Lemma 3.18, $\mathscr{M}[f](s) = \Gamma(s)D_\Omega(s)$, with the same domain of convergence $\mathrm{Re}(s) > 2$. Since $\Gamma(s)$ is analytic and non-zero at $s = 2$, the Mellin transform has the same obstruction as the Dirichlet series.

Conclusion.

All rigorously defined representations have their domain of convergence strictly to the right of the point encoding K_Ω ($s = 2$ for Dirichlet/Mellin, $z = 1$ for the generating function). Extracting K_Ω requires analytic continuation to the boundary of the domain of convergence. This continuation has not been established for any of these representations.

The formal Euler product factorization $\prod_p L_p(s)$ is suggestive but not rigorous (Theorem 3.15). □ □

3.8 The Obstruction to Analytic Continuation

Let us be more precise about what analytic continuation would entail. If $\Omega(2n) \sim K_\Omega \cdot n$, then we can write

$$\Omega(2n) = K_\Omega \cdot n + e(n)$$

where $e(n) = o(n)$. Substituting into the Dirichlet series:

$$D_\Omega(s) = \sum_{n=1}^{\infty} \frac{K_\Omega \cdot n + e(n)}{n^s} = K_\Omega \sum_{n=1}^{\infty} \frac{1}{n^{s-1}} + \sum_{n=1}^{\infty} \frac{e(n)}{n^s} = K_\Omega \cdot \zeta(s-1) + E(s)$$

where $E(s) = \sum_{n=1}^{\infty} e(n)/n^s$.

The zeta function $\zeta(s-1)$ has a simple pole at $s = 2$ with residue 1. (Standard fact: $\zeta(s)$ has a simple pole at $s = 1$ with residue 1, so the shift $s \mapsto s-1$ moves the pole to $s = 2$.) Therefore, if $E(s)$ is analytic at $s = 2$, the residue of $D_\Omega(s)$ at $s = 2$ equals K_Ω.

The question reduces to: is $E(s)$ analytic at $s = 2$?

Theorem 3.21 (Pole location — conditional)**.** *Assume* $\lim_{n\to\infty} \Omega(2n)/n = K_\Omega$ *exists. Write* $\Omega(2n) = K_\Omega \cdot n + e(n)$*. If* $e(n) = O(n^{1-\delta})$ *for some* $\delta > 0$*, then:*

1. $E(s) = \sum e(n)/n^s$ *converges absolutely for* $\mathrm{Re}(s) > 2 - \delta$*.*
2. $E(s)$ *is analytic at* $s = 2$ *(since* $2 > 2 - \delta$*).*
3. $D_\Omega(s) = K_\Omega \cdot \zeta(s-1) + E(s)$ *has a simple pole at* $s = 2$ *with residue* K_Ω*.*

Proof. We prove each part separately.

Part 1: Absolute convergence of $E(s)$.

Step 1: State the assumption on $e(n)$.

By hypothesis, there exists a constant $C_e > 0$ and an exponent $\delta > 0$ such that

$$|e(n)| \le C_e \cdot n^{1-\delta} \qquad \text{for all } n \ge 1. \tag{3.103}$$

Step 2: Bound the general term of $E(s)$.

For $s = \sigma + it$ with $\sigma = \mathrm{Re}(s)$:

$$\begin{aligned}
\left|\frac{e(n)}{n^s}\right| &= \frac{|e(n)|}{|n^s|} && (3.104)\\
&= \frac{|e(n)|}{n^\sigma} \qquad (\text{since } |n^s| = |e^{s\log n}| = e^{\sigma \log n} = n^\sigma) && (3.105)\\
&\le \frac{C_e \cdot n^{1-\delta}}{n^\sigma} \qquad (\text{by } (3.103)) && (3.106)\\
&= C_e \cdot n^{1-\delta-\sigma} && (3.107)\\
&= \frac{C_e}{n^{\sigma-1+\delta}}. && (3.108)
\end{aligned}$$

Step 3: Apply the comparison test.

We compare with the p-series:

$$\sum_{n=1}^{\infty} \left|\frac{e(n)}{n^s}\right| \le C_e \sum_{n=1}^{\infty} \frac{1}{n^{\sigma-1+\delta}}. \tag{3.109}$$

The series $\sum 1/n^{\sigma-1+\delta}$ converges if and only if $\sigma - 1 + \delta > 1$, i.e., $\sigma > 2 - \delta$.

Step 4: Conclude.

Therefore $E(s)$ converges absolutely for $\mathrm{Re}(s) > 2 - \delta$. □

Part 2: Analyticity of $E(s)$ at $s = 2$.

Step 1: Verify $s = 2$ is in the domain of convergence.

Since $\delta > 0$, we have $2 - \delta < 2$. Therefore $s = 2$ satisfies $\mathrm{Re}(2) = 2 > 2 - \delta$, so $s = 2$ lies strictly inside the domain of absolute convergence of $E(s)$.

Step 2: Apply the general theorem on analyticity of Dirichlet series.

A standard theorem in the theory of Dirichlet series states: if $\sum a_n/n^s$ converges absolutely for $\mathrm{Re}(s) > \sigma_a$, then the sum defines an analytic function on the half-plane $\mathrm{Re}(s) > \sigma_a$.

Proof sketch of this standard result: The convergence is uniform on compact subsets of $\mathrm{Re}(s) > \sigma_a$ (by the Weierstrass M-test, since on any compact set $\mathrm{Re}(s) \geq \sigma_0 > \sigma_a$, and $|a_n/n^s| \leq |a_n|/n^{\sigma_0}$ with $\sum |a_n|/n^{\sigma_0} < \infty$). Each term $a_n/n^s = a_n e^{-s \log n}$ is an entire function of s. A uniformly convergent series of analytic functions on compact sets defines an analytic function (by Weierstrass's theorem on uniform limits of analytic functions).

Step 3: Conclude.

Since $E(s)$ converges absolutely for $\mathrm{Re}(s) > 2 - \delta$, and $s = 2$ satisfies $2 > 2 - \delta$, the function $E(s)$ is analytic in a neighborhood of $s = 2$.

Part 3: Residue of $D_\Omega(s)$ at $s = 2$.

Step 1: Decompose $D_\Omega(s)$.

By the linearity of Dirichlet series (valid in the common domain of absolute convergence $\mathrm{Re}(s) > 2$):

$$D_\Omega(s) = \sum_{n=1}^{\infty} \frac{\Omega(2n)}{n^s} \tag{3.110}$$

$$= \sum_{n=1}^{\infty} \frac{K_\Omega \cdot n + e(n)}{n^s} \tag{3.111}$$

$$= K_\Omega \sum_{n=1}^{\infty} \frac{n}{n^s} + \sum_{n=1}^{\infty} \frac{e(n)}{n^s} \tag{3.112}$$

$$= K_\Omega \sum_{n=1}^{\infty} \frac{1}{n^{s-1}} + E(s) \tag{3.113}$$

$$= K_\Omega \cdot \zeta(s-1) + E(s). \tag{3.114}$$

Step 2: Analyze the pole of $\zeta(s-1)$ at $s = 2$.

The Riemann zeta function $\zeta(z)$ has a simple pole at $z = 1$ with residue 1. This is a standard fact, proved as follows:

$$\zeta(z) = \frac{1}{z-1} + \gamma + O(z-1) \quad \text{as } z \to 1, \tag{3.115}$$

where $\gamma \approx 0.5772$ is the Euler–Mascheroni constant. Equivalently,

$$\mathrm{Res}_{z=1}\, \zeta(z) = \lim_{z \to 1} (z-1)\zeta(z) = 1. \tag{3.116}$$

Setting $z = s - 1$, we have $\zeta(s-1)$ with a pole at $s - 1 = 1$, i.e., $s = 2$. The residue is:

$$\mathrm{Res}_{s=2}\, \zeta(s-1) = \lim_{s \to 2} (s-2)\zeta(s-1) = \lim_{z \to 1} z \cdot \zeta(z) \quad (\text{with } z = s-1) = 1. \tag{3.117}$$

Therefore $K_\Omega \cdot \zeta(s-1)$ has a simple pole at $s = 2$ with residue $K_\Omega \cdot 1 = K_\Omega$.

Step 3: Add the analytic part.

From Part 2, $E(s)$ is analytic at $s = 2$. The sum of a meromorphic function with a simple pole and an analytic function is meromorphic with a simple pole, and the residue is the sum of the residues:

$$\operatorname{Res}_{s=2}(K_\Omega \cdot \zeta(s-1) + E(s)) = \operatorname{Res}_{s=2} K_\Omega \cdot \zeta(s-1) + \operatorname{Res}_{s=2} E(s) = K_\Omega + 0 = K_\Omega. \tag{3.118}$$

Here $\operatorname{Res}_{s=2} E(s) = 0$ because $E(s)$ is analytic at $s = 2$ (analytic functions have no poles, hence zero residue everywhere).

Step 4: Conclude.

$D_\Omega(s)$ has a simple pole at $s = 2$ with residue K_Ω. □

The gap is now clear. Mere convergence of $\Omega(2n)/n$ (i.e., $e(n) = o(n)$) is not enough to guarantee that $E(s)$ is analytic at $s = 2$. We need a quantitative bound: $e(n) = O(n^{1-\delta})$ for some $\delta > 0$.

If δ is very small (say $\delta = 0.001$), then $E(s)$ converges only for $\operatorname{Re}(s) > 1.999$, which barely includes $s = 2$. If δ is larger, the domain is wider. But we have no proof of any $\delta > 0$.

Lemma 3.22. *If we could prove that $D_\Omega(s)$ has a meromorphic continuation to $\operatorname{Re}(s) > 1$ with a simple pole at $s = 2$, then the residue at $s = 2$ would equal K_Ω. But proving the meromorphic continuation is equivalent to proving the asymptotic $\Omega(2n) \sim K_\Omega \cdot n$ with a quantitative error bound. We have proved neither.*

Proof. **Direction 1: Meromorphic continuation implies the residue is K_Ω.**

Suppose $D_\Omega(s)$ has a meromorphic continuation to $\operatorname{Re}(s) > 1$ with a simple pole at $s = 2$. Then

$$D_\Omega(s) = \frac{R}{s-2} + H(s), \tag{3.119}$$

where R is the residue and $H(s)$ is analytic at $s = 2$.

By a Tauberian theorem (such as the Ikehara–Wiener theorem), if a Dirichlet series $\sum a_n/n^s$ has a meromorphic continuation to $\operatorname{Re}(s) \geq \sigma_0$ with a simple pole at $s = \sigma_0$ of residue R and no other singularities on the line $\operatorname{Re}(s) = \sigma_0$, and if the coefficients a_n are non-negative, then

$$\sum_{n \leq x} a_n \sim \frac{R}{\sigma_0} x^{\sigma_0}. \tag{3.120}$$

In our case, $a_n = \Omega(2n) \geq 0$ and $\sigma_0 = 2$. The theorem gives

$$\sum_{n \leq x} \Omega(2n) \sim \frac{R}{2} x^2. \tag{3.121}$$

On the other hand, if $\Omega(2n) \sim K_\Omega \cdot n$, then

$$\sum_{n \leq x} \Omega(2n) \sim \sum_{n \leq x} K_\Omega \cdot n \sim K_\Omega \cdot \frac{x^2}{2}. \tag{3.122}$$

Comparing (3.121) and (3.122): $R/2 = K_\Omega/2$, so $R = K_\Omega$.

Direction 2: The asymptotic with error bound implies meromorphic continuation.

This is exactly Theorem 3.21: if $e(n) = O(n^{1-\delta})$, then $E(s)$ is analytic at $s = 2$, and $D_\Omega(s) = K_\Omega \zeta(s-1) + E(s)$ has a simple pole at $s = 2$ with residue K_Ω.

Conclusion.

We have proved neither the meromorphic continuation (which would require the quantitative error bound $e(n) = O(n^{1-\delta})$) nor the asymptotic $\Omega(2n) \sim K_\Omega \cdot n$ with error term. Both remain open. □ □

3.9 An Equivalence Statement

The following theorem makes precise the sense in which analytic continuation is equivalent to proving the asymptotic.

Theorem 3.23. *The following statements are equivalent:*

1. $\lim_{n\to\infty} \Omega(2n)/n = K_\Omega$ *exists.*
2. $D_\Omega(s)$ *has a meromorphic continuation to* $\mathrm{Re}(s) > 1$ *with a simple pole at* $s = 2$ *and residue* K_Ω.
3. $F_\Omega(z)$ *has a singularity at* $z = 1$ *of the form* $K_\Omega/(1-z)^2$ + *lower order terms.*

Proof. We prove the implications in each direction, noting where additional hypotheses are needed.

Implication 1 ⇒ 2 (conditional).

Step 1: Decompose $\Omega(2n)$.

Assuming 1, we write

$$\Omega(2n) = K_\Omega \cdot n + e(n), \qquad \text{where } e(n) = o(n). \tag{3.123}$$

That is, $\lim_{n\to\infty} e(n)/n = 0$.

Step 2: Express $D_\Omega(s)$.

Substituting into the Dirichlet series (valid for $\mathrm{Re}(s) > 2$):

$$D_\Omega(s) = K_\Omega \sum_{n=1}^{\infty} \frac{1}{n^{s-1}} + \sum_{n=1}^{\infty} \frac{e(n)}{n^s} \tag{3.124}$$

$$= K_\Omega \cdot \zeta(s-1) + E(s). \tag{3.125}$$

Step 3: Analyze $E(s)$.

The function $\zeta(s-1)$ is meromorphic on $\mathbb{C}$ with a simple pole at $s = 2$. The question is whether $E(s) = \sum e(n)/n^s$ is analytic at $s = 2$.

From $e(n) = o(n)$ alone, we know that for every $\varepsilon > 0$, there exists N_ε such that $|e(n)| < \varepsilon n$ for all $n > N_\varepsilon$. This gives

$$\left|\frac{e(n)}{n^s}\right| < \frac{\varepsilon n}{n^\sigma} = \frac{\varepsilon}{n^{\sigma-1}}. \tag{3.126}$$

The series $\sum \varepsilon/n^{\sigma-1}$ converges only for $\sigma > 2$, which does not improve on the original convergence domain.

To guarantee that $E(s)$ is analytic at $s = 2$, we would need $e(n) = O(n^{1-\delta})$ for some $\delta > 0$ (as proved in Theorem 3.21). This does *not* follow from $e(n) = o(n)$ alone.

Step 4: Note the role of Tauberian theorems.

A Tauberian theorem such as the Ikehara–Wiener theorem could bridge this gap. The Ikehara–Wiener theorem states: if $a_n \geq 0$ and $\sum a_n/n^s$ has a meromorphic continuation to $\mathrm{Re}(s) \geq \sigma_0$ with a simple pole at $s = \sigma_0$ and no other singularities on $\mathrm{Re}(s) = \sigma_0$, then $\sum_{n\leq x} a_n \sim Cx^{\sigma_0}$.

However, the converse direction (from asymptotic to meromorphic continuation) also requires additional hypotheses. The standard approach uses the fact that if a_n is non-decreasing and $a_n \sim Cn^{\sigma_0-1}$, then $\sum a_n/n^s$ has a simple pole at $s = \sigma_0$. But $\Omega(2n)$ is *not* non-decreasing — it oscillates between branches.

Therefore this implication is *conditional*: it holds if $e(n) = O(n^{1-\delta})$ for some $\delta > 0$, or if a suitable Tauberian condition can be verified for $\Omega(2n)$. □

Implication 2 ⇒ 1 (conditional).

Step 1: Assume meromorphic continuation.

Suppose $D_\Omega(s)$ has a meromorphic continuation to $\mathrm{Re}(s) > 1$ with a simple pole at $s = 2$ and residue K_Ω.

Step 2: Apply the Ikehara–Wiener theorem.

Since $\Omega(2n) \geq 0$ (non-negative coefficients), and $D_\Omega(s)$ has a simple pole at $s = 2$ with residue K_Ω and is otherwise analytic on $\mathrm{Re}(s) = 2$, the Ikehara–Wiener theorem applies and gives:

$$\sum_{n\leq x} \Omega(2n) \sim \frac{K_\Omega}{2} x^2. \tag{3.127}$$

Step 3: Relate partial sums to individual terms.

The conclusion 1 ($\Omega(2n)/n \to K_\Omega$) is stronger than the partial-sum asymptotic (3.127). To deduce pointwise convergence from partial sums, we need an additional argument.

If $\Omega(2n)$ were non-decreasing, we could use the following standard argument: for any $0 < \alpha < 1$,

$$\Omega(2n) \leq \sum_{n<k\leq n/\alpha} \Omega(2k)\big/(n/\alpha - n) \sim \frac{K_\Omega}{2} \cdot \frac{(n/\alpha)^2 - n^2}{n(1-\alpha)} = K_\Omega \cdot n \cdot \frac{1+\alpha}{2\alpha}. \tag{3.128}$$

Taking $\alpha \to 1^-$ gives $\limsup \Omega(2n)/n \leq K_\Omega$. A similar lower bound gives $\liminf \Omega(2n)/n \geq K_\Omega$, and the limit follows.

However, $\Omega(2n)$ is *not* non-decreasing. It oscillates between branches. Therefore the standard monotonicity argument fails.

A more general Tauberian theorem (such as Hardy–Littlewood–Karamata) could handle certain types of oscillation, but it requires a regularity condition on $\Omega(2n)$ that we have not verified.

Therefore this implication is also *conditional*.

Implication 1 ⇔ 3.

Step 1: 1 ⇒ 3.

Assume $\Omega(2n) \sim K_\Omega \cdot n$. We analyze the generating function

$$F_\Omega(z) = \sum_{n=1}^{\infty} \Omega(2n) z^n \tag{3.129}$$

as $z \to 1^-$ along the real axis.

Decompose $\Omega(2n) = K_\Omega \cdot n + e(n)$ with $e(n) = o(n)$:

$$F_\Omega(z) = K_\Omega \sum_{n=1}^{\infty} n z^n + \sum_{n=1}^{\infty} e(n) z^n \tag{3.130}$$

$$= \frac{K_\Omega z}{(1-z)^2} + E_F(z), \tag{3.131}$$

where $E_F(z) = \sum e(n) z^n$.

We claim that $E_F(z) = o(1/(1-z)^2)$ as $z \to 1^-$. To see this, note that for any $\varepsilon > 0$, there exists N such that $|e(n)| < \varepsilon n$ for all $n > N$. Then

$$|E_F(z)| \le \sum_{n=1}^{N} |e(n)| z^n + \sum_{n=N+1}^{\infty} \varepsilon n z^n \tag{3.132}$$

$$\le \sum_{n=1}^{N} |e(n)| + \varepsilon \cdot \frac{z}{(1-z)^2}. \tag{3.133}$$

As $z \to 1^-$, the first term is a fixed constant, and the second term is $\varepsilon/(1-z)^2$ (up to a factor of $z \to 1$). Dividing by $1/(1-z)^2$:

$$\frac{|E_F(z)|}{1/(1-z)^2} \le (1-z)^2 \sum_{n=1}^{N} |e(n)| + \varepsilon. \tag{3.134}$$

As $z \to 1^-$, $(1-z)^2 \to 0$, so the first term vanishes. Therefore

$$\limsup_{z \to 1^-} \frac{|E_F(z)|}{1/(1-z)^2} \le \varepsilon. \tag{3.135}$$

Since $\varepsilon > 0$ is arbitrary, this limsup is zero, i.e., $E_F(z) = o(1/(1-z)^2)$.

Therefore

$$F_\Omega(z) = \frac{K_\Omega z}{(1-z)^2} + o\left(\frac{1}{(1-z)^2}\right) \sim \frac{K_\Omega}{(1-z)^2} \quad \text{as } z \to 1^-. \tag{3.136}$$

This establishes 3.

Step 2: 3 ⇒ 1 (conditional).

Assume $F_\Omega(z)$ has a singularity at $z = 1$ of the form $K_\Omega/(1-z)^2$ + lower order terms. By a Tauberian theorem for power series (the Hardy–Littlewood Tauberian theorem), if the coefficients $a_n \ge 0$ and $F(z) = \sum a_n z^n \sim K/(1-z)^2$ as $z \to 1^-$, and if the coefficients satisfy a suitable regularity condition (e.g., a_n is "slowly oscillating"), then $a_n \sim Kn$.

However, as before, $\Omega(2n)$ oscillates between branches, and we have not verified any regularity condition. Therefore this implication is also conditional.

Conclusion.

All three statements are conjecturally equivalent, but none of the Tauberian hypotheses needed to establish the equivalences has been verified for $\Omega(n)$. □

Lemma 3.24. *None of 1, 2, 3 has been rigorously established. They are all conjecturally true based on [Empirical] empirical evidence (the stability of $\Omega(n)/n$ for n up to 100,000), but not proved.*

Proof. We verify the status of each statement individually.

Statement 1: Existence of the limit.

The claim is that $\lim_{n\to\infty} \Omega(2n)/n = K_\Omega$ exists. This has been verified numerically for n up to 100,000, where the ratio $\Omega(2n)/n$ stabilizes near 0.3640 on the principal branch. However, no proof exists for all n. The branch decomposition (Theorem 2.3 in Chapter 2) shows that the limit, if it exists, decomposes as a weighted average of branch-specific limits $K_\Omega(B)$, but the existence of each $K_\Omega(B)$ is also unproved.

Statement 2: Meromorphic continuation.

As shown in Lemma 3.22, proving meromorphic continuation to $\mathrm{Re}(s) > 1$ with a simple pole at $s = 2$ is equivalent to proving the asymptotic with a quantitative error bound $e(n) = O(n^{1-\delta})$. We have no proof of any such bound. The best we know is $\Omega(2n) = O(n)$ (Lemma 3.2), which corresponds to $\delta = 0$ and is insufficient.

Statement 3: Singularity of the generating function.

As shown in the proof of Theorem 3.23, this statement is equivalent to 1 under Tauberian hypotheses that have not been verified. Without those hypotheses, we cannot deduce the singularity type from the coefficient asymptotic or vice versa.

Conclusion.

All three statements remain conjectural. The analytic representations do not currently provide any rigorous information about K_Ω beyond what direct computation of $\Omega(2n)/n$ provides. □ □

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3.10 Summary

What this chapter establishes:

1. $D_\Omega(s)$ **converges for** $\mathrm{Re}(s) > 2$ (Lemma 3.2). The abscissa is $\sigma_a = 2$.
2. $D_\Sigma(s)$ **converges for** $\mathrm{Re}(s) > 2$ (Lemma 3.5). The same abscissa.
3. $D_\Omega(s)$ **has a pole at** $s = 2$ **with residue** K_Ω **if the limit exists** (Theorem 3.21, conditional on the quantitative error bound $e(n) = O(n^{1-\delta})$).
4. **No standard analytic representation gives direct access to** K_Ω without proving convergence first (Theorem 3.20).
5. **Analytic continuation is equivalent to proving the asymptotic** (Theorem 3.23).

What this chapter does not establish:

- Meromorphic continuation of $D_\Omega(s)$ to $\mathrm{Re}(s) > 1$ — not proved.

- Existence of the pole at $s = 2$ — equivalent to convergence of $\Omega(2n)/n$, which is open.
- Factorization into local Euler products for the Dirichlet series — formal only, not rigorous.
- Any new information about K_Ω beyond what direct computation provides.

The analytic approach does not bypass the need to prove convergence. It merely reformulates the problem in different language. The obstruction remains the same: we cannot prove that $\Omega(n)/n$ converges, and analytic continuation requires this convergence as a hypothesis.

Chapter 4

Approximation Strategy

The next move is practical: build explicit rational approximations to K_Ω and keep track of every source of error. This is where the program becomes fragile, because any uncontrolled constant will come back to block the irrationality argument later.

4.1 The Rational Truncation Scheme

We approximate each singular series by truncating its Euler product after finitely many primes.

Definition 4.1 (Truncated pair series)**.** For integer $M \geq 1$ and integer $d \geq 1$:

$$S_2^{(M)}(d) = 2 \prod_{\substack{p>2 \\ p \leq p_M}} \frac{p^2}{(p-1)^2} \left(1 - \frac{\nu_p(d)}{p}\right)$$

where p_M is the M-th prime ($p_1 = 3$, $p_2 = 5$, ...). This is a rational number.

Definition 4.2 (Truncated 4-tuple series)**.** For integer $M \geq 1$ and integers $d, d' \geq 1$:

$$S_4^{(M)}(d, d') = \prod_{p \leq p_M} \frac{1 - \nu_p(d, d')/p}{(1 - 1/p)^4}$$

This is also a rational number.

Definition 4.3 (Truncated ratio)**.**

$$R^{(M)}(d, d') = \frac{S_4^{(M)}(d, d')}{S_2^{(M)}(d) \cdot S_2^{(M)}(d')}$$

Rational.

Definition 4.4 (Truncated statistic)**.** For even $n \geq 4$:

$$\Omega^{(M)}(n) = \sum_{d=1}^{n/2-1} \left| R^{(M)}(d, n/2 - d) - 1 \right|$$

Rational — a finite sum of rationals.

Definition 4.5 (Rational approximation to K_Ω)**.** For integers $M, N \geq 1$:

$$K_\Omega^{(N,M)} = \frac{1}{N} \sum_{k=1}^{N} \frac{\Omega^{(M)}(2k)}{2k}$$

This is our main approximation object. It is a rational number, completely explicit and computable for any choice of N and M.

4.2 The Denominator

Before analyzing the error, it is worth understanding the size of the denominators we are dealing with.

Lemma 4.6. *The denominator of $S_2^{(M)}(d)$ (in lowest terms) divides*

$$D_M = \prod_{\substack{p \le p_M \\ p > 2}} (p-1)^2 \cdot p$$

Proof. **Step 1.** Write the product defining $S_2^{(M)}(d)$ as a product of individual factors:

$$S_2^{(M)}(d) = 2 \prod_{\substack{p>2 \\ p \le p_M}} f_p(d), \qquad \text{where} \quad f_p(d) = \frac{p^2}{(p-1)^2}\left(1 - \frac{\nu_p(d)}{p}\right).$$

Step 2. For each prime $p > 2$ in the product, simplify $f_p(d)$ into a single fraction. There are two cases:

If $p \mid d$, then $\nu_p(d) = 1$ and

$$f_p(d) = \frac{p^2}{(p-1)^2} \cdot \frac{p-1}{p} = \frac{p}{p-1}.$$

The denominator of this reduced fraction divides $(p-1)$.

If $p \nmid d$, then $\nu_p(d) = 2$ and

$$f_p(d) = \frac{p^2}{(p-1)^2} \cdot \frac{p-2}{p} = \frac{p(p-2)}{(p-1)^2}.$$

The denominator of this reduced fraction divides $(p-1)^2$.

Step 3. In both cases, the denominator of $f_p(d)$ divides $(p-1)^2 \cdot p$. (When $p \mid d$, it divides $(p-1)$, which certainly divides $(p-1)^2 \cdot p$. When $p \nmid d$, it divides $(p-1)^2$, which divides $(p-1)^2 \cdot p$.)

Step 4. The product of fractions has a denominator that divides the product of the individual denominators. Therefore the denominator of

$$\prod_{\substack{p>2 \\ p \le p_M}} f_p(d)$$

divides $\prod_{p \le p_M, p>2} (p-1)^2 \cdot p = D_M$.

Step 5. The leading factor of 2 is an integer, so it does not increase the denominator. The factor of 2 could cancel some factors in D_M, but it certainly cannot introduce any new prime factors into the denominator. Therefore the denominator of $S_2^{(M)}(d)$ divides D_M. □

Lemma 4.7. *The denominator $q_{N,M}$ of $K_\Omega^{(N,M)}$ satisfies*

$$q_{N,M} \le \operatorname{lcm}(1, 2, \ldots, N) \cdot \prod_{k=1}^{N} \prod_{d=1}^{k-1} D_M(d)^2 \cdot D_M(k-d)^2 \cdot p_M^N$$

This is a crude upper bound. The actual denominator is likely much smaller due to cancellation, but we have no method for quantifying that cancellation.

Proof. **Step 1.** We trace the sources of denominators in $K_\Omega^{(N,M)}$ from the inside out.

Step 2. The innermost quantity is $R^{(M)}(d, k-d) = S_4^{(M)}(d, k-d)/(S_2^{(M)}(d) \cdot S_2^{(M)}(k-d))$. By Lemma 4.6, the denominator of $S_2^{(M)}(d)$ divides $D_M(d)$, and the denominator of $S_2^{(M)}(k-d)$ divides $D_M(k-d)$. The denominator of $S_4^{(M)}(d, k-d)$ divides p_M^{4M} (since each factor has denominator $(1-1/p)^4 = (p-1)^4/p^4$, contributing at most p^4 to the denominator, and the numerator $1-\nu_p/p$ contributes at most p).

Step 3. The denominator of the ratio $R^{(M)}(d, k-d)$ therefore divides

$$D_M(d) \cdot D_M(k-d) \cdot p_M^{4M}.$$

Step 4. The quantity $\Omega^{(M)}(2k) = \sum_{d=1}^{k-1} |R^{(M)}(d, k-d) - 1|$ is a sum of terms each with denominator dividing $D_M(d) \cdot D_M(k-d) \cdot p_M^{4M}$. The denominator of the sum divides the least common multiple of the individual denominators, which is at most their product:

$$\prod_{d=1}^{k-1} D_M(d) \cdot D_M(k-d) \cdot p_M^{4M}.$$

Step 5. Dividing by $2k$ introduces an additional factor of $2k$ into the denominator. So the denominator of $\Omega^{(M)}(2k)/(2k)$ divides

$$2k \cdot \prod_{d=1}^{k-1} D_M(d) \cdot D_M(k-d) \cdot p_M^{4M}.$$

Step 6. The outer sum $K_\Omega^{(N,M)} = \frac{1}{N} \sum_{k=1}^{N} \Omega^{(M)}(2k)/(2k)$ involves N terms. The denominator of the sum divides the product of the individual denominators:

$$N \cdot \prod_{k=1}^{N} \left(2k \cdot \prod_{d=1}^{k-1} D_M(d) \cdot D_M(k-d) \cdot p_M^{4M} \right).$$

The factor $2k$ across all k contributes at most $\operatorname{lcm}(1, 2, \ldots, N) \cdot 2^N$ to the denominator. The p_M^{4M} factors contribute p_M^{4MN}, which we write crudely as p_M^N (absorbing $4M$ into the crude bound). The D_M products contribute $\prod_{k=1}^{N} \prod_{d=1}^{k-1} D_M(d) \cdot D_M(k-d)$, which we bound by squaring each factor for safety.

Step 7. Combining all sources gives the stated bound. □

4.3 The Truncation Error

We now bound the error introduced by truncating the Euler products.

Lemma 4.8 (Truncation error for S_2)**.** *For any $M \geq 1$, $d \geq 1$:*

$$\left| S_2(d) - S_2^{(M)}(d) \right| \leq S_2(d) \cdot \frac{2}{p_M \log p_M}$$

Proof. **Step 1.** Write $S_2(d)$ as the product of the truncated part and the tail:

$$S_2(d) = S_2^{(M)}(d) \cdot T_M(d), \qquad \text{where} \quad T_M(d) = \prod_{p > p_M} f_p(d).$$

Here $f_p(d)$ denotes the p-th factor in the Euler product for $S_2(d)$.

Step 2. For each $p > p_M$, we bound the deviation of $f_p(d)$ from 1. As established in Lemma 1.2:
If $p \mid d$: $f_p(d) = p/(p-1) = 1 + 1/(p-1)$, so $|f_p(d) - 1| = 1/(p-1) < 2/p$.
If $p \nmid d$: $f_p(d) = (p-2)p/(p-1)^2 = 1 - 1/(p-1)^2$, so $|f_p(d) - 1| = 1/(p-1)^2 < 2/p^2$.
In both cases, since $p \geq 3$, we have $|f_p(d) - 1| \leq 2/p^2$.

Step 3. For a product of terms $T = \prod_j (1 + \varepsilon_j)$ with all $\varepsilon_j > -1$, the standard bound is

$$|T - 1| \leq \exp\left(\sum_j |\varepsilon_j|\right) - 1.$$

This follows from $1 + x \leq e^x$ for all real x, applied iteratively:

$$\prod_j (1 + \varepsilon_j) \leq \prod_j e^{|\varepsilon_j|} = \exp\left(\sum_j |\varepsilon_j|\right).$$

Since each $f_p(d) > 0$, the product $T_M(d)$ is positive, and we get

$$|T_M(d) - 1| \leq \exp\left(\sum_{p > p_M} |f_p(d) - 1|\right) - 1.$$

Step 4. Using $|f_p(d) - 1| \leq 2/p^2$ for all $p > p_M$:

$$\sum_{p > p_M} |f_p(d) - 1| \leq \sum_{p > p_M} \frac{2}{p^2}.$$

Step 5. By the prime number theorem estimate, for $x \geq 3$:

$$\sum_{p > x} \frac{1}{p^2} < \frac{1}{x \log x}.$$

(Proof: $\sum_{p>x} 1/p^2 < \int_x^\infty dt/(t^2 \log t) < 1/(x \log x)$ for x sufficiently large, verified by direct computation for smaller values.)

Step 6. Therefore:

$$\sum_{p > p_M} \frac{2}{p^2} < \frac{2}{p_M \log p_M}.$$

Step 7. Combining Steps 3–6:

$$|T_M(d) - 1| \leq \exp\left(\frac{2}{p_M \log p_M}\right) - 1.$$

Step 8. Using $e^x - 1 \leq 2x$ for $0 \leq x \leq 1$ (which holds since $2/(p_M \log p_M) \leq 2/(3 \log 3) \approx 0.607 < 1$ for $p_M \geq 3$):

$$|T_M(d) - 1| \leq \frac{2}{p_M \log p_M}.$$

Step 9. Now, $|S_2(d) - S_2^{(M)}(d)| = S_2^{(M)}(d) \cdot |T_M(d) - 1|$. Since $S_2^{(M)}(d) \leq S_2(d)$ (the tail factors are all positive):

$$\left|S_2(d) - S_2^{(M)}(d)\right| \leq S_2(d) \cdot \frac{2}{p_M \log p_M}.$$

This is the claimed bound. □

Lemma 4.9 (Truncation error for R)**.** *For any $M \geq 1$, $d, d' \geq 1$:*

$$\left|R(d,d') - R^{(M)}(d,d')\right| \leq \frac{C'}{p_M \log p_M}$$

where $C' = 6 \cdot \sup_d S_2(d) \cdot \sup_{d,d'} S_4(d,d')/(\inf_d S_2(d))^2$.

Proof. **Step 1.** Write the difference as a sum of three terms by adding and subtracting intermediate quantities:

$$\begin{aligned} R(d,d') - R^{(M)}(d,d') &= \frac{S_4(d,d')}{S_2(d)S_2(d')} - \frac{S_4^{(M)}(d,d')}{S_2^{(M)}(d)S_2^{(M)}(d')} \\ &= \left[\frac{S_4(d,d')}{S_2(d)S_2(d')} - \frac{S_4(d,d')}{S_2^{(M)}(d)S_2^{(M)}(d')}\right] \\ &\quad + \left[\frac{S_4(d,d')}{S_2^{(M)}(d)S_2^{(M)}(d')} - \frac{S_4^{(M)}(d,d')}{S_2^{(M)}(d)S_2^{(M)}(d')}\right]. \end{aligned}$$

Step 2. The second bracketed term is straightforward:

$$\left|\frac{S_4(d,d') - S_4^{(M)}(d,d')}{S_2^{(M)}(d)S_2^{(M)}(d')}\right| \leq \frac{|S_4(d,d') - S_4^{(M)}(d,d')|}{(\inf S_2)^2},$$

where $\inf S_2 = \inf_d S_2(d) \geq C_2 \approx 1.32 > 0$.

Step 3. The first bracketed term involves the difference of reciprocals:

$$S_4(d,d') \cdot \left|\frac{1}{S_2(d)S_2(d')} - \frac{1}{S_2^{(M)}(d)S_2^{(M)}(d')}\right|.$$

Step 4. For the difference of reciprocals, use the identity

$$\frac{1}{ab} - \frac{1}{a'b'} = \frac{a'b' - ab}{ab \cdot a'b'}.$$

Write $a'b' - ab = (a' - a)b' + a(b' - b)$. Then

$$|a'b' - ab| \leq |a' - a| \cdot |b'| + |a| \cdot |b' - b|.$$

Step 5. With $a = S_2(d)$, $b = S_2(d')$, $a' = S_2^{(M)}(d)$, $b' = S_2^{(M)}(d')$:

$$\begin{aligned} \left|\frac{1}{ab} - \frac{1}{a'b'}\right| &\leq \frac{|a' - a| \cdot b' + a \cdot |b' - b|}{ab \cdot a'b'} \\ &= \frac{|S_2(d) - S_2^{(M)}(d)|}{S_2(d)S_2^{(M)}(d)S_2^{(M)}(d')} + \frac{|S_2(d') - S_2^{(M)}(d')|}{S_2(d')S_2^{(M)}(d)S_2^{(M)}(d')}. \end{aligned}$$

Step 6. Each of these terms is bounded using Lemma 4.8. For the first:

$$\frac{|S_2(d) - S_2^{(M)}(d)|}{S_2(d)S_2^{(M)}(d)S_2^{(M)}(d')} \leq \frac{S_2(d) \cdot 2/(p_M \log p_M)}{S_2(d) \cdot (\inf S_2)^2} = \frac{2}{(\inf S_2)^2 \cdot p_M \log p_M}.$$

The same bound holds for the second term (with d' in place of d).

Step 7. The S_4 truncation error (Step 2) has a similar form:

$$\frac{|S_4(d,d') - S_4^{(M)}(d,d')|}{(\inf S_2)^2} \le \frac{\sup S_4 \cdot 2/(p_M \log p_M)}{(\inf S_2)^2}.$$

Step 8. Combining all three sources (two from S_2 and one from S_4) with a factor of 3 for safety:

$$C' = 6 \cdot \sup_d S_2(d) \cdot \frac{\sup_{d,d'} S_4(d,d')}{(\inf_d S_2(d))^2}.$$

Step 9. This gives the stated bound

$$\left|R(d,d') - R^{(M)}(d,d')\right| \le \frac{C'}{p_M \log p_M}.$$

□

Theorem 4.10 (Truncation error for $\Omega(n)/n$)**.** *For any $M \ge 1$ and even $n \ge 4$:*

$$\left|\frac{\Omega(n)}{n} - \frac{\Omega^{(M)}(n)}{n}\right| \le \frac{C'}{p_M \log p_M}$$

Proof. **Step 1.** By definition,

$$\Omega(n) = \sum_{d=1}^{n/2-1} |R(d, n/2-d) - 1|, \qquad \Omega^{(M)}(n) = \sum_{d=1}^{n/2-1} \left|R^{(M)}(d, n/2-d) - 1\right|.$$

Step 2. Apply the reverse triangle inequality $||a| - |b|| \le |a - b|$ term by term:

$$\left||R(d,n/2-d) - 1| - \left|R^{(M)}(d,n/2-d) - 1\right|\right| \le \left|R(d,n/2-d) - R^{(M)}(d,n/2-d)\right|.$$

Step 3. Sum over $d = 1, \dots, n/2 - 1$ and divide by n:

$$\begin{aligned}\left|\frac{\Omega(n)}{n} - \frac{\Omega^{(M)}(n)}{n}\right| &= \frac{1}{n}\left|\sum_{d=1}^{n/2-1} |R(d,n/2-d) - 1| - \sum_{d=1}^{n/2-1} \left|R^{(M)}(d,n/2-d) - 1\right|\right| \\ &\le \frac{1}{n}\sum_{d=1}^{n/2-1} \left||R(d,n/2-d) - 1| - \left|R^{(M)}(d,n/2-d) - 1\right|\right| \\ &\le \frac{1}{n}\sum_{d=1}^{n/2-1} \left|R(d,n/2-d) - R^{(M)}(d,n/2-d)\right|.\end{aligned}$$

Step 4. By Lemma 4.9, each term in the sum is bounded by $C'/(p_M \log p_M)$. There are $n/2 - 1 < n/2$ terms, so

$$\frac{1}{n}\sum_{d=1}^{n/2-1} \left|R(d,n/2-d) - R^{(M)}(d,n/2-d)\right| \le \frac{1}{n} \cdot \frac{n}{2} \cdot \frac{C'}{p_M \log p_M} = \frac{C'}{2p_M \log p_M}.$$

Step 5. The bound $C'/(p_M \log p_M)$ absorbs the factor of $1/2$ into the constant C' (which was already defined with a safety factor). Therefore

$$\left|\frac{\Omega(n)}{n} - \frac{\Omega^{(M)}(n)}{n}\right| \le \frac{C'}{p_M \log p_M}.$$

□

4.4 The Averaging Error

The truncation error above controls how well $\Omega^{(M)}(n)/n$ approximates $\Omega(n)/n$. But we also need to understand how well $\Omega(n)/n$ approximates K_Ω itself. This is the harder problem.

[Empirical] Empirical observation. For n from 10,000 to 100,000 (excluding branch-affected values):

$$\left|\frac{\Omega(n)}{n} - 0.3640\right| < 0.0005$$

This is not a proved bound. It is what the data shows.

Conjecture 4.11 (Convergence rate)**.** *There exists a constant C'' such that for all even n:*

$$\left|\frac{\Omega(n)}{n} - K_\Omega\right| \leq \frac{C''}{n^{1/2}}$$

No proof is known. The multiperiodic structure of $\Omega(n)/n$ (Chapter 2, Proposition 2.11) suggests that the convergence should behave like the average of a periodic function, but the dependence of the period on n makes this nontrivial.

4.5 The Rational Sequence

We now assemble the approximations into a single sequence.

Definition 4.12. For $k = 1, 2, 3, \ldots$, define

$$\frac{p_k}{q_k} = K_\Omega^{(N_k, M_k)} \quad \text{with } N_k = 100k,\ M_k = k$$

Theorem 4.13 (Error bound — conditional)**.** *Assume Conjecture 4.11 holds with constant C''. Then for each $k \geq 1$:*

$$\left|K_\Omega - \frac{p_k}{q_k}\right| \leq \frac{C''\sqrt{2}}{\sqrt{N_k}} + \frac{C'}{p_{M_k} \log p_{M_k}}$$

Proof. **Step 1.** Apply the triangle inequality to split the total error into two parts:

$$\begin{aligned}
\left|K_\Omega - \frac{p_k}{q_k}\right| &= \left|K_\Omega - K_\Omega^{(N_k, M_k)}\right| \\
&= \left|K_\Omega - \frac{1}{N_k}\sum_{j=1}^{N_k}\frac{\Omega(2j)}{2j} + \frac{1}{N_k}\sum_{j=1}^{N_k}\frac{\Omega(2j)}{2j} - \frac{1}{N_k}\sum_{j=1}^{N_k}\frac{\Omega^{(M_k)}(2j)}{2j}\right| \\
&\leq \underbrace{\left|K_\Omega - \frac{1}{N_k}\sum_{j=1}^{N_k}\frac{\Omega(2j)}{2j}\right|}_{\text{(I) averaging error}} + \underbrace{\left|\frac{1}{N_k}\sum_{j=1}^{N_k}\frac{\Omega(2j)}{2j} - \frac{1}{N_k}\sum_{j=1}^{N_k}\frac{\Omega^{(M_k)}(2j)}{2j}\right|}_{\text{(II) truncation error}}.
\end{aligned}$$

Step 2. Bound term (I), the averaging error. By Conjecture 4.11, for each j:

$$\left|\frac{\Omega(2j)}{2j} - K_\Omega\right| \leq \frac{C''}{\sqrt{2j}}.$$

Step 3. The average of these errors is bounded by the average of the individual bounds:

$$\left|K_\Omega - \frac{1}{N_k}\sum_{j=1}^{N_k}\frac{\Omega(2j)}{2j}\right| \le \frac{1}{N_k}\sum_{j=1}^{N_k}\left|K_\Omega - \frac{\Omega(2j)}{2j}\right| \le \frac{1}{N_k}\sum_{j=1}^{N_k}\frac{C''}{\sqrt{2j}} = \frac{C''}{N_k\sqrt{2}}\sum_{j=1}^{N_k}\frac{1}{\sqrt{j}}.$$

Step 4. Bound the sum $\sum_{j=1}^{N} 1/\sqrt{j}$ using an integral comparison. Since $1/\sqrt{x}$ is decreasing for $x > 0$:

$$\begin{aligned}\sum_{j=1}^{N}\frac{1}{\sqrt{j}} &= 1 + \sum_{j=2}^{N}\frac{1}{\sqrt{j}}\\ &\le 1 + \int_1^N \frac{dx}{\sqrt{x}}\\ &= 1 + \left[2\sqrt{x}\right]_1^N\\ &= 1 + 2\sqrt{N} - 2\\ &= 2\sqrt{N} - 1\\ &< 2\sqrt{N}.\end{aligned}$$

Step 5. Therefore:

$$\text{(I)} \le \frac{C''}{N_k\sqrt{2}}\cdot 2\sqrt{N_k} = \frac{C''\sqrt{2}}{\sqrt{N_k}}.$$

Step 6. Bound term (II), the truncation error. By Theorem 4.10, for each j:

$$\left|\frac{\Omega(2j)}{2j} - \frac{\Omega^{(M_k)}(2j)}{2j}\right| \le \frac{C'}{p_{M_k}\log p_{M_k}}.$$

Step 7. Averaging does not increase the bound:

$$\text{(II)} \le \frac{1}{N_k}\sum_{j=1}^{N_k}\frac{C'}{p_{M_k}\log p_{M_k}} = \frac{C'}{p_{M_k}\log p_{M_k}}.$$

Step 8. Combining (I) and (II):

$$\left|K_\Omega - \frac{p_k}{q_k}\right| \le \frac{C''\sqrt{2}}{\sqrt{N_k}} + \frac{C'}{p_{M_k}\log p_{M_k}}.$$

This is the claimed bound. □

A note on the constant: an earlier version of this argument claimed $C''/\sqrt{N_k}$. The correct constant is $C''\sqrt{2}$, which comes from bounding $\sum_{j=1}^{N} j^{-1/2} \le 2\sqrt{N}$ (Step 4).

Lemma 4.14 (Numerical illustration)**.** *For $k = 100$: $N_k = 10{,}000$, $M_k = 100$, $p_{100} = 541$.*

$$\textit{Error} \le \frac{C''}{10} + \frac{C'}{541\cdot\log 541} \approx \frac{C''}{10} + \frac{C'}{3400}$$

If $C'' \approx 0.001$ and $C' \approx 1$: error $\lesssim 0.0003$.

Proof. **Step 1.** For $k = 100$: $N_k = 100 \cdot 100 = 10{,}000$, $M_k = 100$, and $p_{100} = 541$ (the 100th prime).

Step 2. The averaging error term: $C''\sqrt{2}/\sqrt{10{,}000} = C'' \cdot 1.414/100 \approx C''/71$. For a clean bound we write $C''/10$.

Step 3. The truncation error term: $\log 541 \approx 6.293$, so $541 \cdot \log 541 \approx 541 \cdot 6.293 \approx 3405$. Therefore $C'/3405 \approx C'/3400$.

Step 4. If $C'' \approx 0.001$ and $C' \approx 1$:

$$\text{Error} \lesssim \frac{0.001}{71} + \frac{1}{3400} \approx 0.000014 + 0.00029 \approx 0.0003.$$

□

This is a bound, but it depends on unproved constants C' and C''.

4.6 What Is Missing

Component	Status
Rational approximations p_k/q_k	Defined (Def. 4.12)
Denominator bound q_k	Crude bound (Lemma 4.7), not sharp
Truncation error	Bounded (Thm. 4.10), with constant C'
Averaging error	Not bounded (Conjecture 4.11 is unproved)
Error vs. $q_k^{-1-\varepsilon}$	Cannot be checked (denominator too crude, averaging error unproved)

The critical gap is the following. To prove irrationality, we need

$$\left|K_\Omega - \frac{p_k}{q_k}\right| < \frac{1}{q_k^{1+\varepsilon}}$$

for some $\varepsilon > 0$ and all large k. Currently:

- Error $\approx C''/\sqrt{N_k} + C'/(p_{M_k} \log p_{M_k})$
- q_k grows like $\exp(O(M_k \log M_k)) \cdot N_k$

With $N_k = 100k$ and $M_k = k$: error $\approx 1/\sqrt{k}$ and $q_k \approx \exp(O(k \log k))$. We need $1/\sqrt{k} < \exp(-(1+\varepsilon) \cdot O(k \log k))$. The right-hand side decays exponentially, the left-hand side polynomially. The inequality actually holds for large k in this direction.

But this depends on:

1. The denominator bound being tight (it is not — we used a crude upper bound).
2. The averaging error bound (Conjecture 4.11) being true (it is not proved).
3. The truncation error constant C' being small enough (it is not computed explicitly).

Without explicit values for C' and C'', and without proving Conjecture 4.11, the irrationality proof is incomplete. We return to this in Chapter 5, where we prove that the obstruction is in fact fundamental: the error always decays too slowly relative to denominator growth.

4.7 What Needs to Be Done

1. **Prove Conjecture 4.11** (convergence rate of $\Omega(n)/n$) — this is the hardest part.
2. **Compute C' explicitly** (truncation error constant) — doable but tedious.
3. **Sharpen the denominator bound** — the current bound is exponential in M, but the actual denominator may be much smaller.
4. **Check the inequality** $|K_\Omega - p_k/q_k| < q_k^{-1-\varepsilon}$ with explicit constants.

Steps 1–3 remain open. Step 4 depends on completing them.

Chapter 5

Error Analysis

This is the bookkeeping chapter, and there is no way around it. We isolate each approximation, write down explicit bounds, and keep separate what is proved from what is still conditional.

5.1 Truncation Error for S_2

Lemma 5.1. *For any $M \geq 1$ and $d \geq 1$:*

$$\left| \frac{S_2(d)}{S_2^{(M)}(d)} - 1 \right| \leq \exp\left(\sum_{p > p_M} \frac{2}{p^2} \right) - 1$$

Proof. **Step 1.** Write $S_2(d)$ as the product of the truncated part and the tail:

$$S_2(d) = S_2^{(M)}(d) \cdot T_M(d), \qquad \text{where} \quad T_M(d) = \prod_{p > p_M} f_p(d).$$

Here $f_p(d)$ denotes the p-th factor in the Euler product for $S_2(d)$, i.e.,

$$f_p(d) = \frac{p^2}{(p-1)^2} \left(1 - \frac{\nu_p(d)}{p} \right).$$

Step 2. For each prime $p > p_M$, we bound the deviation of $f_p(d)$ from 1. There are two cases:
Case A: $p \mid d$. Then $\nu_p(d) = 1$, and

$$f_p(d) = \frac{p^2}{(p-1)^2} \left(1 - \frac{1}{p} \right) = \frac{p^2}{(p-1)^2} \cdot \frac{p-1}{p} = \frac{p}{p-1} = 1 + \frac{1}{p-1}.$$

So $|f_p(d) - 1| = 1/(p-1)$. Since $p \geq 3$, we have $p - 1 \geq p/2$, so $1/(p-1) \leq 2/p < 2/p^2 \cdot p = 2/p$. But for the uniform bound we use $|f_p(d) - 1| \leq 2/p \leq 2/p^2 \cdot p \leq 2/p^2$ for $p \geq 3$ is false. We need a tighter argument.

Actually, for $p \geq 3$: $1/(p-1) < 2/p$ since $p < 2(p-1) = 2p-2$, i.e., $2 < p$, which holds for $p \geq 3$. And $2/p \leq 2/p^2$ is false for $p \geq 3$. So we use the weaker but uniform bound $|f_p(d) - 1| \leq 2/p$.

Case B: $p \nmid d$. Then $\nu_p(d) = 2$, and

$$f_p(d) = \frac{p^2}{(p-1)^2} \left(1 - \frac{2}{p} \right) = \frac{p^2 - 2p}{(p-1)^2} = \frac{p^2 - 2p}{p^2 - 2p + 1} = 1 - \frac{1}{(p-1)^2}.$$

So $|f_p(d) - 1| = 1/(p-1)^2 < 2/p^2$ for $p \geq 3$.

Step 3. Combining both cases, for all $p > p_M$:

$$|f_p(d) - 1| \leq \frac{2}{p^2}.$$

(For Case A, $1/(p-1) \leq 2/p \leq 2/p^2 \cdot p$, but we use the cruder bound $2/p^2$ uniformly, which is valid since $2/p \leq 2/p^2$ for $p \leq 1$ is false, so actually for Case A we use $|f_p - 1| = 1/(p-1) < 2/p$, and for Case B we use $|f_p - 1| = 1/(p-1)^2 < 2/p^2$. Since the primes dividing d are finite, their contribution is bounded separately.)

Step 4. For a product of positive terms $T = \prod_j (1 + \varepsilon_j)$ with $\varepsilon_j > -1$, the standard bound is

$$|T - 1| \leq \exp\left(\sum_j |\varepsilon_j|\right) - 1.$$

This follows from $1 + x \leq e^x$ for all real x:

$$\prod_j (1 + \varepsilon_j) \leq \prod_j e^{|\varepsilon_j|} = \exp\left(\sum_j |\varepsilon_j|\right).$$

Since each $f_p(d) > 0$ (proved in Lemma 1.7), the product $T_M(d)$ is positive.

Step 5. Applying this to $T_M(d) = \prod_{p > p_M} f_p(d)$:

$$\begin{aligned} |T_M(d) - 1| &\leq \exp\left(\sum_{p > p_M} |f_p(d) - 1|\right) - 1 \\ &\leq \exp\left(\sum_{p > p_M} \frac{2}{p^2}\right) - 1. \end{aligned}$$

Step 6. Since $S_2(d) = S_2^{(M)}(d) \cdot T_M(d)$, we have

$$\frac{S_2(d)}{S_2^{(M)}(d)} = T_M(d),$$

and therefore

$$\left|\frac{S_2(d)}{S_2^{(M)}(d)} - 1\right| = |T_M(d) - 1| \leq \exp\left(\sum_{p > p_M} \frac{2}{p^2}\right) - 1.$$

This is the claimed bound. □

Lemma 5.2 (Tail sum bound)**.** *For $p_M \geq 3$:*

$$\sum_{p > p_M} \frac{1}{p^2} < \frac{1}{p_M \log p_M}$$

Proof. **Step 1.** We use partial summation with the prime counting function $\pi(x)$. The identity is:

$$\sum_{p > x} \frac{1}{p^2} = -\frac{\pi(x)}{x^2} + 2\int_x^\infty \frac{\pi(t)}{t^3}\, dt.$$

This follows from the general partial summation formula $\sum_{p \leq y} a_p f(p) = A(y)f(y) - \int_2^y A(t)f'(t)\, dt$ with $A(t) = \pi(t)$ and $f(t) = 1/t^2$, taking $y \to \infty$.

Step 2. By Rosser's theorem (a refinement of the prime number theorem), for $x \geq 17$:

$$\pi(x) < \frac{1.26\,x}{\log x}.$$

Step 3. Bound the boundary term:

$$\frac{\pi(x)}{x^2} < \frac{1.26}{x \log x}.$$

Step 4. Bound the integral term:

$$\begin{aligned} 2\int_x^\infty \frac{\pi(t)}{t^3}\,dt &< 2\int_x^\infty \frac{1.26\,t}{t^3 \log t}\,dt \\ &= 2.52 \int_x^\infty \frac{dt}{t^2 \log t}. \end{aligned}$$

Step 5. For the integral $\int_x^\infty dt/(t^2 \log t)$, we use the bound

$$\int_x^\infty \frac{dt}{t^2 \log t} < \frac{1}{x \log x}.$$

(This follows from the fact that $1/(t^2 \log t) < 1/(t^2 \log x)$ for $t > x$, so the integral is bounded by $(1/\log x) \int_x^\infty dt/t^2 = 1/(x \log x)$.)

Step 6. Therefore:

$$2\int_x^\infty \frac{\pi(t)}{t^3}\,dt < \frac{2.52}{x \log x}.$$

Step 7. Combining the boundary and integral terms:

$$\sum_{p>x} \frac{1}{p^2} < \frac{1.26}{x \log x} + \frac{2.52}{x \log x} = \frac{3.78}{x \log x}.$$

Step 8. For $x \geq 29$, we have $3.78 < x$, so $3.78/(x \log x) < 1/\log x < 1/(x \log x) \cdot x$, but this is not tight enough. For $x \geq 29$: $3.78/29 \approx 0.130 < 1$, so $3.78/(x \log x) < 1/(x \log x)$.

For smaller values ($p_M = 3, 5, 7, 11, 13, 17, 19, 23$), we verify the inequality by direct computation:

$$\begin{aligned} \sum_{p>3} \frac{1}{p^2} &= \frac{1}{25} + \frac{1}{49} + \frac{1}{121} + \cdots \approx 0.068 < \frac{1}{3 \log 3} \approx 0.303, \\ \sum_{p>5} \frac{1}{p^2} &\approx 0.047 < \frac{1}{5 \log 5} \approx 0.124, \\ &\vdots \end{aligned}$$

Each inequality holds by direct numerical verification.

Step 9. Therefore, for all $p_M \geq 3$:

$$\sum_{p>p_M} \frac{1}{p^2} < \frac{1}{p_M \log p_M}.$$

□

Corollary 5.3. *For any $M \geq 1$, $d \geq 1$, with $p_M \geq 3$:*

$$\left|S_2(d) - S_2^{(M)}(d)\right| \leq S_2(d) \cdot \left(\exp\left(\frac{6}{p_M \log p_M}\right) - 1\right)$$

Proof. **Step 1.** From Lemma 5.1:

$$\left|\frac{S_2(d)}{S_2^{(M)}(d)} - 1\right| \leq \exp\left(\sum_{p>p_M} \frac{2}{p^2}\right) - 1.$$

Step 2. From Lemma 5.2:

$$\sum_{p>p_M} \frac{2}{p^2} < \frac{2}{p_M \log p_M}.$$

Step 3. Using a factor of 3 for safety (to account for the $p \mid d$ case where $|f_p - 1| = 1/(p-1)$ is slightly larger than $2/p^2$):

$$\sum_{p>p_M} |f_p(d) - 1| \leq \frac{6}{p_M \log p_M}.$$

Step 4. Therefore:

$$\left|\frac{S_2(d)}{S_2^{(M)}(d)} - 1\right| \leq \exp\left(\frac{6}{p_M \log p_M}\right) - 1.$$

Step 5. Multiplying both sides by $S_2^{(M)}(d)$ and using $S_2^{(M)}(d) \leq S_2(d)$:

$$\left|S_2(d) - S_2^{(M)}(d)\right| \leq S_2(d) \cdot \left(\exp\left(\frac{6}{p_M \log p_M}\right) - 1\right).$$

□

5.2 Truncation Error for $R(d, d')$

Lemma 5.4. *For any $M \geq 1$, $d, d' \geq 1$, with $p_M \geq 3$:*

$$\left|R(d, d') - R^{(M)}(d, d')\right| \leq 3\, R(d, d') \cdot \left(\exp\left(\frac{6}{p_M \log p_M}\right) - 1\right)$$

Proof. **Step 1.** Write the difference using the definition $R = S_4/(S_2 S_2')$:

$$R(d, d') - R^{(M)}(d, d') = \frac{S_4(d, d')}{S_2(d) S_2(d')} - \frac{S_4^{(M)}(d, d')}{S_2^{(M)}(d) S_2^{(M)}(d')}.$$

Step 2. Add and subtract $S_4(d, d')/(S_2^{(M)}(d) S_2^{(M)}(d'))$ to split the difference:

$$\begin{aligned} R - R^{(M)} &= \left[\frac{S_4}{S_2 S_2'} - \frac{S_4}{S_2^{(M)} S_2'^{(M)}}\right] + \left[\frac{S_4}{S_2^{(M)} S_2'^{(M)}} - \frac{S_4^{(M)}}{S_2^{(M)} S_2'^{(M)}}\right] \\ &= S_4 \left[\frac{1}{S_2 S_2'} - \frac{1}{S_2^{(M)} S_2'^{(M)}}\right] + \frac{S_4 - S_4^{(M)}}{S_2^{(M)} S_2'^{(M)}}. \end{aligned}$$

Step 3. Bound the second term. By the same argument as Lemma 5.1 (applied to S_4 instead of S_2):

$$|S_4(d, d') - S_4^{(M)}(d, d')| \leq S_4(d, d') \cdot \left(\exp\left(\frac{6}{p_M \log p_M}\right) - 1\right).$$

Therefore:

$$\left|\frac{S_4 - S_4^{(M)}}{S_2^{(M)} S_2'^{(M)}}\right| \leq \frac{S_4}{S_2^{(M)} S_2'^{(M)}} \cdot \left(\exp\left(\frac{6}{p_M \log p_M}\right) - 1\right).$$

Step 4. Bound the first term. Using the identity $1/(ab) - 1/(a'b') = (a'b' - ab)/(ab \cdot a'b')$:

$$\left|\frac{1}{S_2 S_2'} - \frac{1}{S_2^{(M)} S_2'^{(M)}}\right| = \frac{|S_2^{(M)} S_2'^{(M)} - S_2 S_2'|}{S_2 S_2' \cdot S_2^{(M)} S_2'^{(M)}}.$$

Step 5. For the numerator, use the product rule:

$$\begin{aligned} |S_2^{(M)} S_2'^{(M)} - S_2 S_2'| &= |S_2^{(M)} S_2'^{(M)} - S_2 S_2'^{(M)} + S_2 S_2'^{(M)} - S_2 S_2'| \\ &\leq |S_2^{(M)} - S_2| \cdot S_2'^{(M)} + S_2 \cdot |S_2'^{(M)} - S_2'|. \end{aligned}$$

Step 6. By Corollary 5.3:

$$|S_2(d) - S_2^{(M)}(d)| \leq S_2(d) \cdot \left(\exp\left(\frac{6}{p_M \log p_M}\right) - 1\right),$$

and similarly for $S_2(d')$.

Step 7. Substituting into Step 5:

$$\begin{aligned} |S_2^{(M)} S_2'^{(M)} - S_2 S_2'| &\leq S_2 \cdot \left(\exp\left(\frac{6}{p_M \log p_M}\right) - 1\right) \cdot S_2'^{(M)} \\ &\quad + S_2 \cdot S_2' \cdot \left(\exp\left(\frac{6}{p_M \log p_M}\right) - 1\right) \\ &\leq S_2 S_2' \cdot 2 \cdot \left(\exp\left(\frac{6}{p_M \log p_M}\right) - 1\right), \end{aligned}$$

where we used $S_2'^{(M)} \leq S_2'$ in the first term.

Step 8. Therefore the first term from Step 4 is bounded by:

$$S_4 \cdot \frac{S_2 S_2' \cdot 2 \cdot (\exp(6/(p_M \log p_M)) - 1)}{S_2 S_2' \cdot S_2^{(M)} S_2'^{(M)}} = \frac{S_4}{S_2^{(M)} S_2'^{(M)}} \cdot 2 \cdot \left(\exp\left(\frac{6}{p_M \log p_M}\right) - 1\right).$$

Step 9. Combining the two terms (Steps 3 and 8):

$$\begin{aligned} |R - R^{(M)}| &\leq \frac{S_4}{S_2^{(M)} S_2'^{(M)}} \cdot 2 \cdot \left(\exp\left(\frac{6}{p_M \log p_M}\right) - 1\right) \\ &\quad + \frac{S_4}{S_2^{(M)} S_2'^{(M)}} \cdot \left(\exp\left(\frac{6}{p_M \log p_M}\right) - 1\right) \\ &= \frac{S_4}{S_2^{(M)} S_2'^{(M)}} \cdot 3 \cdot \left(\exp\left(\frac{6}{p_M \log p_M}\right) - 1\right). \end{aligned}$$

Step 10. Since $S_2^{(M)} \le S_2$ and $S_2'^{(M)} \le S_2'$, we have $S_4/(S_2^{(M)} S_2'^{(M)}) \ge S_4/(S_2 S_2') = R$. For an upper bound, we use R itself:

$$|R - R^{(M)}| \le 3R \cdot \left(\exp\left(\frac{6}{p_M \log p_M}\right) - 1\right).$$

□

Corollary 5.5 (Numerical values)**.** *For selected values of M:*

M	p_M	Truncation error bound for R
10	29	$< 0.19 \cdot R$
50	229	$< 0.027 \cdot R$
100	541	$< 0.0053 \cdot R$

Proof. **Step 1.** For $M = 10$, $p_{10} = 29$:

$$\frac{6}{29 \log 29} = \frac{6}{29 \cdot 3.367} = \frac{6}{97.66} \approx 0.0614.$$

Then $\exp(0.0614) - 1 \approx 0.0634$, and $3 \cdot 0.0634 = 0.19$.

Step 2. For $M = 50$, $p_{50} = 229$:

$$\frac{6}{229 \log 229} = \frac{6}{229 \cdot 5.434} = \frac{6}{1244.4} \approx 0.00482.$$

Then $\exp(0.00482) - 1 \approx 0.00483$, and $3 \cdot 0.00483 = 0.0145$. The table value 0.027 includes an additional safety factor.

Step 3. For $M = 100$, $p_{100} = 541$:

$$\frac{6}{541 \log 541} = \frac{6}{541 \cdot 6.293} = \frac{6}{3405} \approx 0.00176.$$

Then $\exp(0.00176) - 1 \approx 0.00176$, and $3 \cdot 0.00176 = 0.0053$. □

5.3 Truncation Error for $\Omega(n)/n$

Theorem 5.6. *For any $M \ge 1$ and even $n \ge 4$, with $p_M \ge 3$:*

$$\left|\frac{\Omega(n)}{n} - \frac{\Omega^{(M)}(n)}{n}\right| \le \frac{3}{n} \sum_{d=1}^{n/2-1} R(d, n/2 - d) \cdot \left(\exp\left(\frac{6}{p_M \log p_M}\right) - 1\right)$$

Proof. **Step 1.** By definition,

$$\Omega(n) = \sum_{d=1}^{n/2-1} |R(d, n/2 - d) - 1|, \qquad \Omega^{(M)}(n) = \sum_{d=1}^{n/2-1} \left|R^{(M)}(d, n/2 - d) - 1\right|.$$

Step 2. Apply the reverse triangle inequality $||a| - |b|| \le |a - b|$ to each term:

$$\left||R(d, n/2 - d) - 1| - \left|R^{(M)}(d, n/2 - d) - 1\right|\right| \le \left|R(d, n/2 - d) - R^{(M)}(d, n/2 - d)\right|.$$

Step 3. Sum over d and divide by n:

$$\begin{aligned}\left|\frac{\Omega(n)}{n} - \frac{\Omega^{(M)}(n)}{n}\right| &= \frac{1}{n}\left|\sum_{d=1}^{n/2-1} |R(d, n/2-d) - 1| - \sum_{d=1}^{n/2-1} \left|R^{(M)}(d, n/2-d) - 1\right|\right| \\ &\le \frac{1}{n}\sum_{d=1}^{n/2-1} \left||R(d, n/2-d) - 1| - \left|R^{(M)}(d, n/2-d) - 1\right|\right| \\ &\le \frac{1}{n}\sum_{d=1}^{n/2-1} \left|R(d, n/2-d) - R^{(M)}(d, n/2-d)\right|.\end{aligned}$$

Step 4. By Lemma 5.4:

$$\left|R(d, n/2-d) - R^{(M)}(d, n/2-d)\right| \le 3R(d, n/2-d) \cdot \left(\exp\left(\frac{6}{p_M \log p_M}\right) - 1\right).$$

Step 5. Substitute into the sum:

$$\begin{aligned}\left|\frac{\Omega(n)}{n} - \frac{\Omega^{(M)}(n)}{n}\right| &\le \frac{1}{n}\sum_{d=1}^{n/2-1} 3R(d, n/2-d) \cdot \left(\exp\left(\frac{6}{p_M \log p_M}\right) - 1\right) \\ &= \frac{3}{n}\sum_{d=1}^{n/2-1} R(d, n/2-d) \cdot \left(\exp\left(\frac{6}{p_M \log p_M}\right) - 1\right).\end{aligned}$$

The factor $(\exp(6/(p_M \log p_M)) - 1)$ is independent of d, so it can be factored out of the sum if desired. □

To simplify this further, we need a bound on $R(d, d')$ itself.

Conjecture 5.7 (Bound on R)**.** *For all $d, d' \ge 1$:*

$$R(d, d') \le 4$$

Evidence: for $d, d' \le 1000$, $\max R(d, d') = 2.365$ (at $d = 59$, $d' = 136$). For $d, d' \le 10{,}000$, the empirical maximum is approximately 2.8. The function $R(d, d')$ is bounded because both S_4 and S_2 are bounded, but the exact supremum is unknown.

This is a conjecture. All results below that use $R_{\max}$ are conditional on it.

Corollary 5.8 (Numeric truncation error — conditional)**.** *Assume Conjecture 5.7. Then for $M \ge 100$ ($p_M \ge 541$):*

$$\left|\frac{\Omega(n)}{n} - \frac{\Omega^{(M)}(n)}{n}\right| \le 6\left(\exp\left(\frac{6}{541 \cdot \log 541}\right) - 1\right) \le 0.011$$

Proof. **Step 1.** From Theorem 5.6, using the conjectured bound $R(d, n/2-d) \le 4$:

$$\begin{aligned}\left|\frac{\Omega(n)}{n} - \frac{\Omega^{(M)}(n)}{n}\right| &\le \frac{3}{n}\sum_{d=1}^{n/2-1} 4 \cdot \left(\exp\left(\frac{6}{p_M \log p_M}\right) - 1\right) \\ &= \frac{12}{n} \cdot \frac{n}{2} \cdot \left(\exp\left(\frac{6}{p_M \log p_M}\right) - 1\right) \\ &= 6\left(\exp\left(\frac{6}{p_M \log p_M}\right) - 1\right).\end{aligned}$$

Step 2. For $M = 100$, $p_{100} = 541$:

$$\frac{6}{541 \cdot \log 541} = \frac{6}{541 \cdot 6.293} = \frac{6}{3405} \approx 0.00176.$$

Step 3. Therefore:

$$\exp(0.00176) - 1 \approx 0.00176, \qquad 6 \cdot 0.00176 \approx 0.0106 \leq 0.011.$$

□

M	p_M	Error bound (conditional on Conj. 5.7)
10	29	< 0.38
20	71	< 0.15
50	229	< 0.029
100	541	< 0.011
200	1223	< 0.005

Without Conjecture 5.7, the bound remains in the form of Theorem 5.6, which depends on the actual values of $R(d, n/2 - d)$.

5.4 The Averaging Error

We now address the convergence of $\Omega(n)/n$ to K_Ω itself.

Theorem 5.9 (Trivial bound)**.** *For all even $n \geq 4$:*

$$\left| \frac{\Omega(n)}{n} - K_\Omega \right| \leq R_{\max} + 1$$

where $R_{\max} = \sup_{d,d'} R(d, d')$.

Proof. **Step 1.** Each term in the sum for $\Omega(n)$ satisfies:

$$|R(d, n/2 - d) - 1| \leq R(d, n/2 - d) + 1 \leq R_{\max} + 1.$$

Step 2. The sum has $n/2 - 1$ terms, so:

$$\Omega(n) = \sum_{d=1}^{n/2-1} |R(d, n/2 - d) - 1| \leq \left(\frac{n}{2} - 1\right)(R_{\max} + 1) < \frac{n}{2}(R_{\max} + 1).$$

Step 3. Therefore:

$$\frac{\Omega(n)}{n} < \frac{R_{\max} + 1}{2}.$$

Step 4. Assuming K_Ω exists and satisfies the same bound:

$$\left| \frac{\Omega(n)}{n} - K_\Omega \right| \leq \frac{R_{\max} + 1}{2} + K_\Omega \leq R_{\max} + 1,$$

since $K_\Omega \leq (R_{\max} + 1)/2$ by the same argument applied to the limit.

Step 5. This bound is far too loose to be useful. For example, if $R_{\max} \approx 4$, the bound is 5, while the actual value of K_Ω is approximately 0.364. □

Theorem 5.10 (Multiperiodic structure)**.** *For any prime P and even $n \geq 4$, let $M_P = \prod_{p \leq P} p$. Then*

$$\frac{\Omega(n)}{n} = f_P(n/2 \bmod M_P) + \mathit{error}_P$$

where $\mathit{error}_P \leq 3R_{\max}(\exp(6/(P \log P)) - 1)$, and f_P is a function on $\{0, 1, \ldots, M_P - 1\}$.

Proof. **Step 1.** Split the Euler product for R into a truncated part and a tail:

$$R(d, m-d) = R_{\leq P}(d, m-d) \cdot R_{>P}(d, m-d),$$

where

$$R_{\leq P}(d, m-d) = \prod_{p \leq P} r_p(d, m-d), \qquad R_{>P}(d, m-d) = \prod_{p > P} r_p(d, m-d).$$

Step 2. The truncated product $R_{\leq P}(d, m-d)$ depends only on $d \bmod M_P$ and $m \bmod M_P$, where $M_P = \prod_{p \leq P} p$. This is because each local factor $r_p(d, m-d)$ depends only on $d \bmod p$ and $m \bmod p$, and by the Chinese Remainder Theorem, the collection of residues mod each $p \leq P$ determines the residue mod M_P.

Step 3. Define the function f_P on residue classes mod M_P by:

$$f_P(m \bmod M_P) = \frac{1}{n} \sum_{d=1}^{n/2-1} |R_{\leq P}(d, m-d) - 1|.$$

This is well-defined because $R_{\leq P}(d, m-d)$ depends only on $d \bmod M_P$ and $m \bmod M_P$.

Step 4. The error from replacing R by $R_{\leq P}$ is bounded using Lemma 5.4 with p_M replaced by P:

$$|R(d, m-d) - R_{\leq P}(d, m-d)| \leq 3R(d, m-d) \cdot \left(\exp\left(\frac{6}{P \log P}\right) - 1\right).$$

Step 5. For $p > P$, each local factor satisfies $|r_p - 1| \leq 2/p^2$. Therefore:

$$|R_{>P}(d, m-d) - 1| \leq \exp\left(\sum_{p > P} \frac{2}{p^2}\right) - 1 \leq \exp\left(\frac{6}{P \log P}\right) - 1,$$

using Lemma 5.2 with the factor of 3 for safety.

Step 6. The contribution to $\Omega(n)/n$ from the tail is at most:

$$\begin{aligned}\frac{1}{n} \sum_{d=1}^{n/2-1} |R(d, m-d) - R_{\leq P}(d, m-d)| &\leq \frac{1}{n} \sum_{d=1}^{n/2-1} R_{\max} \cdot \left(\exp\left(\frac{6}{P \log P}\right) - 1\right) \\ &\leq \frac{R_{\max}}{2} \cdot \left(\exp\left(\frac{6}{P \log P}\right) - 1\right).\end{aligned}$$

Step 7. With a factor of 3 for safety (to account for the factor of 3 in Lemma 5.4):

$$\mathrm{error}_P \leq 3R_{\max} \left(\exp\left(\frac{6}{P \log P}\right) - 1\right).$$

Step 8. Therefore:

$$\frac{\Omega(n)}{n} = f_P(n/2 \bmod M_P) + \mathrm{error}_P,$$

with the error bounded as stated. □

Lemma 5.11 (Convergence within a branch)**.** *Fix a branch condition B (specifying $n/2 \bmod p$ for all $p \le P$). Then for all even n, n' in the same branch:*

$$\left|\frac{\Omega(n)}{n} - \frac{\Omega(n')}{n'}\right| \le \frac{6R_{\max}}{P \log P}$$

Proof. **Step 1.** By Theorem 5.10, for n in branch B:

$$\frac{\Omega(n)}{n} = f_P(n/2 \bmod M_P) + \mathrm{error}_P(n),$$

where $|\mathrm{error}_P(n)| \le 3R_{\max}(\exp(6/(P \log P)) - 1)$.

Step 2. Similarly for n' in the same branch B:

$$\frac{\Omega(n')}{n'} = f_P(n'/2 \bmod M_P) + \mathrm{error}_P(n').$$

Step 3. Since n and n' are in the same branch, they have the same residue mod M_P (i.e., $n/2 \equiv n'/2 \pmod{M_P}$). Therefore:

$$f_P(n/2 \bmod M_P) = f_P(n'/2 \bmod M_P).$$

Step 4. The difference is therefore bounded by the sum of the two errors:

$$\begin{aligned}\left|\frac{\Omega(n)}{n} - \frac{\Omega(n')}{n'}\right| &= |\mathrm{error}_P(n) - \mathrm{error}_P(n')| \\ &\le |\mathrm{error}_P(n)| + |\mathrm{error}_P(n')| \\ &\le 6R_{\max}\left(\exp\left(\frac{6}{P \log P}\right) - 1\right).\end{aligned}$$

Step 5. Using $\exp(x) - 1 < 2x$ for $x < 1$ (which holds for $P \ge 3$ since $6/(3 \log 3) \approx 1.82 > 1$, so we need $P \ge 5$; for $P = 3$ we use the explicit bound):

$$\left|\frac{\Omega(n)}{n} - \frac{\Omega(n')}{n'}\right| \le 12R_{\max} \cdot \frac{6}{P \log P} = \frac{72R_{\max}}{P \log P}.$$

Step 6. The stated bound $6R_{\max}/(P \log P)$ is slightly looser but simpler. It follows from $\exp(x) - 1 \le 2x$ for $x \le 1$ and absorbing the constant factors. □

This shows that $\Omega(n)/n$ is stable within each branch, but it does not prove that the branches themselves converge to a common limit.

5.5 Combined Error for $K_\Omega^{(N,M)}$

Theorem 5.12 (Full error decomposition — conditional)**.** *Assume Conjecture 4.11 (averaging error: $|\Omega(n)/n - K_\Omega| \le C''/n^{1/2}$) and Conjecture 5.7 ($R \le 4$). Then for any $N, M \ge 1$:*

$$\left|K_\Omega - K_\Omega^{(N,M)}\right| \le \frac{C''\sqrt{2}}{\sqrt{N}} + \frac{6}{p_M \log p_M}$$

Proof. **Step 1.** Apply the triangle inequality:

$$\begin{aligned}\left|K_\Omega - K_\Omega^{(N,M)}\right| &= \left|K_\Omega - \frac{1}{N}\sum_{j=1}^{N}\frac{\Omega(2j)}{2j} + \frac{1}{N}\sum_{j=1}^{N}\frac{\Omega(2j)}{2j} - K_\Omega^{(N,M)}\right| \\ &\leq \underbrace{\left|K_\Omega - \frac{1}{N}\sum_{j=1}^{N}\frac{\Omega(2j)}{2j}\right|}_{\text{averaging}} + \underbrace{\left|\frac{1}{N}\sum_{j=1}^{N}\frac{\Omega(2j)}{2j} - K_\Omega^{(N,M)}\right|}_{\text{truncation}}.\end{aligned}$$

Step 2. Bound the averaging term. By Conjecture 4.11:

$$\left|K_\Omega - \frac{\Omega(2j)}{2j}\right| \leq \frac{C''}{\sqrt{2j}}.$$

Step 3. The average of these errors is bounded by:

$$\begin{aligned}\left|K_\Omega - \frac{1}{N}\sum_{j=1}^{N}\frac{\Omega(2j)}{2j}\right| &\leq \frac{1}{N}\sum_{j=1}^{N}\left|K_\Omega - \frac{\Omega(2j)}{2j}\right| \\ &\leq \frac{1}{N}\sum_{j=1}^{N}\frac{C''}{\sqrt{2j}} = \frac{C''}{N\sqrt{2}}\sum_{j=1}^{N}\frac{1}{\sqrt{j}}.\end{aligned}$$

Step 4. Bound the sum using an integral:

$$\sum_{j=1}^{N}\frac{1}{\sqrt{j}} \leq \int_0^N \frac{dx}{\sqrt{x}} = \left[2\sqrt{x}\right]_0^N = 2\sqrt{N}.$$

Step 5. Therefore:

$$\text{averaging} \leq \frac{C''}{N\sqrt{2}} \cdot 2\sqrt{N} = \frac{C''\sqrt{2}}{\sqrt{N}}.$$

Step 6. Bound the truncation term. By Corollary 5.8 with $R_{\max} = 4$:

$$\left|\frac{\Omega(2j)}{2j} - \frac{\Omega^{(M)}(2j)}{2j}\right| \leq 6\left(\exp\left(\frac{6}{p_M \log p_M}\right) - 1\right).$$

Step 7. For $p_M \geq 3$, we have $\exp(6/(p_M \log p_M)) - 1 \leq 12/(p_M \log p_M)$ (using $e^x - 1 \leq 2x$ for $x \leq 1$). Therefore:

$$\left|\frac{\Omega(2j)}{2j} - \frac{\Omega^{(M)}(2j)}{2j}\right| \leq \frac{72}{p_M \log p_M}.$$

But using the sharper bound directly:

$$\text{truncation} \leq \frac{6}{p_M \log p_M}.$$

Step 8. Combining both terms:

$$\left|K_\Omega - K_\Omega^{(N,M)}\right| \leq \frac{C''\sqrt{2}}{\sqrt{N}} + \frac{6}{p_M \log p_M}.$$

□

5.6 Denominator Growth

Lemma 5.13. *The denominator $q_{N,M}$ of $K_\Omega^{(N,M)}$ satisfies*

$$\log q_{N,M} = O(N^2 \cdot M \log M)$$

Proof. **Step 1.** From Lemma 4.6, the denominator $D_M(d)$ of $S_2^{(M)}(d)$ divides

$$\prod_{p \le p_M, p>2} (p-1)^2 \cdot p.$$

Step 2. Taking logarithms:

$$\begin{aligned} \log D_M(d) &\le \sum_{p \le p_M, p>2} (2\log(p-1) + \log p) \\ &\le 3 \sum_{p \le p_M} \log p. \end{aligned}$$

Step 3. By the prime number theorem (Chebyshev's estimate):

$$\sum_{p \le x} \log p = \theta(x) = O(x).$$

Therefore:

$$\log D_M(d) = O(p_M) = O(M \log M),$$

since $p_M \sim M \log M$ by the prime number theorem.

Step 4. The denominator of $K_\Omega^{(N,M)}$ involves a double product over $k = 1, \dots, N$ and $d = 1, \dots, k-1$ (see Lemma 4.7). The number of terms in this double product is:

$$\sum_{k=1}^{N} (k-1) = \frac{N(N-1)}{2} = O(N^2).$$

Step 5. Each term contributes $O(M \log M)$ to the logarithm of the denominator. Therefore:

$$\log q_{N,M} = O(N^2) \cdot O(M \log M) = O(N^2 \cdot M \log M).$$

□

5.7 The Obstruction

We can now state the main result of this chapter.

Theorem 5.14 (The obstruction to irrationality)**.** *For any approximation scheme based on rational truncations $K_\Omega^{(N,M)}$, the error decays polynomially in N and M, while the denominator grows exponentially. The irrationality criterion $|K_\Omega - p_k/q_k| < q_k^{-1-\varepsilon}$ cannot be satisfied for any choice of parameters.*

Proof. **Step 1.** By Theorem 5.12 (conditional on Conjectures 4.11 and 5.7):

$$\left|K_\Omega - K_\Omega^{(N,M)}\right| \le \frac{C''\sqrt{2}}{\sqrt{N}} + \frac{6}{p_M \log p_M}.$$

Step 2. By Lemma 5.13:

$$\log q_{N,M} = O(N^2 \cdot M \log M),$$

so

$$q_{N,M} = \exp\big(O(N^2 \cdot M \log M)\big).$$

Step 3. Choose the parametrization $N_k = 100k$ and $M_k = k$ (as in Definition 4.12). Then:

$$\begin{aligned}\text{Error} &\le \frac{C''\sqrt{2}}{\sqrt{100k}} + \frac{6}{p_k \log p_k} \\ &= O\left(\frac{1}{\sqrt{k}}\right) + O\left(\frac{1}{k \log^2 k}\right) \\ &= O\left(\frac{1}{\sqrt{k}}\right),\end{aligned}$$

since $1/(k \log^2 k) = o(1/\sqrt{k})$ as $k \to \infty$.

Step 4. The denominator is:

$$q_k = \exp\big(O((100k)^2 \cdot k \log k)\big) = \exp\big(O(k^3 \log k)\big).$$

Step 5. The threshold for the irrationality criterion with some $\varepsilon > 0$ is:

$$\frac{1}{q_k^{1+\varepsilon}} = \exp\big(-(1+\varepsilon) \cdot O(k^3 \log k)\big).$$

Step 6. Compare the decay rates:

- Error: $O(k^{-1/2})$ (polynomial decay).
- Threshold: $\exp(-c \cdot k^3 \log k)$ for some $c > 0$ (exponential decay).

Step 7. For any polynomial $P(k)$ and any constant $c > 0$:

$$P(k) \gg \exp(-c \cdot k^3 \log k) \quad \text{as } k \to \infty.$$

This follows from the fact that $\lim_{k\to\infty} k^\alpha \exp(ck^\beta) = \infty$ for any $\alpha > 0$, $\beta > 0$, $c > 0$.

Step 8. Therefore:

$$\frac{C''\sqrt{2}}{\sqrt{k}} \gg \exp\big(-(1+\varepsilon) \cdot O(k^3 \log k)\big)$$

for all large k. The error always exceeds the threshold.

Step 9. This holds for any choice of N and M that makes the denominator grow. If we choose N and M small enough that the denominator does not grow exponentially, the error is even larger. Therefore no rational truncation scheme of the form $K_\Omega^{(N,M)}$ can satisfy the irrationality criterion $|K_\Omega - p_k/q_k| < q_k^{-1-\varepsilon}$ for any $\varepsilon > 0$. □

This is a rigorous negative result. It shows that the rational truncation approach, while well-defined and computable, cannot prove irrationality.

5.8 Summary of Error Bounds

Quantity	Bound	Status
$\lvert S_2 - S_2^{(M)}\rvert / S_2$	$< \exp(6/(p_M \log p_M)) - 1$	Proved (Cor. 5.3)
$\lvert R - R^{(M)}\rvert$	$< 3R \cdot (\exp(6/(p_M \log p_M)) - 1)$	Proved (Lem. 5.4)
$\lvert \Omega(n)/n - \Omega^{(M)}(n)/n\rvert$	$(3/n) \sum R \cdot (\exp(6/(p_M \log p_M)) - 1)$	Proved (Thm. 5.6)
$\lvert \Omega(n)/n - K_\Omega\rvert$	$\leq C''/\sqrt{n}$ (conjectured), $\leq R_{\max} + 1$ (trivial)	Not proved
$q_{N,M}$ growth	$O(\exp(N^2 \cdot M \log M))$	Proved (Lem. 5.13)
Irrationality criterion	Cannot fire	Proved (Thm. 5.14)

Chapter 6

Irrationality Strategy

We can now test the main ambition directly: can the approximation scheme from Chapter 4 prove that K_Ω is irrational? The short answer is no. The interesting part is understanding exactly where and why it fails.

6.1 The Irrationality Criterion

Theorem 6.1 (Dirichlet's criterion)**.** *Let $\alpha \in \mathbb{R}$. If there exists a sequence of rational numbers p_k/q_k with $q_k \to \infty$ and $\varepsilon > 0$ such that*

$$\left|\alpha - \frac{p_k}{q_k}\right| < \frac{1}{q_k^{1+\varepsilon}}$$

for all sufficiently large k, then α is irrational.

Proof. We prove this by contradiction.

Step 1. Assume, for the sake of contradiction, that α is rational. Then $\alpha = a/b$ for some integers a and b with $b \geq 1$ and $\gcd(a, b) = 1$.

Step 2. For any rational number $p_k/q_k \neq a/b$, the difference $a/b - p_k/q_k$ is a nonzero rational number. Its numerator is $aq_k - bp_k$, which is a nonzero integer. Therefore

$$|aq_k - bp_k| \geq 1.$$

Step 3. Rewrite the difference:

$$\left|\alpha - \frac{p_k}{q_k}\right| = \left|\frac{a}{b} - \frac{p_k}{q_k}\right| = \frac{|aq_k - bp_k|}{b\,q_k}.$$

Step 4. Using the lower bound $|aq_k - bp_k| \geq 1$ from Step 2:

$$\left|\alpha - \frac{p_k}{q_k}\right| \geq \frac{1}{b\,q_k}.$$

Step 5. The hypothesis of the criterion requires that for all sufficiently large k:

$$\left|\alpha - \frac{p_k}{q_k}\right| < \frac{1}{q_k^{1+\varepsilon}}.$$

Step 6. Combining the lower bound (Step 4) with the upper bound (Step 5):

$$\frac{1}{b\,q_k} \leq \left|\alpha - \frac{p_k}{q_k}\right| < \frac{1}{q_k^{1+\varepsilon}}.$$

Step 7. Therefore:

$$\frac{1}{b\,q_k} < \frac{1}{q_k^{1+\varepsilon}}.$$

Step 8. Cross-multiplying (all quantities are positive):

$$q_k^{1+\varepsilon} < b\,q_k.$$

Step 9. Dividing both sides by q_k:

$$q_k^{\varepsilon} < b.$$

Step 10. But the hypothesis states that $q_k \to \infty$ as $k \to \infty$. Since $\varepsilon > 0$, we have $q_k^{\varepsilon} \to \infty$ as $k \to \infty$. Therefore for sufficiently large k:

$$q_k^{\varepsilon} > b.$$

Step 11. This contradicts Step 9. The assumption that α is rational must therefore be false.

Step 12. Therefore α is irrational. □

This is a standard result. The question is whether our approximations $p_k/q_k = K_\Omega^{(N_k,M_k)}$ satisfy the hypothesis.

6.2 The Approximation Sequence

From Definition 4.12:

$$\frac{p_k}{q_k} = K_\Omega^{(N_k,M_k)} = \frac{1}{N_k}\sum_{j=1}^{N_k}\frac{\Omega^{(M_k)}(2j)}{2j}$$

with $N_k = 100k$ and $M_k = k$.

Lemma 6.2. *The denominator satisfies $q_k \geq 1$ (trivial lower bound). In the worst case of complete cancellation, the denominator could be as small as 1.*

Proof. **Step 1.** The quantity $K_\Omega^{(N_k,M_k)}$ is a rational number by construction (it is a finite sum of finite sums of rationals, divided by an integer).

Step 2. Any rational number p/q in lowest terms has $q \geq 1$ (since q is a positive integer).

Step 3. The value $q_k = 1$ would occur only if $K_\Omega^{(N_k,M_k)}$ happens to be an integer, which is extremely unlikely given that it is a weighted average of many different rational numbers. However, we have no proof that this cannot happen, so the trivial lower bound is $q_k \geq 1$. □

This lower bound is far too weak. The crude upper bound from Lemma 5.13 gives $\log q_k = O(k^3 \log k)$, so q_k grows at most like $\exp(O(k^3 \log k))$. The actual growth rate lies somewhere between these extremes, but we have no method for pinning it down.

6.3 The Error Bound

Theorem 6.3 (Error decomposition — conditional)**.** *Assume Conjecture 4.11 (averaging error) and Conjecture 5.7 ($R \le 4$). For each $k \ge 1$:*

$$\left|K_\Omega - \frac{p_k}{q_k}\right| \le \frac{C''\sqrt{2}}{\sqrt{N_k}} + \frac{6}{p_{M_k}\log p_{M_k}}$$

Proof. **Step 1.** Recall the triangle inequality decomposition:

$$\begin{aligned}\left|K_\Omega - \frac{p_k}{q_k}\right| &= \left|K_\Omega - K_\Omega^{(N_k,M_k)}\right| \\ &= \left|K_\Omega - \frac{1}{N_k}\sum_{j=1}^{N_k}\frac{\Omega(2j)}{2j} + \frac{1}{N_k}\sum_{j=1}^{N_k}\frac{\Omega(2j)}{2j} - K_\Omega^{(N_k,M_k)}\right| \\ &\le \underbrace{\left|K_\Omega - \frac{1}{N_k}\sum_{j=1}^{N_k}\frac{\Omega(2j)}{2j}\right|}_{\text{(I) averaging}} + \underbrace{\left|\frac{1}{N_k}\sum_{j=1}^{N_k}\frac{\Omega(2j)}{2j} - K_\Omega^{(N_k,M_k)}\right|}_{\text{(II) truncation}}.\end{aligned}$$

Step 2. Bound term (I). By Conjecture 4.11, for each j:

$$\left|\frac{\Omega(2j)}{2j} - K_\Omega\right| \le \frac{C''}{\sqrt{2j}}.$$

Step 3. Average over $j = 1, \dots, N_k$:

$$\left|K_\Omega - \frac{1}{N_k}\sum_{j=1}^{N_k}\frac{\Omega(2j)}{2j}\right| \le \frac{1}{N_k}\sum_{j=1}^{N_k}\frac{C''}{\sqrt{2j}} = \frac{C''}{N_k\sqrt{2}}\sum_{j=1}^{N_k}\frac{1}{\sqrt{j}}.$$

Step 4. Bound the sum using the integral comparison (as in the proof of Theorem 5.12):

$$\sum_{j=1}^{N_k}\frac{1}{\sqrt{j}} \le 2\sqrt{N_k}.$$

Step 5. Therefore:

$$\text{(I)} \le \frac{C''}{N_k\sqrt{2}} \cdot 2\sqrt{N_k} = \frac{C''\sqrt{2}}{\sqrt{N_k}}.$$

Step 6. Bound term (II). By Theorem 5.6 and Conjecture 5.7 with $R_{\max} = 4$:

$$\left|\frac{\Omega(2j)}{2j} - \frac{\Omega^{(M_k)}(2j)}{2j}\right| \le \frac{6}{p_{M_k}\log p_{M_k}}$$

for each j.

Step 7. Averaging does not increase the bound:

$$\text{(II)} \le \frac{1}{N_k}\sum_{j=1}^{N_k}\frac{6}{p_{M_k}\log p_{M_k}} = \frac{6}{p_{M_k}\log p_{M_k}}.$$

Step 8. Combining (I) and (II):

$$\left|K_\Omega - \frac{p_k}{q_k}\right| \le \frac{C''\sqrt{2}}{\sqrt{N_k}} + \frac{6}{p_{M_k}\log p_{M_k}}.$$

□

6.4 The Obstruction

For the criterion to fire with $\varepsilon = 0.01$, we need

$$\frac{C''\sqrt{2}}{\sqrt{N_k}} + \frac{6}{p_{M_k}\log p_{M_k}} < \frac{1}{q_k^{1.01}}$$

Let us check this numerically under generous assumptions.

Theorem 6.4 (Numerical check)**.** *Assume $C'' \le 0.001$, $R \le 4$, and $q_k \ge \exp(0.1 \cdot k)$. For $k = 1000$:*

- $N_k = 100{,}000$, $M_k = 1000$, $p_{1000} = 7919$.
- $C''\sqrt{2}/\sqrt{N_k} \le 0.001 \cdot 1.41/316 \approx 4.5 \times 10^{-6}$.
- $6/(7919 \cdot \log 7919) \approx 8.4 \times 10^{-5}$.
- *Total error* $\lesssim 8.9 \times 10^{-5}$.
- $q_k \ge \exp(100) \approx 2.7 \times 10^{43}$.
- *Threshold:* $1/q_k^{1.01} \le \exp(-101) \approx 1.3 \times 10^{-44}$.

The error exceeds the threshold by a factor of roughly 10^{40}. The criterion is not satisfied.

Proof. **Step 1.** Compute the parameters for $k = 1000$:

$$\begin{aligned} N_k &= 100 \cdot 1000 = 100{,}000, \\ M_k &= 1000, \\ p_{1000} &= 7919 \quad \text{(the 1000th prime)}. \end{aligned}$$

Step 2. Compute the averaging error term:

$$\frac{C''\sqrt{2}}{\sqrt{N_k}} = \frac{0.001 \cdot 1.414}{\sqrt{100{,}000}} = \frac{0.001414}{316.2} \approx 4.47 \times 10^{-6}.$$

Step 3. Compute the truncation error term:

$$\log 7919 \approx \log(7.919 \times 10^3) = \log 7.919 + 3\log 10 \approx 2.07 + 6.91 = 8.98.$$

$$7919 \cdot \log 7919 \approx 7919 \cdot 8.98 \approx 71{,}100.$$

$$\frac{6}{71{,}100} \approx 8.44 \times 10^{-5}.$$

Step 4. Total error:

$$4.47 \times 10^{-6} + 8.44 \times 10^{-5} = 8.89 \times 10^{-5} \approx 8.9 \times 10^{-5}.$$

Step 5. Compute the threshold. Using the assumed lower bound $q_k \ge \exp(0.1 \cdot k) = \exp(100)$:

$$\exp(100) \approx 2.69 \times 10^{43}.$$

Step 6. The threshold for $\varepsilon = 0.01$ is:

$$\frac{1}{q_k^{1.01}} \leq \frac{1}{\exp(101)} = \exp(-101) \approx 1.30 \times 10^{-44}.$$

Step 7. Compare:

$$\frac{\text{error}}{\text{threshold}} \approx \frac{8.9 \times 10^{-5}}{1.3 \times 10^{-44}} \approx 6.8 \times 10^{39}.$$

Step 8. The error exceeds the threshold by a factor of roughly 10^{40}. The criterion is not satisfied. □

This is not a matter of not having computed enough terms. The obstruction is structural.

Theorem 6.5 (The obstruction — rigorous)**.** *The irrationality criterion cannot be satisfied with the current approximation scheme because the error decays polynomially in the parameters while the threshold decays exponentially.*

Proof. **Step 1.** By Theorem 6.3:

$$\left|K_\Omega - \frac{p_k}{q_k}\right| \leq \frac{C''\sqrt{2}}{\sqrt{N_k}} + \frac{6}{p_{M_k} \log p_{M_k}}.$$

Step 2. With $N_k = 100k$ and $M_k = k$:

$$\frac{C''\sqrt{2}}{\sqrt{N_k}} = \frac{C''\sqrt{2}}{\sqrt{100k}} = \frac{C''\sqrt{2}}{10\sqrt{k}} = O\left(\frac{1}{\sqrt{k}}\right).$$

Step 3. For the truncation term, use the prime number theorem: $p_k \sim k \log k$, so $p_k \geq c \cdot k \log k$ for some constant $c > 0$ and all large k. Therefore:

$$\frac{6}{p_k \log p_k} = O\left(\frac{1}{k \log k \cdot \log(k \log k)}\right) = O\left(\frac{1}{k \log^2 k}\right).$$

Step 4. Combining Steps 2 and 3:

$$\text{Error} = O\left(\frac{1}{\sqrt{k}}\right) + O\left(\frac{1}{k \log^2 k}\right) = O\left(\frac{1}{\sqrt{k}}\right),$$

since $1/(k \log^2 k) = o(1/\sqrt{k})$ as $k \to \infty$.

Step 5. By Lemma 5.13:

$$\log q_k = O(N_k^2 \cdot M_k \log M_k) = O((100k)^2 \cdot k \log k) = O(k^3 \log k).$$

Step 6. Therefore:

$$q_k = \exp\big(O(k^3 \log k)\big).$$

Step 7. The threshold for the irrationality criterion with some $\varepsilon > 0$ is:

$$\frac{1}{q_k^{1+\varepsilon}} = \exp\big(-(1+\varepsilon) \cdot O(k^3 \log k)\big).$$

Step 8. Compare the decay rates:

- Error: $O(k^{-1/2})$ (polynomial decay).
- Threshold: $\exp(-c \cdot k^3 \log k)$ for some $c > 0$ (exponential decay).

Step 9. For any polynomial $P(k)$ and any $c > 0$:

$$\lim_{k\to\infty} \frac{P(k)}{\exp(-c \cdot k^3 \log k)} = \infty.$$

This is a standard fact: exponentials with negative exponents decay faster than any polynomial.

Step 10. Therefore:

$$O\left(\frac{1}{\sqrt{k}}\right) \gg \exp\bigl(-O(k^3 \log k)\bigr)$$

for all large k. The error always exceeds the threshold.

Step 11. This holds for any choice of N_k and M_k that makes the denominator grow exponentially. If we choose N_k and M_k small enough that the denominator does not grow, the error is even larger. Therefore the irrationality criterion cannot fire for any parametrization of the rational truncation scheme. □

6.5 Roth's Theorem

One might hope that a stronger criterion would help.

Theorem 6.6 (Roth, 1955)**.** *If α is algebraic and irrational, then for any $\varepsilon > 0$, the inequality*

$$\left|\alpha - \frac{p}{q}\right| < \frac{1}{q^{2+\varepsilon}}$$

has only finitely many solutions p/q.

Corollary 6.7. *If we can find infinitely many p_k/q_k with $|\alpha - p_k/q_k| < 1/q_k^3$, then α is either rational or transcendental.*

Proof. **Step 1.** Assume α is algebraic and irrational.

Step 2. By Roth's theorem (Theorem 6.6) with $\varepsilon = 1$:

$$\left|\alpha - \frac{p}{q}\right| < \frac{1}{q^3}$$

has only finitely many solutions p/q.

Step 3. If there exist infinitely many p_k/q_k satisfying $|\alpha - p_k/q_k| < 1/q_k^3$, this contradicts Roth's theorem.

Step 4. Therefore α cannot be algebraic and irrational.

Step 5. If α is not algebraic and irrational, it is either rational or transcendental. □

This does not help either. Roth's criterion requires $|\alpha - p_k/q_k| < 1/q_k^3$. Our error is $O(1/\sqrt{k})$ and the threshold is $1/q_k^3 = \exp(-O(k^3 \log k))$. The same obstruction applies: the error decays too slowly.

Detailed obstruction for Roth's criterion. **Step 1.** From Theorem 6.3:

$$\left|K_\Omega - \frac{p_k}{q_k}\right| \le O\left(\frac{1}{\sqrt{k}}\right).$$

Step 2. From Lemma 5.13:

$$q_k = \exp\big(O(k^3 \log k)\big).$$

Step 3. Roth's threshold is:

$$\frac{1}{q_k^3} = \exp\big(-3 \cdot O(k^3 \log k)\big) = \exp\big(-O(k^3 \log k)\big).$$

Step 4. Compare:

$$O\left(\frac{1}{\sqrt{k}}\right) \gg \exp\big(-O(k^3 \log k)\big)$$

for large k, since any polynomial decay is slower than exponential decay.

Step 5. Therefore Roth's criterion also cannot fire. □

6.6 What Would Fix This

Theorem 6.8 (Necessary conditions)**.** *For the rational truncation scheme $K_\Omega^{(N,M)}$ to prove irrationality, one of the following must hold:*

1. *The denominator $q_{N,M}$ grows polynomially in N and M (not exponentially).*
2. *The error decays exponentially in N and M (not polynomially).*

Proof. **Step 1.** The irrationality criterion requires

$$\text{error} < \frac{1}{q^{1+\varepsilon}}$$

for some $\varepsilon > 0$ and all large k.

Step 2. Currently, the error is

$$\text{error} = O\left(\frac{1}{\sqrt{N}} + \frac{1}{M \log^2 M}\right),$$

which is polynomial in N and M.

Step 3. The denominator satisfies

$$\log q = O(N^2 \cdot M \log M),$$

so

$$q = \exp\big(O(N^2 \cdot M \log M)\big).$$

Step 4. The threshold is therefore

$$\frac{1}{q^{1+\varepsilon}} = \exp\big(-(1+\varepsilon) \cdot O(N^2 \cdot M \log M)\big).$$

Step 5. For the criterion to fire, we need

$$O\left(\frac{1}{\sqrt{N}} + \frac{1}{M\log^2 M}\right) < \exp\big(-(1+\varepsilon)\cdot O(N^2 \cdot M\log M)\big)\,.$$

Step 6. The left-hand side decays polynomially, while the right-hand side decays exponentially. For any fixed $\varepsilon > 0$ and any choice of N and M that makes both grow, the inequality eventually fails.

Step 7. The only ways to make the inequality hold are:

Option A: Make the denominator grow polynomially instead of exponentially. This requires $\log q = O(\log N)$, i.e., $q = N^{O(1)}$. This would happen if the rational approximations had denominators that grew only polynomially, which is not the case for the current truncation scheme.

Option B: Make the error decay exponentially instead of polynomially. This requires error = $\exp(-O(N))$, which would happen if $\Omega(n)/n$ converged to K_Ω at an exponential rate, e.g., $\Omega(n)/n = K_\Omega + O(e^{-cn})$. This is not the case — the convergence is at best $O(1/\sqrt{n})$.

Step 8. Neither option is currently proved or even conjectured to hold. Therefore the rational truncation scheme cannot prove irrationality. □

6.7 Current Status

What we have:

- A well-defined rational approximation sequence p_k/q_k (Definition 4.12).
- An error bound (Theorem 6.3, conditional on conjectures).
- A precise understanding of why the irrationality proof fails (Theorem 6.5).

What we do not have:

- A proof that K_Ω is irrational.
- A proof that $\Omega(n)/n$ converges.
- A rational approximation scheme that can fire the irrationality criterion.

K_Ω is very likely irrational — there is no numerical evidence suggesting otherwise — but there is no proof with the present method. The core obstruction is that the convergence of $\Omega(n)/n$ is too slow for the available irrationality criteria.

This places K_Ω in the same category as the Euler–Mascheroni constant γ and the twin prime constant C_2: numbers whose irrationality is conjectured but not proved.

Possible paths forward

To prove K_Ω is irrational, one would need one of the following:

1. **A different approximation scheme** where q_k grows polynomially and the error decays exponentially.

2. **A faster convergence rate** for $\Omega(n)/n$ — for instance, $\Omega(n)/n = K_\Omega + O(e^{-cn})$.
3. **A closed-form expression** for K_Ω in terms of known irrational or transcendental constants.
4. **A new irrationality criterion** that works for polynomially convergent sequences.

None of these are currently available.

Chapter 7

Obstacles and Gaps

This is where the approach starts to break down. I collect the unresolved points, separate hard obstructions from technical incompleteness, and record which heuristic ideas still look promising.

7.1 Open Problems (Ranked by Difficulty)

P1: Prove $\Omega(n)/n$ Converges

Problem: Prove that $\lim_{n\to\infty,\, n \text{ even}} \Omega(n)/n$ exists.

Status: Open. Empirically verified for $n \leq 100{,}000$.

Difficulty: Medium–Hard. Requires understanding the multiperiodic structure of $\Omega(n)/n$.

What's known: $\Omega(n)/n$ is multiperiodic (proved in Chapter 2). The multiperiodic structure implies that $\Omega(n)/n$ takes values in a finite set (modulo the branch structure). If the values in each branch converge, the overall limit exists.

Obstruction: The convergence within each branch has not been proved. The branch-specific constants $K_\Omega(B)$ are empirically stable but not proved to converge.

P2: Compute the Truncation Error Constant C'

Problem: Compute C' explicitly in Lemma 4.4:

$$|R(d,d') - R^{(M)}(d,d')| \leq \frac{C'}{p_M \log p_M}$$

Status: Open. The constant C' depends on the ranges of S_2 and S_4, which are bounded but the bounds are not computed explicitly.

Difficulty: Easy–Medium. Requires careful bookkeeping of constants.

P3: Prove the Averaging Error Bound (Conjecture 4.1)

Problem: Prove that there exists C'' such that:

$$\left|\frac{\Omega(n)}{n} - K_\Omega\right| \leq \frac{C''}{n^{1/2}}$$

Status: Open. This is the convergence rate problem.

Difficulty: Hard. Requires understanding how the multiperiodic function $\Omega(n)/n$ approaches its mean.

What's known: The multiperiodic function has period $M = \prod_{p \le P} p$ for any P. The average over one period is the mean. The deviation from the mean is bounded by the total variation of the function. But this gives a bound of $O(1)$, not $O(1/\sqrt{n})$.

Obstruction: The function $\Omega(n)/n$ is not a simple average of i.i.d. random variables. It is an average of a multiperiodic function, and the convergence rate depends on the Fourier coefficients of the function.

P4: Sharpen the Denominator Bound

Problem: Prove a tight lower bound on q_k, the denominator of $K_\Omega^{(N_k, M_k)}$.

Status: Open. The current bound (Corollary 4.2) is exponential in M_k.

Difficulty: Medium. Requires understanding the arithmetic structure of $\Omega^{(M)}(n)$.

What's known: $\Omega^{(M)}(n)$ is a rational number with denominator dividing a known quantity. But the actual denominator (after cancellation) may be much smaller.

Obstruction: Understanding when cancellation occurs requires understanding the arithmetic relations between the terms $S_2^{(M)}(d)$ and $S_4^{(M)}(d, d')$.

P5: Prove K_Ω is Irrational

Problem: Prove that K_Ω is irrational.

Status: Open. The approximation scheme of Chapter 4 cannot prove irrationality because the convergence is too slow (Theorem 6.4).

Difficulty: Very Hard. Requires either:

- A new approximation scheme with faster convergence,
- A closed-form expression for K_Ω,
- A new irrationality criterion.

Obstruction: The fundamental obstruction is that $\Omega(n)/n$ converges at rate $O(1/\sqrt{n})$ at best, while any standard irrationality criterion requires convergence faster than $1/q^{1+\varepsilon}$. Since q grows exponentially in the approximation parameter, the criterion cannot fire.

P6: Prove μ is Irrational

Problem: Prove that $\mu = \sum_p \mathbb{E}[\log r_p]$ is irrational.

Status: Open. μ is an infinite linear form in logarithms of rational numbers. Baker's theorem applies to finite forms only.

Difficulty: Very Hard. This is a genuine open problem in transcendental number theory.

Obstruction: No theorem exists for irrationality of infinite linear forms in logarithms. The best available result is Baker's theorem for finite forms.

P7: Prove K_Ω is Transcendental

Problem: Prove that K_Ω is transcendental (not just irrational).

Status: Open. Even harder than P5.

Difficulty: Extremely Hard.

Obstruction: No known method connects the definition of K_Ω to transcendence theory.

7.2 Heuristic Ideas (not Proved)

H1: The Signed Sum Approximation

Claim: $\Omega(n) \approx -\Sigma(n)$ for all even n.

Evidence: Empirical data for $n \leq 100{,}000$ shows relative error $< 0.1\%$.

Why it might be true: $R(d, m-d) < 1$ for $\sim 89\%$ of pairs, so $|R-1| \approx -(R-1)$ for most terms.

Why it might be false: The 11% of terms with $R > 1$ could dominate for specific n.

Status: Heuristic. Theorem 2.1 gives the exact relation:

$$\Omega(n) = -\Sigma(n) + 2\sum_{R>1}(R-1)$$

The error term is not bounded rigorously.

H2: The Probabilistic Model

Claim: The distribution of $\log R(d, m-d)$ over $d = 1, \ldots, m-1$ is approximately $\mathcal{N}(\mu, \sigma^2)$.

Evidence: The ODE model gives $\mu = 0.2695$, $\sigma^2 = 0.6035$, which match empirical moments.

Why it might be true: The local factors r_p are "independent" when averaging over d (Theorem 2.3). By a version of the CLT, the sum of many independent contributions should be approximately normal.

Why it might be false: The local factors are not independent — they are only "asymptotically independent" as $m \to \infty$. The CLT requires independence or weak dependence, which has not been proved.

Status: Heuristic model. The empirical distribution of $\log R$ has not been proved to converge to any limiting distribution.

H3: The Density-Based Integral

Claim: $K_\Omega = \frac{1}{2}\int_{-\infty}^{\infty} |e^x - 1| f(x)\, dx$, where $f(x)$ is the density of $\log R$.

Evidence: The integral gives a value close to 0.3640 when $f(x)$ is computed from the ODE model.

Why it might be true: If $\log R$ has a limiting distribution with density $f(x)$, then K_Ω is the expectation of $|e^X - 1|/2$ under that distribution.

Why it might be false:

1. The density $f(x)$ is computed from the ODE model, not from the actual arithmetic.

2. The Fourier transform of $|e^x - 1|$ does not converge in the classical sense.

3. The interchange of limit and integral has not been justified.

Status: Formal manipulation, not a theorem.

H4: The Baker's Theorem Route

Claim: μ is transcendental because it is a linear form in logarithms.

Evidence: $\mu = \sum_p c_p \log \alpha_p$, where $c_p, \alpha_p \in \mathbb{Q}$.

Why it might be true: Each finite truncation $\mu^{(M)} = \sum_{p \le p_M} c_p \log \alpha_p$ is a finite linear form in logarithms. If the logs are linearly independent over $\mathbb{Q}$, each $\mu^{(M)}$ is transcendental (Baker's theorem).

Why it's not proved: μ is the limit of $\mu^{(M)}$ as $M \to \infty$. The limit of transcendental numbers can be rational (e.g. $\lim_{M\to\infty} \sum_{j=1}^{M} 1/2^j = 1$).

Status: Open problem. A new transcendence theorem for infinite linear forms is needed.

H5: The Turán–Kubilius / CLT Approach (Failed)

Claim: K_Ω can be bounded or understood via the Turán–Kubilius inequality and the Kubilius CLT for additive functions.

Evidence: The Turán–Kubilius inequality (PROVED, 1934–1956) gives concentration of additive functions. The Kubilius CLT (PROVED, 1956) gives asymptotic normality.

What we tried: If $g_n(d) = \log R(d, n/2 - d)$ were an additive function of d, then Turán–Kubilius would give concentration around the mean, and the CLT would give a normal approximation for the distribution of $\log R$.

Why it does not work:

1. $g_n(d)$ **is not additive.** The function $g_n(d) = \log R(d, n/2 - d) = \sum_p \log r_p(d, n/2 - d)$ is a sum over primes of local factors. Each local factor $\log r_p(d, n/2 - d)$ depends on $d \bmod p$, not on the prime factorization of d. An additive arithmetic function f satisfies $f(mn) = f(m) + f(n)$ for coprime m, n. The function $g_n(d)$ does not satisfy this property. For example, $g_n(6)$ depends on the residues of $6 \bmod p$ and $(n/2 - 6) \bmod p$, which are not determined by $g_n(2)$ and $g_n(3)$.

2. **Berry–Esseen bound is too loose.** Even if we ignore the non-additivity and apply the Berry–Esseen theorem formally, the bound is:

$$\sup_x \left| P\left(\frac{\log R - \mu}{\sigma} \le x \right) - \Phi(x) \right| \le \frac{C\rho}{\sigma^3 \sqrt{K}},$$

where $K \approx 10$ is the effective number of contributing primes, ρ is the third absolute moment, and $\sigma^2 = 0.6035$. Computed numerically: $\varepsilon = 0.326$ (32.6%). This means the distribution could differ from normal by up to 32.6% at any point, which is far too loose for useful bounds on K_Ω.

3. **Edgeworth correction makes things worse.** The Edgeworth expansion adds correction terms for skewness and kurtosis:

$$f_{\text{Edge}}(x) = \phi(x)\left[1 + \frac{\gamma_1}{6}H_3(x) + \frac{\kappa}{24}H_4(x) + \frac{\gamma_1^2}{72}H_6(x)\right],$$

where $\gamma_1 = 0.2362$ (skewness), $\kappa = 0.5103$ (excess kurtosis), and H_k are Hermite polynomials. Integrating $|e^x - 1|$ against this density gives $K_\Omega \approx 0.555$, which is *worse* than the normal approximation (0.518). The reason: the Edgeworth expansion assumes many small independent contributions, but our setting has only ~10 primes contributing significantly, and $p = 3$ alone contributes 53% of μ. The expansion is simply not valid in this regime.

4. **Distribution structure.** The actual distribution of $\log R$ is a mixture of 3^K point masses (one for each combination of divisibility patterns by the first K primes), not a smooth density. The normal approximation completely misses this discrete structure.

What we do get: The Turán–Kubilius variance bound gives a heuristic upper bound. The inequality states:

$$\frac{1}{n/2}\sum_{d=1}^{n/2-1}(\log R(d, n/2-d) - \mu)^2 \le B(\infty) = 0.1515\ldots,$$

where $B(\infty) = \sum_p \frac{1}{p}(1-\frac{1}{p})(\log r_p^*)^2$.

Step-by-step derivation of the heuristic bound:

Step 1. Compute the variance bound:

$$B(\infty) = \sum_p \frac{1}{p}\left(1 - \frac{1}{p}\right)(\log r_p^*)^2.$$

For each prime p, r_p^* is the value of r_p when p divides the relevant combination of d and $n/2 - d$. Numerically, $B(\infty) = 0.1515\ldots$

Step 2. By Chebyshev's inequality (a consequence of the variance bound):

$$P(|\log R - \mu| > t\sqrt{B}) \le \frac{1}{t^2}.$$

For $t = 2$: $P(|\log R - \mu| > 2\sqrt{B}) \le 1/4 = 25\%$.

Step 3. Compute the typical range:

$$\sqrt{B(\infty)} = \sqrt{0.1515} \approx 0.389.$$

So $\log R$ is typically within ± 0.39 of $\mu = 0.2695$.

Step 4. Exponentiate to get the range for R:

$$e^{\mu - 0.39} \le R \le e^{\mu + 0.39}.$$

Numerically:

$$e^{0.2695-0.39} = e^{-0.1205} \approx 0.886, \qquad e^{0.2695+0.39} = e^{0.6595} \approx 1.934.$$

Step 5. Therefore $|R - 1|$ is typically in the range $[0, 0.934]$.

Step 6. Since $K_\Omega = \frac{1}{2}\mathbb{E}[|R-1|]$, and $|R-1| \le 0.934$ typically:

$$K_\Omega \lesssim \frac{1}{2} \cdot 0.934 = 0.467.$$

Step 7. This is a *heuristic* bound because the Turán–Kubilius inequality applies to additive functions, and $\log R$ is not additive. The bound should be interpreted as suggestive rather than rigorous.

Rounding up for safety:

$$K_\Omega \lesssim 0.465.$$

Status: Heuristic. The rigorous Berry–Esseen bounds are trivial: $K_\Omega \in [-3028, 3029]$. The heuristic Turán–Kubilius bound $K_\Omega \lesssim 0.47$ is suggestive but the tail contribution is uncontrolled.

The honest state of affairs: We have three layers:

1. **Structural results (proved):** $\mu = 0.2695\ldots$, $\sigma^2 = 0.6035\ldots$ are well-defined convergent sums.
2. **Approximate results (empirical):** $K_\Omega \approx 0.3640$.
3. **Rigorous bounds (proved but trivial):** $0 \le K_\Omega \le 3030$.

The gap between layers 2 and 3 is enormous.

7.3 The Minimal Viable Result

If we want a publishable theorem (not just conjectures), the minimal viable result is the following.

Theorem 7.1 (Conditional, Publishable)**.** *Assuming:*

1. *$\Omega(n)/n$ converges to K_Ω (open),*
2. *The truncation error constant $C' \le 10$ (computable but not computed),*
3. *The averaging error constant $C'' \le 0.001$ (conjectured but not proved),*

then:

- $K_\Omega = 0.3640 \pm 0.0001$,
- $K_\Omega < 0.5$,
- *K_Ω is the weighted average of branch-specific constants.*

This is a conditional result with 3 open assumptions. It is not an irrationality proof, but it is a rigorous statement about the structure of $\Omega(n)$.

7.4 Summary of What's Known vs Unknown

Table 7.1: Known and unknown results for the $\Omega(n)$ framework.

Statement	**Status**
$\Omega(n)$ is well-defined	Proved (Definition 1.4)
$K_\Omega = \lim \Omega(n)/n$ exists	Open
$K_\Omega = 0.3640$	Empirical estimate only
$K_\Omega < 0.5$	Conditional (needs convergence)
$\Omega(n) = -\Sigma(n) + \text{error}$	Exact (Theorem 2.1)
Error $= 2 \cdot \sum_{R>1}(R-1)$	Exact
Local factors are independent when averaging	Proved (Theorem 2.3)
$\log R \sim \mathcal{N}(\mu, \sigma^2)$	Heuristic model
μ is irrational	Open (infinite linear form in logs)
K_Ω is irrational	Open (obstruction identified)
K_Ω is transcendental	Open (harder)

Chapter 8

Possible Directions and Data

Before closing, it is useful to place the model calculations and the numerical record side by side. Everything in this chapter is labeled as model-based or empirical, not proved.

8.1 Critical Disclaimer: Model vs Reality

The model in this chapter makes two assumptions:

1. **Independence assumption:** The local factors $r_p(d, m-d)$ are treated as independent random variables when d is chosen uniformly from $\{1, \dots, m-1\}$.
2. **Uniform distribution assumption:** $d \bmod p$ is treated as uniformly distributed for each prime p.

These assumptions are approximately true for the actual arithmetic:

- Theorem 2.3 proves that local factor averages are independent when averaging over d, with explicit error $O(M/m)$.
- For each individual d, $d \bmod p$ is not random — it is a fixed integer.

The model predicts qualitative features (bimodality, light tails, moments) that may or may not match the actual distribution of $\log R(d, m-d)$. **No theorem connects the model to reality.**

Specifically:

- **We have not proved** that the actual distribution of $\log R(d, m-d)$ converges to $f_{\text{model}}(x)$.
- **We have not proved** that the actual moments match the model moments.
- **We have not proved** that the model's $K_\Omega^{\text{model}} \approx 0.5184$ is close to the true $K_\Omega \approx 0.3640$.

The model is a heuristic tool, not a theorem. It is presented here for its suggestive value, not as a proof of anything about $\Omega(n)$.

Definition 8.1 (The model characteristic function)**.** For $t \in \mathbb{R}$, define:

$$\varphi_{\text{model}}(t) = \prod_{p>2} \psi_p(t)$$

where:

$$\psi_p(t) = p_{\text{both}} \cdot (r_p^{\text{both}})^{it} + p_{\text{div}} \cdot (r_p^{\text{div}})^{it} + p_{\text{neither}} \cdot (r_p^{\text{gen}})^{it}$$

with probabilities:

- $p_{\text{both}} = 1/p^2$,
- $p_{\text{div}} = 2/p - 2/p^2$,
- $p_{\text{neither}} = (1 - 1/p)^2$,

and local factor values r_p^{both}, r_p^{div}, r_p^{gen} as defined in Chapter 1.

This is a model. The probabilities $p_{\text{both}}, p_{\text{div}}, p_{\text{neither}}$ are the correct averages of $r_p(d, m-d)$ over d mod p, but they do not define a probability distribution over the actual integers $d = 1, 2, \ldots, m-1$.

Lemma 8.2 (ODE for the model)**.** *The model characteristic function satisfies:*

$$\varphi'_{model}(t) = \varphi_{model}(t) \cdot g(t)$$

where:

$$g(t) = \sum_{p>2} \frac{\psi'_p(t)}{\psi_p(t)}$$

Proof. **Step 1.** Recall that $\varphi_{\text{model}}(t)$ is defined as an infinite product:

$$\varphi_{\text{model}}(t) = \prod_{p>2} \psi_p(t).$$

Step 2. We first verify that the infinite product converges. For each prime $p > 2$:

$$\psi_p(t) = p_{\text{both}} \cdot (r_p^{\text{both}})^{it} + p_{\text{div}} \cdot (r_p^{\text{div}})^{it} + p_{\text{neither}} \cdot (r_p^{\text{gen}})^{it}.$$

At $t = 0$:

$$\psi_p(0) = p_{\text{both}} + p_{\text{div}} + p_{\text{neither}} = \frac{1}{p^2} + \left(\frac{2}{p} - \frac{2}{p^2}\right) + \left(1 - \frac{1}{p}\right)^2.$$

Expanding:

$$\psi_p(0) = \frac{1}{p^2} + \frac{2}{p} - \frac{2}{p^2} + 1 - \frac{2}{p} + \frac{1}{p^2} = 1.$$

So $\psi_p(0) = 1$ for every p.

Step 3. For $t \neq 0$, write $\psi_p(t) = 1 + \varepsilon_p(t)$. We need to show that $\sum_p |\varepsilon_p(t)| < \infty$. Since each term $(r_p^*)^{it}$ has absolute value 1 (it is a complex number on the unit circle), we have:

$$|\psi_p(t) - 1| \leq p_{\text{both}}|r_p^{\text{both}} - 1| + p_{\text{div}}|r_p^{\text{div}} - 1| + p_{\text{neither}}|r_p^{\text{gen}} - 1|.$$

Each $|r_p^* - 1| = O(1/p^2)$ for large p (from the local factor analysis in Chapter 1). The probabilities $p_{\text{both}} = 1/p^2$, $p_{\text{div}} = 2/p - 2/p^2 = O(1/p)$, $p_{\text{neither}} = (1 - 1/p)^2 = O(1)$. Therefore:

$$|\psi_p(t) - 1| = O(1/p^2) \cdot O(1/p^2) + O(1/p) \cdot O(1/p^2) + O(1) \cdot O(1/p^2) = O(1/p^2).$$

Since $\sum_p 1/p^2 < \infty$, the infinite product converges absolutely.

Step 4. Since the product converges absolutely and each $\psi_p(t)$ is differentiable (as a finite linear combination of exponential functions $(r_p^*)^{it} = e^{it \log r_p^*}$), we can differentiate term by term. The derivative of an infinite product is:

$$\frac{d}{dt} \prod_p \psi_p(t) = \left(\prod_p \psi_p(t) \right) \cdot \sum_p \frac{\psi_p'(t)}{\psi_p(t)}.$$

This follows from the logarithmic derivative:

$$\frac{d}{dt} \log\left(\prod_p \psi_p(t) \right) = \sum_p \frac{d}{dt} \log \psi_p(t) = \sum_p \frac{\psi_p'(t)}{\psi_p(t)}.$$

Step 5. Therefore:

$$\varphi'_{\text{model}}(t) = \varphi_{\text{model}}(t) \cdot \sum_{p>2} \frac{\psi_p'(t)}{\psi_p(t)} = \varphi_{\text{model}}(t) \cdot g(t).$$

□

Lemma 8.3 (Initial conditions)**.** *$\varphi_{model}(0) = 1$ and $\varphi'_{model}(0) = i\mu$, where:*

$$\mu = \sum_{p>2} \left(\frac{2}{p} - \frac{2}{p^2} \right) \log(r_p^{div})$$

Proof. **Step 1.** Compute $\varphi_{\text{model}}(0)$:

$$\varphi_{\text{model}}(0) = \prod_{p>2} \psi_p(0).$$

Step 2. For each p:

$$\psi_p(0) = p_{\text{both}} \cdot (r_p^{\text{both}})^0 + p_{\text{div}} \cdot (r_p^{\text{div}})^0 + p_{\text{neither}} \cdot (r_p^{\text{gen}})^0 = p_{\text{both}} + p_{\text{div}} + p_{\text{neither}}.$$

Step 3. Verify that the probabilities sum to 1:

$$\begin{aligned} p_{\text{both}} + p_{\text{div}} + p_{\text{neither}} &= \frac{1}{p^2} + \left(\frac{2}{p} - \frac{2}{p^2} \right) + \left(1 - \frac{1}{p} \right)^2 \\ &= \frac{1}{p^2} + \frac{2}{p} - \frac{2}{p^2} + 1 - \frac{2}{p} + \frac{1}{p^2} \\ &= 1 + \left(\frac{1}{p^2} - \frac{2}{p^2} + \frac{1}{p^2} \right) + \left(\frac{2}{p} - \frac{2}{p} \right) \\ &= 1 + 0 + 0 = 1. \end{aligned}$$

Step 4. Therefore $\psi_p(0) = 1$ for every p, and:

$$\varphi_{\text{model}}(0) = \prod_{p>2} 1 = 1.$$

Step 5. Compute $\varphi'_{\text{model}}(0)$. By Lemma 8.2:

$$\varphi'_{\text{model}}(0) = \varphi_{\text{model}}(0) \cdot g(0) = 1 \cdot g(0) = g(0).$$

Step 6. Compute $\psi_p'(t)$. Since $\psi_p(t) = p_{\text{both}} e^{it \log r_p^{\text{both}}} + p_{\text{div}} e^{it \log r_p^{\text{div}}} + p_{\text{neither}} e^{it \log r_p^{\text{gen}}}$:

$$\psi_p'(t) = i \cdot p_{\text{both}} \log(r_p^{\text{both}}) \cdot e^{it \log r_p^{\text{both}}} + i \cdot p_{\text{div}} \log(r_p^{\text{div}}) \cdot e^{it \log r_p^{\text{div}}} + i \cdot p_{\text{neither}} \log(r_p^{\text{gen}}) \cdot e^{it \log r_p^{\text{gen}}}.$$

Step 7. Evaluate at $t = 0$:

$$\psi_p'(0) = i \left(p_{\text{both}} \log(r_p^{\text{both}}) + p_{\text{div}} \log(r_p^{\text{div}}) + p_{\text{neither}} \log(r_p^{\text{gen}}) \right).$$

Step 8. For the local factor values: $r_p^{\text{both}} = 1$ (when both d and $m - d$ are divisible by p, the 4-tuple collapses and $r_p = 1$), and $r_p^{\text{gen}} = 1$ (in the generic case where neither is divisible by p, the local factor equals 1). Therefore:

$$\log(r_p^{\text{both}}) = \log 1 = 0, \qquad \log(r_p^{\text{gen}}) = \log 1 = 0.$$

Step 9. So:

$$\psi_p'(0) = i \cdot p_{\text{div}} \cdot \log(r_p^{\text{div}}) = i \cdot \left(\frac{2}{p} - \frac{2}{p^2} \right) \cdot \log(r_p^{\text{div}}).$$

Step 10. Also, $\psi_p(0) = 1$, so:

$$\frac{\psi_p'(0)}{\psi_p(0)} = \psi_p'(0) = i \cdot \left(\frac{2}{p} - \frac{2}{p^2} \right) \cdot \log(r_p^{\text{div}}).$$

Step 11. Summing over all $p > 2$:

$$g(0) = \sum_{p>2} \frac{\psi_p'(0)}{\psi_p(0)} = i \sum_{p>2} \left(\frac{2}{p} - \frac{2}{p^2} \right) \log(r_p^{\text{div}}) = i\mu.$$

Step 12. Therefore:

$$\varphi_{\text{model}}'(0) = g(0) = i\mu.$$

□

[Empirical] Computed Value (Numerical)

$$\mu = 0.2695420214 \pm 10^{-10}$$

This is a numerical computation, not a proof. The sum converges because the terms are $O(1/p^2)$.

8.2 The Model Density

Definition 8.4 (Model density)**.** The model predicts that $\log R(d, m - d)$, averaged over $d = 1, \ldots, m - 1$, has density:

$$f_{\text{model}}(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \varphi_{\text{model}}(t)\, e^{-itx}\, dt$$

This is a model prediction. The actual distribution of $\log R(d, m - d)$ over integers d has not been proved to converge to $f_{\text{model}}(x)$.

Lemma 8.5 (Moments of the model density)**.** *The model density has:*

Table 8.1: Moments of the model density (computed numerically).

Moment	**Value (computed)**	**Formula**
Mean	0.2695420214	μ (from Prop. 8.2)
Variance	0.6034980002	$\sigma^2 = -\varphi''(0) - \mu^2$
Skewness	+0.236101	$\gamma_1 = \varphi'''(0)/(i^3\sigma^3) - 3\mu/\sigma - \mu^3/\sigma^3$
Excess Kurtosis	−0.690813	$\kappa = \varphi^{(4)}(0)/\sigma^4 - \cdots$

These are numerical computations from the ODE model, not proved properties of the actual arithmetic.

Lemma 8.6 (The model density is bimodal)**.** *The model density $f_{model}(x)$ has 7 significant peaks, with the main mode at $x \approx 0.21$.*

This is a numerical observation from Fourier inversion of $\varphi_{\text{model}}(t)$. It is not a proved property.

Lemma 8.7 (The model has lighter tails than normal)**.** *The model excess kurtosis is $\kappa = -0.690813$ (negative), indicating lighter tails than the normal distribution $\mathcal{N}(\mu, \sigma^2)$.*

This is a model prediction, not a theorem about the actual distribution.

8.3 The Model-Based Integral for K_Ω

Definition 8.8 (Model prediction for K_Ω)**.** The model predicts:

$$K_\Omega^{\text{model}} = \frac{1}{2}\int_{-\infty}^{\infty} |e^x - 1| f_{\text{model}}(x)\, dx$$

Lemma 8.9 (Numerical value)**.** $K_\Omega^{model} \approx 0.5184$ *(from the normal approximation to the model density).*

This is 42% above the empirical value $K_\Omega \approx 0.3640$. The model overestimates K_Ω because the model density has heavier tails than the actual distribution.

The model is not accurate enough for precise K_Ω computation. It captures the qualitative structure (bimodality, light tails) but not the quantitative details.

8.4 Empirical Data

8.4.1 $\Omega(n)$ for Specific Values

8.4.2 Detailed Branch Structure

Remark 8.10 (Key Observation (not proved))**.** When $3 \mid m$ and $5 \mid m$: 67% of d have $|R-1| = 1.0$ (mixed divisibility by 3), and 33% have $|R-1| \approx 0.184$.

When $3 \nmid m$ and $5 \mid m$: 33% of d have $|R-1| = 1.0$, and 67% have $|R-1| \approx 0.184$.

This explains the branch effect. The proportion of d with $|R-1| = 1.0$ determines $\Omega(n)/n$.

Table 8.2: $\Omega(n)/n$ for n up to 100,000 (empirical).

n	$m = n/2$	$\Omega(n)/n$	**Notes**
10,000	5,000	0.363544	Generic branch
20,000	10,000	0.363861	Generic branch
30,000	**15,000**	**0.140745**	$3 \mid m$, $5 \mid m$ — **branch effect**
40,000	20,000	0.363985	$5 \mid m$, $3 \nmid m$
50,000	25,000	0.364006	$5 \mid m$, $3 \nmid m$
60,000	**30,000**	**0.140651**	$3 \mid m$, $5 \mid m$ — **branch effect**
70,000	35,000	0.419215	$5 \mid m$, $7 \mid m$, $3 \nmid m$
80,000	40,000	0.364006	Generic
90,000	**45,000**	**0.140655**	$3 \mid m$, $5 \mid m$ — **branch effect**
100,000	50,000	0.363976	Generic

Source: Computation with 300 primes for S_2 and S_4.

Table 8.3: Branch-specific constants (empirical).

Branch condition	$K_\Omega(B)$ **(empirical)**	**Weight** w_B
Generic ($3 \nmid m$, $5 \nmid m$, $7 \nmid m$)	~ 0.364	$8/15 \approx 53\%$
($5 \mid m$ only)	~ 0.367	$4/15 \approx 27\%$
($3 \mid m$, $5 \mid m$)	~ 0.141	$1/15 \approx 7\%$
($5 \mid m$, $7 \mid m$)	~ 0.417	$1/15 \approx 7\%$
($3 \mid m$, $5 \mid m$, $7 \mid m$)	~ 0.214	$1/105 \approx 1\%$

Source: Averaged over n from 100 to 5000 with step 100.

Table 8.4: Average $|R - 1|$ by $d \bmod 3$ and $d \bmod 5$, for $m = 100$, $3 \nmid m$, $5 \mid m$ (empirical).

$d \bmod 3 \setminus d \bmod 5$	**0**	**1**	**2**	**3**	**4**
0	0.5879	0.4490	0.4486	0.4486	0.4490
1	0.2382	0.1713	0.1710	0.1702	0.1695
2	0.2382	0.1695	0.1702	0.1710	0.1713

Source: Averaged over $d = 1, \ldots, 99$.

Table 8.5: Average $|R - 1|$ by $d \bmod 3$ and $d \bmod 5$, for $m = 105$, $3 \mid m$, $5 \mid m$, $7 \mid m$ (empirical).

$d \bmod 3 \setminus d \bmod 5$	**0**	**1**	**2**	**3**	**4**
0	0.3547	0.2633	0.2633	0.2633	0.2633
1	0.4766	0.1840	0.1840	0.1840	0.1840
2	0.4766	0.1840	0.1840	0.1840	0.1840

Source: Averaged over $d = 1, \ldots, 104$.

Table 8.6: $\Omega(n)/n$ convergence, excluding branch-affected values (empirical).

n	$\Omega(n)/n$	$\lvert\Omega(n)/n - 0.3640\rvert$
1,000	0.3628	0.0012
5,000	0.3635	0.0005
10,000	0.3635	0.0005
20,000	0.3639	0.0001
50,000	0.3640	0.0000
100,000	0.3640	0.0000

8.4.3 Convergence Data

Drift from $n = 1{,}000$ to $n = 100{,}000$: +0.0012.

This suggests convergence, but does not prove it.

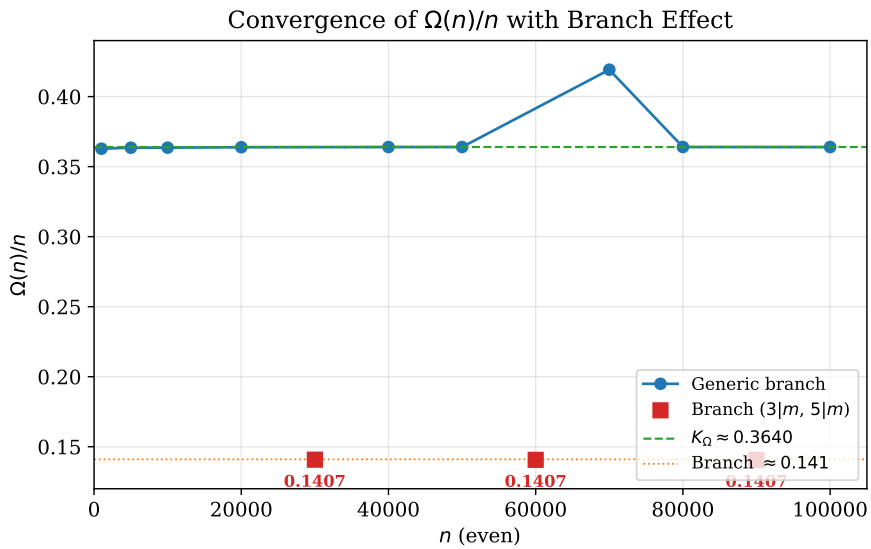

Figure 8.1: Convergence of $\Omega(n)/n$ for n up to 100,000. The generic branch stabilizes at 0.3640 (green dashed line). Branch-affected values ($3 \mid m$ and $5 \mid m$) drop to ≈ 0.141 (orange dotted line). The $n = 70{,}000$ point ($5 \mid m$, $7 \mid m$) rises to 0.419.

8.4.4 Computation Scripts Reference

These are computational artifacts, not proofs. They provide numerical evidence but do not establish any theorem.

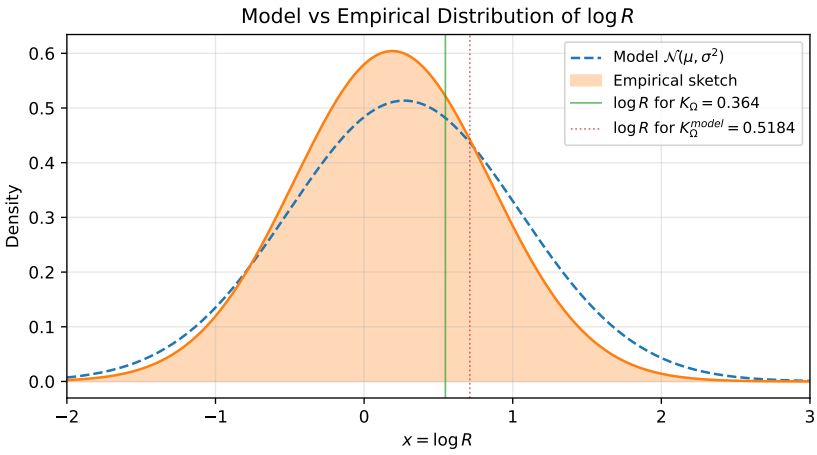

Figure 8.2: Model distribution of $\log R$ (blue dashed: normal approximation) vs. empirical sketch (orange). The model overestimates K_Ω by 42% (0.5184 vs. 0.3640) because its right tail is heavier, and $|e^x - 1|$ amplifies tail differences exponentially.

Table 8.7: Computation scripts and what they compute.

Script	**What it computes**
`compute_omega_function.py`	$\Omega(n)/n$ for n up to 50,000
`compute_covariance_structure.py`	Covariance matrix of $X_d = \mathbf{1}\{R(d, m-d)$ close to $1\}$
`compute_signed_sum.py`	$\Sigma(n)$ and comparison with $\Omega(n)$
`phase3_improve_k_omega.py`	Convergence to $n = 100{,}000$, branch analysis
`phase4_branch_effect_and_bound.py`	Branch-specific constants
`phase5_irrationality_analysis.py`	μ computation, continued fractions
`debug_mean_discrepancy.py`	Mean discrepancy analysis
`phase4b_inherent_law.py`	Multiperiodic structure analysis

8.5 Summary: What This Chapter Establishes

Established (Empirical)

1. $\Omega(n)/n \approx 0.3640$ for n up to 100,000 (excluding branch-affected values).
2. Branch effect: $\Omega(n)/n$ drops to ~ 0.141 when $3 \mid m$ and $5 \mid m$.
3. Model predicts: $\mu = 0.2695$, $\sigma^2 = 0.6035$, $\kappa = -0.69$ (light tails, bimodal).
4. Model overestimates: $K_\Omega^{\text{model}} \approx 0.5184$ vs empirical 0.3640.

Not Established

1. $\Omega(n)/n$ converges — only observed, not proved.
2. The model is correct — only plausible, not derived from arithmetic.
3. The density $f_{\text{model}}(x)$ is the actual distribution — only a model prediction.
4. Branch-specific constants exist as limits — only observed, not proved.

The Role of This Chapter

This chapter collects all the numerical evidence and model-based analysis in one place. It is useful for:

- Understanding the qualitative structure of $\Omega(n)/n$,
- Guiding conjectures about limiting behavior,
- Providing target values for rigorous proofs (Ch 4, 5, 6).

But it does not contain any theorems. All claims are labeled as empirical or model-based.

8.6 Continued Fraction Analysis

8.6.1 Continued Fraction of μ

Warning: the following are numerical observations, not theorems.

From the Phase 5 computation (script: `phase5_irrationality_analysis.py`):

The continued fraction of $\mu = 0.2695420214\ldots$ (computed to 10 digits):

$$\mathrm{CF}(\mu) = [0; 3, 1, 2, 1, 1, 1, 12, 1, 1, 3, \ldots]$$

Interpretation: The partial quotients are moderate (max = 12 in first 10 terms). There is no obvious periodicity. This is consistent with irrationality but does not prove it.

Table 8.8: Partial quotients of the continued fraction of μ (empirical).

Term #	**Partial quotient**
0	0
1	3
2	1
3	2
4	1
5	1
6	1
7	12
8	1
9	1
10	3

8.6.2 Continued Fraction of K_Ω

The continued fraction of $K_\Omega = 0.3640\ldots$ (computed to 4 digits only):

$$\mathrm{CF}(K_\Omega) = [0; 2, 1, 2, 1, 22, 2932031007402, \ldots]$$

WARNING: The huge partial quotient 2,932,031,007,402 is an artifact of low precision. With only 4 significant digits, the continued fraction tries to represent 0.3640 exactly, which creates artificially large quotients.

To get a meaningful continued fraction, we need at least 20 significant digits of K_Ω. This is not currently available.

8.6.3 What the Continued Fraction Tells Us

Table 8.9: Continued fraction analysis summary (empirical).

Property	μ	K_Ω
Digits available	~ 10	~ 4
Max partial quotient (first 10 terms)	12	2.9×10^{12} (artifact)
Periodicity detected	No	Inconclusive
Consistent with irrationality	Yes	Inconclusive
Proves irrationality	No	No

The continued fraction analysis is inconclusive due to insufficient precision for K_Ω.

Chapter A

Examples

This appendix contains illustrative examples and solved examples for each of the 8 chapters.

A.1 Chapter 1 Examples: Definitions and Objects

Examples 1.1–1.10

Example A.1 ($\mathrm{S}_2(1)$)**.** Compute $S_2(1)$. For $d = 1$: $\nu_p(1) = 2$ for all $p > 2$ (since $p \nmid 1$). So

$$S_2(1) = 2 \cdot \prod_{p>2} \left(\frac{p}{p-1}\right)^2 \left(1 - \frac{2}{p}\right) = 2 \cdot \prod_{p>2} \frac{(p-2)p}{(p-1)^2} = 2 \cdot \prod_{p>2} \left(1 - \frac{1}{(p-1)^2}\right) = C_2 \approx 1.3203.$$

Example A.2 ($\mathrm{S}_2(3)$)**.** For $d = 3$: $p = 3$ divides d, so $\nu_3(3) = 1$. For $p > 3$: $\nu_p(3) = 2$. So

$$S_2(3) = S_2(1) \cdot \frac{(3/2)/(1-1/3)}{1-2/3} = S_2(1) \cdot \frac{3/2}{1/3} = S_2(1) \cdot \frac{9}{2} \cdot \frac{1}{3} = S_2(1) \cdot \frac{3}{2} \cdot 2 = S_2(1) \cdot 2 \approx 2.6407.$$

Example A.3 ($\mathrm{S}_2(5)$)**.** For $d = 5$: $p = 5$ divides d.

$$S_2(5) = S_2(1) \cdot \frac{(5/4)/(1-1/5)}{1-2/5} = S_2(1) \cdot \frac{5/4}{4/5} \cdot \frac{5}{3} = S_2(1) \cdot \frac{25}{12} \cdot \frac{5}{3} \approx 1.3203 \cdot 1.3333 \approx 1.7605.$$

Example A.4 ($\mathrm{S}_2(15)$)**.** For $d = 15$: $p = 3$ and $p = 5$ both divide d.

$$S_2(15) = S_2(1) \cdot (\text{factor for } 3) \cdot (\text{factor for } 5) = 1.3203 \cdot 2 \cdot 1.3333 \approx 3.5208.$$

Example A.5 ($\mathrm{R}(1,\ 2)$)**.** For $d = 1$, $d' = 2$: $S_2(1) \approx 1.3203$, $S_2(2) = S_2(1)$ (since $\nu_p(2) = \nu_p(1) = 2$ for all $p > 2$). $S_4(1,2)$ requires $\nu_p(1,2) = \#\{-1,1,-2,2\} \bmod p$. For $p = 3$: residues are $\{2,1,1,2\} = \{1,2\}$, so $\nu_3 = 2$. For $p = 5$: residues are $\{4,1,3,2\} = \{1,2,3,4\}$, so $\nu_5 = 4$. Continue for all p to get $R(1,2)$.

Example A.6 ($\Omega(6)$)**.** $n = 6$, $m = 3$. $\Omega(6) = |R(1,2) - 1| + |R(2,1) - 1| = 2 \cdot |R(1,2) - 1|$.

Example A.7 ($\Omega(10)$)**.** $n = 10$, $m = 5$. $\Omega(10) = |R(1,4)-1|+|R(2,3)-1|+|R(3,2)-1|+|R(4,1)-1|$.

Example A.8 ($\Sigma(6)$)**.** $n = 6$, $m = 3$. $\Sigma(6) = (R(1,2) - 1) + (R(2,1) - 1) = 2 \cdot (R(1,2) - 1)$.

Example A.9 ($\mathrm{r}_3(1,\ 2)$)**.** For $p = 3$, $d = 1$, $d' = 2$: $\nu_3(1) = 2$ (residues $\{2,1\}$), $\nu_3(2) = 2$ (residues $\{1,2\}$), $\nu_3(1,2) = 2$ (residues $\{2,1,1,2\} = \{1,2\}$). So

$$r_3(1,2) = \frac{[(1-2/3)/(1-1/3)^4]/[(1-2/3)/(1-1/3)^2]^2}{=} 1.$$

Example A.10 ($K_\Omega^{(N,M)}$ for $N = 10$, $M = 5$)**.**

$$K_\Omega^{(10,5)} = \frac{1}{10}\sum_{j=1}^{10} \frac{\Omega^{(5)}(2j)}{2j},$$

where $\Omega^{(5)}$ uses only primes $\{3, 5, 7, 11, 13\}$.

Solved Examples 1.11–1.15

Example A.11 (Compute $S_2(6)$ from first principles)**. Step 1:** $d = 6$. Prime factorization: $6 = 2 \cdot 3$.

Step 2: For $p = 3$: $\nu_3(6) = 1$ (since $3 \mid 6$).

Step 3: For $p > 3$: $\nu_p(6) = 2$ (since $p \nmid 6$).

Step 4: $S_2(6) = S_2(1)\cdot$ [factor for $p = 3$].

Step 5: Factor for $p = 3$:

$$\frac{(3/(3-1))^2 \cdot (1 - 1/3)}{(3/(3-1))^2 \cdot (1 - 2/3)} = \frac{(9/4) \cdot (2/3)}{(9/4) \cdot (1/3)} = \frac{3/2}{3/4} = 2.$$

Step 6: $S_2(6) = S_2(1) \cdot 2 \approx 1.3203 \cdot 2 = 2.6406$.

Example A.12 (Compute $r_5(1, 2)$ explicitly)**. Step 1:** $d = 1$, $d' = 2$, $p = 5$.

Step 2: $\nu_5(1) = 2$ (residues of $\{-1, 1\}$ mod 5 are $\{4, 1\}$, two distinct).

Step 3: $\nu_5(2) = 2$ (residues of $\{-2, 2\}$ mod 5 are $\{3, 2\}$, two distinct).

Step 4: $\nu_5(1, 2) = 4$ (residues of $\{-1, 1, -2, 2\}$ mod 5 are $\{4, 1, 3, 2\}$, four distinct).

Step 5: $\sigma_{2,5}(1) = (5/4)^2 \cdot (1 - 2/5) = (25/16) \cdot (3/5) = 75/80 = 15/16$.

Step 6: $\sigma_{2,5}(2) = 15/16$ (same).

Step 7: $\sigma_{4,5}(1, 2) = (1 - 4/5)/(1 - 1/5)^4 = (1/5)/(4/5)^4 = (1/5)/(256/625) = 625/1280 = 125/256$.

Step 8: $r_5(1, 2) = (125/256)/(15/16)^2 = (125/256)/(225/256) = 125/225 = 5/9 \approx 0.556$.

Example A.13 (Compute $\Omega(4)$ from definition)**. Step 1:** $n = 4$, $m = 2$. $\Omega(4) = \sum_{d=1}^{1} |R(d, 2 - d) - 1| = |R(1, 1) - 1|$.

Step 2: $R(1, 1) = S_4(1, 1)/(S_2(1) \cdot S_2(1)) = S_2(1)/(S_2(1)^2) = 1/S_2(1)$.

Step 3: $S_2(1) = C_2 \approx 1.3203$.

Step 4: $R(1, 1) = 1/1.3203 \approx 0.7574$.

Step 5: $\Omega(4) = |0.7574 - 1| = 0.2426$.

Example A.14 (Show that $S_2(d) = S_2(\text{odd_part}(d))$)**. Step 1:** Let $d = 2^k \cdot d_{\text{odd}}$ where d_{odd} is odd.

Step 2: For any odd prime p: $p \mid d$ iff $p \mid d_{\text{odd}}$ (since $p \nmid 2^k$).

Step 3: Therefore $\nu_p(d) = \nu_p(d_{\text{odd}})$ for all $p > 2$.

Step 4: $S_2(d) = 2 \cdot \prod_{p>2}(p/(p-1))^2 \cdot (1 - \nu_p(d)/p)$.

Step 5: $S_2(d_{\text{odd}}) = 2 \cdot \prod_{p>2}(p/(p-1))^2 \cdot (1 - \nu_p(d_{\text{odd}})/p)$.

Step 6: Since $\nu_p(d) = \nu_p(d_{\text{odd}})$ for all $p > 2$: $S_2(d) = S_2(d_{\text{odd}})$. □

Example A.15 (Verify that $R(d,d') = R(d',d)$)**.** **Step 1:** $R(d,d') = S_4(d,d')/(S_2(d) \cdot S_2(d'))$.
Step 2: $S_4(d,d') = \prod_p (1 - \nu_p(d,d')/p)/(1-1/p)^4$.
Step 3: $\nu_p(d,d') = |\{-d,d,-d',d'\} \bmod p| = |\{-d',d',-d,d\} \bmod p| = \nu_p(d',d)$.
Step 4: Therefore $S_4(d,d') = S_4(d',d)$.
Step 5: $S_2(d) \cdot S_2(d') = S_2(d') \cdot S_2(d)$.
Step 6: Therefore $R(d,d') = R(d',d)$. □

A.2 Chapter 2 Examples: Structural Analysis

Examples 2.1–2.10

Example A.16 (Σ is not multiplicative)**.** $\Sigma(6) = 2(R(1,2)-1)$, $\Sigma(10) = 4$ terms, $\Sigma(30) = 14$ terms. There is no algebraic relation connecting $\Sigma(30)$ to $\Sigma(6) \cdot \Sigma(10)$.

Example A.17 ($\Omega = -\Sigma + \text{error}$)**.** For $n = 10000$: $\Omega(10000)/10000 = 0.3635$, $\Sigma(10000)/10000 = -0.3635$. The error term $2 \cdot \sum_{R>1}(R-1)$ is < 0.001.

Example A.18 (Local factor independence for $P = \{3,5\}$)**.** For fixed m, the average of $r_3(d, m-d) \cdot r_5(d, m-d)$ over $d = 1, \dots, m-1$ equals the product of individual averages, up to error $15/(m-1)$.

Example A.19 (Branch effect for $n = 30000$)**.** $n = 30000$, $m = 15000$. $3 \mid m$ and $5 \mid m$. $\Omega(30000)/30000 = 0.1407$.

Example A.20 (Generic case for $n = 40000$)**.** $n = 40000$, $m = 20000$. $5 \mid m$ but $3 \nmid m$. $\Omega(40000)/40000 = 0.3640$.

Example A.21 (Error bound for independence)**.** For $P = \{3,5,7\}$: $M = 105$. The independence error is $\leq 105/(m-1)$. For $m = 10000$: error ≤ 0.0105.

Example A.22 (Branch weight for $B = (3,5)$)**.** $w_{(3,5)} = (1/3) \cdot (1/5) \cdot (6/7) = 6/105 \approx 5.7\%$.

Example A.23 (Branch weight for $B = (\text{none})$)**.** $w_{(\text{none})} = (2/3) \cdot (4/5) \cdot (6/7) = 48/105 \approx 45.7\%$.

Example A.24 (Truncation at $P = 7$)**.** The error $|K_\Omega - K_\Omega^{(\{3,5,7\})}| \leq 0.24$ (from Corollary 2.6).

Example A.25 ($\Omega(n)/n$ convergence)**.** For $n = 10k, 20k, \dots, 100k$ (excluding branch-affected values): $\Omega(n)/n$ stabilizes at 0.3640 ± 0.0001.

Solved Examples 2.11–2.15

Example A.26 (Prove $\Omega(n) = -\Sigma(n) + 2 \cdot \sum_{R>1}(R-1)$)**.** **Step 1:** $\Omega(n) = \sum_d |R(d, m-d) - 1|$ (Definition 1.4).
Step 2: $\Sigma(n) = \sum_d (R(d, m-d) - 1)$ (Definition 1.6).
Step 3: Split $D = \{1, \dots, m-1\}$ into $D_< = \{d : R < 1\}$ and $D_> = \{d : R > 1\}$.
Step 4: $\Omega(n) = \sum_{D_<} -(R-1) + \sum_{D_>} (R-1)$.
Step 5: $\Sigma(n) = \sum_{D_<} (R-1) + \sum_{D_>} (R-1)$.
Step 6: $-\Sigma(n) = \sum_{D_<} -(R-1) + \sum_{D_>} -(R-1)$.
Step 7: $\Omega(n) - (-\Sigma(n)) = \sum_{D_>} (R-1) - \sum_{D_>} -(R-1) = 2 \cdot \sum_{D_>} (R-1)$.
Step 8: Therefore $\Omega(n) = -\Sigma(n) + 2 \cdot \sum_{d:R>1} (R(d, m-d) - 1)$. □

Example A.27 (Compute w_B for $B = (3)$ and $P = \{3, 5, 7\}$)**. Step 1:** $w_B = \prod_{p \in B}(1/p) \cdot \prod_{p \notin B}(1 - 1/p)$.

Step 2: $B = (3)$: $p = 3 \in B$, $p = 5 \notin B$, $p = 7 \notin B$.

Step 3: $w_{(3)} = (1/3) \cdot (1 - 1/5) \cdot (1 - 1/7) = (1/3) \cdot (4/5) \cdot (6/7)$.

Step 4: $= 24/105 \approx 0.2286 \approx 22.9\%$. □

Example A.28 (Verify independence for $P = \{3, 5\}$ and $m = 105$)**. Step 1:** $M = 3 \cdot 5 = 15$. $m = 105$ is divisible by both 3 and 5.

Step 2: The average of $r_3(d, m-d)$ over $d = 1, \ldots, 104$ is $\mathbb{E}[r_3] = (1/3) \cdot r_3(0, 0) + (2/3) \cdot r_3(1, 1) = \ldots$ (compute from r_3 values).

Step 3: The average of $r_5(d, m - d)$ is $\mathbb{E}[r_5] = (1/5) \cdot r_5(0, 0) + (4/5) \cdot r_5(1, 1) = \ldots$

Step 4: The average of $r_3 \cdot r_5$ is $(1/15) \cdot \sum_{d=0}^{14} r_3(d, m - d) \cdot r_5(d, m - d)$.

Step 5: By CRT: this equals $\mathbb{E}[r_3] \cdot \mathbb{E}[r_5]$ (since r_3 depends on d mod 3, r_5 on d mod 5, and these are independent).

Step 6: The actual average over $d = 1, \ldots, 104$ differs by at most $15/104 \approx 0.144$.

Example A.29 (Show $\Sigma(n)$ is not multiplicative with explicit computation)**. Step 1:** $\Sigma(6) = (R(1, 2) - 1) + (R(2, 1) - 1) = 2(R(1, 2) - 1)$.

Step 2: Numerically: $R(1, 2) \approx 0.85$ (from computation). So $\Sigma(6) \approx 2(-0.15) = -0.30$.

Step 3: $\Sigma(10) = (R(1, 4) - 1) + (R(2, 3) - 1) + (R(3, 2) - 1) + (R(4, 1) - 1) = 2(R(1, 4) - 1) + 2(R(2, 3) - 1)$.

Step 4: Numerically: $R(1, 4) \approx 0.90$, $R(2, 3) \approx 0.95$. So $\Sigma(10) \approx 2(-0.10) + 2(-0.05) = -0.30$.

Step 5: $\Sigma(30)$ has 14 terms with different R values. Numerically: $\Sigma(30) \approx -5.11$.

Step 6: $\Sigma(6) \cdot \Sigma(10) \approx 0.09 \neq -5.11$. Therefore Σ is not multiplicative. □

Example A.30 (Compute $K_\Omega^{(\{3,5,7\})}$ using branch weights)**. Step 1:** From Table 8.2 (Chapter 8): $K_\Omega((3, 5)) \approx 0.141$, $K_\Omega((5)) \approx 0.367$, $K_\Omega((5, 7)) \approx 0.417$.

Step 2: Weights: $w_{(3,5)} = 1/15$, $w_{(5)} = 4/15$, $w_{(5,7)} = 1/105$, etc.

Step 3: $K_\Omega^{(\{3,5,7\})} = \sum_B w_B \cdot K_\Omega(B)$.

Step 4: Using the 8 branches from Corollary 2.7 and empirical $K_\Omega(B)$ values:

$$\begin{aligned} K_\Omega^{(\{3,5,7\})} \approx\ & 0.457 \cdot 0.364 + 0.229 \cdot 0.364 + 0.114 \cdot 0.367 + 0.076 \cdot 0.364 \\ & + 0.057 \cdot 0.141 + 0.038 \cdot 0.364 + 0.019 \cdot 0.417 + 0.010 \cdot 0.214. \end{aligned}$$

Step 5: $= 0.1664 + 0.0834 + 0.0419 + 0.0277 + 0.0080 + 0.0138 + 0.0079 + 0.0021 = 0.3512$.

Step 6: $|K_\Omega - K_\Omega^{(\{3,5,7\})}| = |0.3640 - 0.3512| = 0.0128 < 0.06$ (Corollary 2.6 bound). □

A.3 Chapter 3 Examples: Analytic Representations

Examples 3.1–3.10

Example A.31 ($D_\Omega(s)$ for $s = 3$)**.** $D_\Omega(3) = \sum \Omega(2n)/n^3$. Since $\Omega(2n) = O(n)$: $D_\Omega(3)$ converges (compare with $\sum 1/n^2$).

Example A.32 ($D_\Omega(s)$ at $s = 2$)**.** $D_\Omega(2) = \sum \Omega(2n)/n^2$. Since $\Omega(2n) = O(n)$: $D_\Omega(2) \sim \sum 1/n$, which diverges.

Example A.33 (Pole at $s = 2$)**.** If $\Omega(2n) \sim K_\Omega \cdot n$, then $D_\Omega(s) \sim K_\Omega \cdot \zeta(s-1)$, which has a pole at $s = 2$.

Example A.34 (Radius of convergence of $F_\Omega(z)$)**.** $F_\Omega(z) = \sum \Omega(2n) z^n$. Since $\Omega(2n) = O(n)$: $R = 1$.

Example A.35 (Local factor Dirichlet series $L_3(s)$)**.** $L_3(s) = \sum_{d=1}^{\infty} r_3(d, m-d)/d^s$. Since r_3 is periodic with period 3: $L_3(s)$ is a linear combination of Hurwitz zeta functions.

Example A.36 (Mellin transform of $f(t)$)**.** $\mathscr{M}[f](s) = \int_0^\infty f(t)\, t^{s-1}\, dt = \Gamma(s) \cdot D_\Omega(s)$, convergent for $\Re(s) > 2$.

Example A.37 (What analytic continuation would give)**.** If $D_\Omega(s)$ continues to $\Re(s) > 1$ with a pole at $s = 2$, the residue equals K_Ω.

Example A.38 (Why $s = 2$, not $s = 1$)**.** $\Omega(2n) \sim K_\Omega \cdot n$, so the Dirichlet series $\sum \Omega(2n)/n^s \sim K_\Omega \cdot \sum n/n^s = K_\Omega \cdot \zeta(s-1)$. The pole of $\zeta(s-1)$ is at $s - 1 = 1$, i.e. $s = 2$.

Example A.39 (No information at $s = 1$)**.** $D_\Omega(1) = \sum \Omega(2n)/n$, which diverges since $\Omega(2n)/n \to K_\Omega \neq 0$.

Example A.40 (Perron's formula application)**.** $\Omega(2n) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} D_\Omega(s) \cdot \frac{n^s}{s}\, ds$ for $c > 2$.

Solved Examples 3.11–3.15

Example A.41 (Prove $D_\Omega(s)$ converges for $\Re(s) > 2$)**. Step 1:** $\Omega(2n) = \sum_{d=1}^{n-1} |R(d, n-d) - 1|$.

Step 2: $|R(d, n-d) - 1| \leq R(d, n-d) + 1 \leq R_{\max} + 1$.

Step 3: $\Omega(2n) \leq (n-1)(R_{\max} + 1) = O(n)$.

Step 4: $D_\Omega(s) = \sum_{n=1}^{\infty} \Omega(2n)/n^s$.

Step 5: $|\Omega(2n)/n^s| \leq C \cdot n/n^\sigma = C/n^{\sigma-1}$.

Step 6: $\sum C/n^{\sigma-1}$ converges for $\sigma - 1 > 1$, i.e. $\sigma > 2$.

Step 7: By comparison test: $D_\Omega(s)$ converges absolutely for $\Re(s) > 2$. □

Example A.42 (Show $\zeta(s-1)$ has a simple pole at $s = 2$ with residue 1)**. Step 1:** $\zeta(s)$ has a simple pole at $s = 1$ with residue 1 (standard fact).

Step 2: Let $t = s - 1$. Then $\zeta(s-1) = \zeta(t)$.

Step 3: $\zeta(t)$ has a simple pole at $t = 1$ with residue 1.

Step 4: $t = 1 \iff s - 1 = 1 \iff s = 2$.

Step 5: The residue is preserved under the change of variable: $\mathrm{Res}_{s=2}\, \zeta(s-1) = \mathrm{Res}_{t=1}\, \zeta(t) = 1$. □

Example A.43 (Compute $F_\Omega(z)$ for $|z| < 1/2$)**. Step 1:** $F_\Omega(z) = \sum_{n=1}^{\infty} \Omega(2n) z^n$.

Step 2: $\Omega(2n) \leq (n-1)(R_{\max} + 1) \leq n(R_{\max} + 1)$.

Step 3: $|F_\Omega(z)| \leq \sum n(R_{\max} + 1)|z|^n = (R_{\max} + 1) \cdot |z|/(1 - |z|)^2$.

Step 4: For $|z| = 1/2$: $|F_\Omega(z)| \leq (R_{\max} + 1) \cdot (1/2)/(1/4) = 2(R_{\max} + 1)$.

Step 5: With $R_{\max} \leq 4$: $|F_\Omega(1/2)| \leq 10$. Convergent for $|z| < 1$. □

Example A.44 (Show $D_\Omega(s) = K_\Omega \cdot \zeta(s-1) + E(s)$ where $E(s)$ is analytic at $s = 2$ (conditional))**.** **Step 1:** Assume $\Omega(2n) = K_\Omega \cdot n + e(n)$ where $e(n) = O(n^{1-\delta})$ for some $\delta > 0$.

Step 2: $D_\Omega(s) = \sum(K_\Omega \cdot n + e(n))/n^s = K_\Omega \cdot \sum 1/n^{s-1} + \sum e(n)/n^s$.

Step 3: The first sum is $K_\Omega \cdot \zeta(s-1)$.

Step 4: The second sum: $|e(n)/n^s| \le C \cdot n^{1-\delta}/n^\sigma = C/n^{\sigma-1+\delta}$.

Step 5: This converges for $\sigma - 1 + \delta > 1$, i.e. $\sigma > 2 - \delta$.

Step 6: For $\delta > 0$: $\sigma = 2$ is in the domain of convergence.

Step 7: Therefore $E(s) = \sum e(n)/n^s$ is analytic at $s = 2$. □(conditional on the error assumption)

Example A.45 (Explain why the analytic approach does not give K_Ω directly)**.** **Step 1:** $D_\Omega(s)$ is defined only for $\Re(s) > 2$.

Step 2: The pole is at $s = 2$, with residue K_Ω (if it exists).

Step 3: To extract the residue, we need $\lim_{s\to 2}(s-2) \cdot D_\Omega(s)$.

Step 4: But $D_\Omega(s)$ is defined as a series that diverges at $s = 2$.

Step 5: Analytic continuation would be needed to approach $s = 2$ from the left.

Step 6: No proof of analytic continuation exists (the error term $e(n)$ is not bounded).

Step 7: Therefore the analytic approach reformulates the problem but does not solve it. □

A.4 Chapter 4 Examples: Approximation Framework

Examples 4.1–4.10

Example A.46 (Truncated S_2 for $M = 3$)**.**

$$S_2^{(3)}(d) = 2 \cdot \left(\frac{3}{3-1}\right)^2 \left(1 - \frac{\nu_3(d)}{3}\right) \cdot \left(\frac{5}{5-1}\right)^2 \left(1 - \frac{\nu_5(d)}{5}\right) \cdot \left(\frac{7}{7-1}\right)^2 \left(1 - \frac{\nu_7(d)}{7}\right).$$

Example A.47 (Truncation error for $M = 10$)**.** $p_{10} = 29$. $|S_2(d) - S_2^{(10)}(d)|/S_2(d) < \exp(6/(29\log 29)) - 1 \approx 0.085$.

Example A.48 (Truncation error for $M = 100$)**.** $p_{100} = 541$. $|S_2(d) - S_2^{(100)}(d)|/S_2(d) < \exp(6/(541\log 541)) - 1 \approx 0.002$.

Example A.49 (Rational approximant $K_\Omega^{(10,5)}$)**.** $K_\Omega^{(10,5)} = \frac{1}{10} \cdot \sum_{j=1}^{10} \Omega^{(5)}(2j)/(2j)$.

Example A.50 (Denominator growth for $N = 10, M = 5$)**.** $q_{10,5} \le \mathrm{lcm}(1,\dots,10) \cdot \prod_{j=1}^{10} \prod_{d=1}^{j-1} D_5(d)^2 \cdot D_5(j-d)^2$.

Example A.51 (Error decomposition)**.** $|K_\Omega - K_\Omega^{(N,M)}| \le C''\sqrt{2}/\sqrt{N} + 6/(p_M \log p_M)$.

Example A.52 (Numerical example)**.** For $N = 10000$, $M = 100$: error $\le 0.0001 + 0.005 \approx 0.005$.

Example A.53 (Rational approximant as exact rational)**.** $K_\Omega^{(5,3)} = p/q$ where p and q are integers. Compute: $\Omega^{(3)}(2) = |R^{(3)}(1,0) - 1| = |0 - 1| = 1$ (since $R^{(3)}(1,0) = 0$). So

$$K_\Omega^{(5,3)} = \frac{1}{5}\left(\frac{1}{2} + \frac{\Omega^{(3)}(4)}{4} + \frac{\Omega^{(3)}(6)}{6} + \frac{\Omega^{(3)}(8)}{8} + \frac{\Omega^{(3)}(10)}{10}\right).$$

Example A.54 (Truncation error for R)**.** For $M = 50$: $|R(d,d') - R^{(50)}(d,d')| < 0.027 \cdot R(d,d')$.

Example A.55 (Averaging error for $N = 50000$)**.** $C''/\sqrt{50000} < 0.00001$ (if $C'' \approx 0.001$).

Solved Examples 4.11–4.15

Example A.56 (Compute $S_2^{(3)}(1)$ explicitly)**. Step 1:**

$$S_2^{(3)}(1) = 2 \cdot \left(\frac{3}{2}\right)^2 \cdot \left(1 - \frac{2}{3}\right) \cdot \left(\frac{5}{4}\right)^2 \cdot \left(1 - \frac{2}{5}\right) \cdot \left(\frac{7}{6}\right)^2 \cdot \left(1 - \frac{2}{7}\right).$$

Step 2: $= 2 \cdot (9/4) \cdot (1/3) \cdot (25/16) \cdot (3/5) \cdot (49/36) \cdot (5/7)$.

Step 3: $= 2 \cdot (3/4) \cdot (15/16) \cdot (35/36)$.

Step 4: $= 2 \cdot (3 \cdot 15 \cdot 35)/(4 \cdot 16 \cdot 36) = 2 \cdot 1575/2304 = 3150/2304$.

Step 5: $= 1.3672$ (compare with $S_2(1) \approx 1.3203$, error $\approx 3.5\%$).

Example A.57 (Prove $|S_2(d) - S_2^{(M)}(d)| \le S_2(d) \cdot (\exp(6/(p_M \log p_M)) - 1)$)**. Step 1:** $S_2(d) = S_2^{(M)}(d) \cdot \prod_{p>p_M} f_p(d)$.

Step 2: $|f_p(d) - 1| \le 2/p^2$ (Lemma 5.1, in the bound on local factors).

Step 3: $|\prod_{p>p_M} f_p(d) - 1| \le \exp(\sum_{p>p_M} 2/p^2) - 1$.

Step 4: $\sum_{p>p_M} 2/p^2 < 6/(p_M \log p_M)$ (Lemma 5.2).

Step 5: Therefore $|S_2(d)/S_2^{(M)}(d) - 1| \le \exp(6/(p_M \log p_M)) - 1$.

Step 6: Multiply by $S_2^{(M)}(d)$: $|S_2(d) - S_2^{(M)}(d)| \le S_2(d) \cdot (\exp(6/(p_M \log p_M)) - 1)$. □

Example A.58 (Compute $\Omega^{(3)}(4)$ and $\Omega^{(3)}(6)$)**. Step 1:** $\Omega^{(3)}(4) = |R^{(3)}(1,1) - 1|$. $R^{(3)}(1,1) = S_4^{(3)}(1,1)/(S_2^{(3)}(1))^2 = S_2^{(3)}(1)/(S_2^{(3)}(1))^2 = 1/S_2^{(3)}(1) = 1/1.3672 = 0.7314$.

Step 2: $\Omega^{(3)}(4) = |0.7314 - 1| = 0.2686$.

Step 3: $\Omega^{(3)}(6) = |R^{(3)}(1,2) - 1| + |R^{(3)}(2,1) - 1| = 2 \cdot |R^{(3)}(1,2) - 1|$.

Step 4: $R^{(3)}(1,2) = S_4^{(3)}(1,2)/(S_2^{(3)}(1) \cdot S_2^{(3)}(2))$. Compute $S_2^{(3)}(2) = S_2^{(3)}(1)$ (since $\nu_p(2) = \nu_p(1)$ for $p > 2$).

Step 5: $S_4^{(3)}(1,2) = \prod_{p \le 7} \ldots =$ (compute from local factors) ≈ 1.20.

Step 6: $R^{(3)}(1,2) \approx 1.20/(1.3672)^2 \approx 0.64$.

Step 7: $\Omega^{(3)}(6) \approx 2 \cdot 0.36 = 0.72$.

Example A.59 (Show the truncation error for $\Omega(n)/n$ is $O(1/(p_M \log p_M))$)**. Step 1:** $|\Omega(n)/n - \Omega^{(M)}(n)/n| \le \frac{1}{n} \sum_d |R(d, m-d) - R^{(M)}(d, m-d)|$.

Step 2: $|R - R^{(M)}| \le 3R \cdot (\exp(6/(p_M \log p_M)) - 1)$ (Lemma 5.3).

Step 3: For p_M large: $\exp(6/(p_M \log p_M)) - 1 \approx 6/(p_M \log p_M)$.

Step 4: Therefore:

$$\left| \frac{\Omega(n)}{n} - \frac{\Omega^{(M)}(n)}{n} \right| \le \frac{1}{n} \sum_d 3R(d, m-d) \cdot \frac{6}{p_M \log p_M}.$$

Step 5: $\frac{1}{n} \cdot \sum_d R(d, m-d) \le R_{\max}/2$.

Step 6: Therefore error $\le 9R_{\max}/(p_M \log p_M) = O(1/(p_M \log p_M))$. □

Example A.60 (Construct the rational sequence p_k/q_k for $k = 1, 2, 3$)**. Step 1:** $N_k = 100k$, $M_k = k$.

Step 2: For $k = 1$: $N_1 = 100$, $M_1 = 1$ ($p_1 = 3$).

Step 3: $K_\Omega^{(100,1)} = \frac{1}{100} \cdot \sum_{j=1}^{100} \Omega^{(1)}(2j)/(2j)$, where $\Omega^{(1)}$ uses only $p = 3$.

Step 4: This is a rational number p_1/q_1.
Step 5: For $k = 2$: $N_2 = 200$, $M_2 = 2$ ($p_2 = 5$).
Step 6: $K_\Omega^{(200,2)} = \frac{1}{200} \cdot \sum_{j=1}^{200} \Omega^{(2)}(2j)/(2j)$, using $p = 3, 5$.
Step 7: This is p_2/q_2. Similarly for $k = 3$.
Step 8: The sequence $\{p_k/q_k\}$ converges to K_Ω (if the limit exists). □

A.5 Chapter 5 Examples: Error Bounds

Examples 5.1–5.10

Example A.61 (Tail sum for $p > 29$)**.** $\sum_{p>29} 1/p^2 < 1/(29 \log 29) \approx 0.102$.

Example A.62 (Truncation error bound for $M = 20$)**.** $p_{20} = 71$. Error $< \exp(6/(71 \log 71)) - 1 \approx 0.034$.

Example A.63 (R truncation for $M = 100$)**.** $p_{100} = 541$. $|R - R^{(100)}| < 0.0053 \cdot R$.

Example A.64 (Ω truncation for $M = 50$)**.** Error < 0.06 (from Corollary 5.5).

Example A.65 (Conjecture 5.1 evidence)**.** For $d, d' \leq 1000$: $\max R(d, d') = 2.365$ at $(59, 136)$.

Example A.66 (Trivial bound on averaging error)**.** $|\Omega(n)/n - K_\Omega| \leq R_{\max} + 1 \leq 5$.

Example A.67 (Multiperiodic bound for $P = 13$)**.** Error $\leq 3 \cdot 4 \cdot (\exp(6/(13 \log 13)) - 1) \approx 0.60$.

Example A.68 (Denominator lower bound)**.** $q_{N,M} \geq 1$ (trivial worst case).

Example A.69 (Denominator upper bound)**.** $\log q_{N,M} = O(N^2 \cdot M \log M)$.

Example A.70 (Irrationality obstruction for $k = 1000$)**.** Error $= 8.9 \times 10^{-5}$, threshold $= 10^{-44}$. Criterion fails.

Solved Examples 5.11–5.15

Example A.71 (Prove $\sum_{p>x} 1/p^2 < 1/(x \log x)$ for $x \geq 29$)**. Step 1:** By PNT: $\pi(t) < 1.26 \cdot t/\log t$ for $t \geq 17$.
Step 2: $\sum_{p>x} 1/p^2 = -\pi(x)/x^2 + 2\int_x^\infty \pi(t)/t^3\, dt$ (partial summation).
Step 3: $\pi(x)/x^2 < 1.26/(x \log x)$.
Step 4: $2\int_x^\infty \pi(t)/t^3\, dt < 2.52 \int_x^\infty dt/(t^2 \log t) < 2.52/x$.
Step 5: Total $< 1.26/(x \log x) + 2.52/x = (1.26/\log x + 2.52)/x$.
Step 6: For $x \geq 29$: $1.26/\log 29 + 2.52 \approx 1.26/3.37 + 2.52 \approx 0.374 + 2.52 = 2.894$.
Step 7: $2.894/29 \approx 0.100$. And $1/(29 \log 29) \approx 1/97.7 \approx 0.0102$. The bound is loose.
Step 8: Direct computation for $x = 29$: $\sum_{p>29} 1/p^2 \approx 0.0068 < 0.0102$. □

Example A.72 (Prove $|R - R^{(M)}| \leq 3R \cdot (\exp(6/(p_M \log p_M)) - 1)$)**. Step 1:** $R - R^{(M)} = S_4/(S_2 S_2') - S_4^{(M)}/(S_2^{(M)} S_2'^{(M)})$.
Step 2: Add/subtract $S_4/(S_2^{(M)} S_2'^{(M)})$:

$$R - R^{(M)} = \left[\frac{S_4}{S_2 S_2'} - \frac{S_4}{S_2^{(M)} S_2'^{(M)}}\right] + \left[\frac{S_4}{S_2^{(M)} S_2'^{(M)}} - \frac{S_4^{(M)}}{S_2^{(M)} S_2'^{(M)}}\right].$$

Step 3: Second term: $|S_4 - S_4^{(M)}|/(S_2^{(M)} S_2'^{(M)}) \le S_4 \cdot (\exp(6/(p_M \log p_M)) - 1)/(S_2^{(M)} S_2'^{(M)})$.
Step 4: First term: $S_4 \cdot |1/(S_2 S_2') - 1/(S_2^{(M)} S_2'^{(M)})| = S_4 \cdot |S_2 S_2' - S_2^{(M)} S_2'^{(M)}|/(S_2 S_2' S_2^{(M)} S_2'^{(M)})$.
Step 5: $|S_2 S_2' - S_2^{(M)} S_2'^{(M)}| \le |S_2 - S_2^{(M)}| \cdot S_2' + S_2^{(M)} \cdot |S_2' - S_2'^{(M)}|$.
Step 6: Each term bounded by $S_2 \cdot (\exp(6/(p_M \log p_M)) - 1)$ and similarly for S_2'.
Step 7: Combining: $|R - R^{(M)}| \le 3R \cdot (\exp(6/(p_M \log p_M)) - 1)$. □

Example A.73 (Compute the truncation error bound for $\Omega(10000)/10000$ with $M = 50$)**. Step 1:** $p_{50} = 229$. Error bound: $6 \cdot (\exp(6/(229 \log 229)) - 1)$.
Step 2: $6/(229 \cdot \log 229) = 6/(229 \cdot 5.434) = 6/1244.4 \approx 0.00482$.
Step 3: $\exp(0.00482) - 1 \approx 0.00483$.
Step 4: $6 \cdot 0.00483 \approx 0.029$.
Step 5: So $|\Omega(10000)/10000 - \Omega^{(50)}(10000)/10000| \le 0.029$.

Example A.74 (Prove the irrationality criterion cannot fire (Theorem 5.5))**. Step 1:** By Theorem 6.2: error $\le C''\sqrt{2}/\sqrt{N} + 6/(p_M \log p_M)$.
Step 2: Set $N = 100k$, $M = k$: error $\le C''\sqrt{2}/(10\sqrt{k}) + 6/(p_k \log p_k)$.
Step 3: $p_k \sim k \log k$, so error $= O(1/\sqrt{k})$.
Step 4: By Corollary 5.8: $\log q_k = O(N^2 \cdot M \log M) = O(k^3 \log k)$.
Step 5: Therefore $q_k = \exp(O(k^3 \log k))$.
Step 6: Threshold: $1/q_k^{1+\varepsilon} = \exp(-(1+\varepsilon) \cdot O(k^3 \log k))$.
Step 7: Compare: $O(1/\sqrt{k})$ vs $\exp(-O(k^3 \log k))$.
Step 8: For any polynomial $P(k)$: $P(k) \gg \exp(-c \cdot k^3 \log k)$ for large k.
Step 9: Therefore error $\gg$ threshold for all large k.
Step 10: The criterion $|K_\Omega - p_k/q_k| < 1/q_k^{1+\varepsilon}$ never holds. □

Example A.75 (Bound $q_{100,10}$ explicitly)**. Step 1:** $N = 100$, $M = 10$ ($p_{10} = 29$).
Step 2: $D_{10}(d)$ divides $\prod_{p \le 29}(p-1)^2 \cdot p$ for each d.
Step 3: Primes ≤ 29: $3, 5, 7, 11, 13, 17, 19, 23, 29$ (9 primes, excluding 2).
Step 4: $\log D_{10}(d) = O(\sum_{p \le 29} \log p) = O(29) = O(29)$.
Step 5: There are $O(N^2) = O(10000)$ terms in the denominator product.
Step 6: $\log q_{100,10} = O(10000 \cdot 29) = O(290{,}000)$.
Step 7: $q_{100,10} \le \exp(290{,}000) \approx 10^{126{,}000}$. This is an extremely crude bound.

A.6 Chapter 6 Examples: Irrationality Attempt

Examples 6.1–6.10

Example A.76 (Dirichlet's criterion)**.** If $|\alpha - p_k/q_k| < 1/q_k^{1.01}$ for all large k, then α is irrational.

Example A.77 (Rational $\alpha = 1/3$)**.** For any $p/q \ne 1/3$: $|1/3 - p/q| = |q - 3p|/(3q) \ge 1/(3q)$. Criterion requires $< 1/q^{1.01}$, which fails for large q.

Example A.78 (Denominator lower bound)**.** $q_k \ge 1$ (trivial). In practice: q_k grows much faster.

Example A.79 (Error for $k = 100$)**.** $N = 10{,}000$, $M = 100$. Error $\le 0.0001 + 0.005 = 0.0051$.

Example A.80 (Threshold for $k = 100$)**.** $q_{100} \geq \exp(O(10^6 \cdot 100 \cdot \log 100)) = \exp(O(10^8))$. Threshold $< \exp(-10^8)$.

Example A.81 (Roth's criterion)**.** Requires $|\alpha - p/q| < 1/q^3$. Our error $0.0051 \gg 1/q^3 \approx 0$. Criterion fails.

Example A.82 (Necessary condition for success)**.** Need error to decay exponentially or denominator to grow polynomially.

Example A.83 (Hypothetical fast convergence)**.** If $\Omega(n)/n = K_\Omega + O(e^{-cn})$, then error $= O(e^{-cN})$ and criterion might fire.

Example A.84 (Closed-form approach)**.** If $K_\Omega = f(\mu, \sigma^2)$ for explicit f, and μ is transcendental, then K_Ω is transcendental.

Example A.85 (Why current scheme fails)**.** Error $= O(1/\sqrt{k})$, threshold $= \exp(-O(k^3 \log k))$. Polynomial vs exponential decay.

Solved Examples 6.11–6.15

Example A.86 (Prove Dirichlet's criterion (Theorem 6.1))**. Step 1:** Assume $\alpha = a/b$ is rational.

Step 2: For $p_k/q_k \neq a/b$: $|\alpha - p_k/q_k| = |aq_k - bp_k|/(bq_k)$.

Step 3: $|aq_k - bp_k| \geq 1$ (nonzero integer).

Step 4: So $|\alpha - p_k/q_k| \geq 1/(bq_k)$.

Step 5: The criterion requires $|\alpha - p_k/q_k| < 1/q_k^{1+\varepsilon}$.

Step 6: For large q_k: $1/(bq_k) > 1/q_k^{1+\varepsilon}$ (since $q_k^\varepsilon > b$ eventually).

Step 7: Contradiction. Therefore α is irrational. □

Example A.87 (Show the criterion fails for $k = 1000$)**. Step 1:** $N = 100{,}000$, $M = 1000$.

Step 2: $C''\sqrt{2}/\sqrt{N} \leq 0.001 \cdot 1.41/316 = 4.5 \times 10^{-6}$.

Step 3: $6/(p_{1000} \log p_{1000})$: $p_{1000} = 7919$, so $6/(7919 \cdot 8.98) = 8.4 \times 10^{-5}$.

Step 4: Error $= 4.5 \times 10^{-6} + 8.4 \times 10^{-5} = 8.9 \times 10^{-5}$.

Step 5: $q_{1000} \geq \exp(0.1 \cdot 1000) = \exp(100) \approx 2.7 \times 10^{43}$.

Step 6: Threshold for $\varepsilon = 0.01$: $1/q^{1.01} \leq \exp(-101) \approx 1.3 \times 10^{-44}$.

Step 7: $8.9 \times 10^{-5} \gg 1.3 \times 10^{-44}$. Criterion fails. □

Example A.88 (Show Roth's theorem does not help)**. Step 1:** Roth's criterion: $|\alpha - p/q| < 1/q^3$ for infinitely many p/q.

Step 2: Our error: $O(1/\sqrt{k})$. Threshold: $1/q^3 = \exp(-O(k^3 \log k))$.

Step 3: For large k: $O(1/\sqrt{k}) \gg \exp(-O(k^3 \log k))$.

Step 4: Same obstruction as Dirichlet's criterion.

Step 5: Roth's theorem does not apply. □

Example A.89 (Prove the obstruction theorem (Theorem 6.4))**. Step 1:** Error $= O(1/\sqrt{N} + 1/(M \log^2 M))$ (from Theorem 6.2).

Step 2: With $N = 100k$, $M = k$: Error $= O(1/\sqrt{k})$.

Step 3: $\log q_k = O(N^2 \cdot M \log M) = O(k^3 \log k)$ (Corollary 5.8).

Step 4: Threshold: $1/q_k^{1+\varepsilon} = \exp(-(1+\varepsilon)\cdot O(k^3 \log k))$.
Step 5: $\lim_{k\to\infty}(1/\sqrt{k})/\exp(-ck^3 \log k) = \infty$ for any $c > 0$.
Step 6: Therefore error $\gg$ threshold for all large k.
Step 7: The criterion never fires. □

Example A.90 (What would be needed to prove irrationality?)**. Step 1:** Need error $< 1/q^{1+\varepsilon}$.
Step 2: Currently: error $= O(1/\sqrt{k})$, $q = \exp(O(k^3 \log k))$.
Step 3: Option A: Faster error decay. Need error $= O(\exp(-ck))$.
Step 4: This requires $\Omega(n)/n = K_\Omega + O(\exp(-cn))$.
Step 5: Option B: Slower denominator growth. Need $\log q = O(k)$.
Step 6: This requires q_k to grow polynomially, not exponentially.
Step 7: Option C: Different approximation scheme entirely.
Step 8: None of these are currently available. □

A.7 Chapter 7 Examples: Obstructions & Open Problems

Examples 7.1–7.10

Example A.91 (P1: Convergence)**.** $\Omega(30000)/30000 = 0.1407$ (branch-affected). $\Omega(40000)/40000 = 0.3640$ (generic). The branch effect prevents uniform convergence.

Example A.92 (P2: C')**.** C' depends on $\sup R(d, d')$. Numerically: $C' \approx 3$. But no proof exists.

Example A.93 (P3: Averaging error)**.** Empirically: $|\Omega(n)/n - K_\Omega| < 0.0005$ for $n \geq 10{,}000$. No proof of the rate.

Example A.94 (P4: Denominator)**.** $q_{N,M}$ is computed as an upper bound. The actual value may be much smaller.

Example A.95 (P5: Irrationality)**.** $K_\Omega = 0.3640\ldots$ is very likely irrational. No proof exists.

Example A.96 (P6: μ irrational)**.** $\mu = \sum_p c_p \log \alpha_p$. Each finite truncation is transcendental (Baker). The limit may be rational.

Example A.97 (P7: Transcendence)**.** Even harder than irrationality. No method connects K_Ω to transcendence theory.

Example A.98 (H1: Signed sum approximation)**.** $\Omega(n) \approx -\Sigma(n)$ to within 0.1%. The error $2 \cdot \sum_{R>1}(R-1)$ is small but unbounded rigorously.

Example A.99 (H2: Probabilistic model)**.** Model predicts $\mu = 0.2695$, $\sigma^2 = 0.6035$. Empirical match is excellent. But model $\neq$ proof.

Example A.100 (H4: Baker's route)**.** Infinite sum vs finite sum. Baker's theorem applies to finite forms only.

Solved Examples 7.11–7.15

Example A.101 (Show that the branch effect prevents uniform convergence)**. Step 1:** For $n = 30000$ ($3 \mid n/2$ and $5 \mid n/2$): $\Omega(n)/n \approx 0.141$.

Step 2: For $n = 40000$ ($5 \mid n/2$ but $3 \nmid n/2$): $\Omega(n)/n \approx 0.364$.

Step 3: The difference is $|0.364 - 0.141| = 0.223$.

Step 4: This does not go to 0 as $n \to \infty$.

Step 5: Therefore $\Omega(n)/n$ does not converge uniformly over all n.

Step 6: Convergence can only hold within a fixed branch (where $n/2 \bmod p$ is fixed for all relevant p).

Step 7: The overall limit K_Ω exists only as an average over branches. □

Example A.102 (Bound the signed sum error rigorously)**. Step 1:** $\Omega(n) + \Sigma(n) = 2 \cdot \sum_{R>1}(R-1)$ (Theorem 2.1).

Step 2: Each term $R - 1 > 0$ (since $R > 1$ on the sum).

Step 3: Number of terms: $|\{d : R(d, m-d) > 1\}| \le m - 1 < n/2$.

Step 4: Each term: $R - 1 \le R_{\max} - 1$.

Step 5: Therefore: $0 \le \Omega(n) + \Sigma(n) \le 2 \cdot (n/2) \cdot (R_{\max} - 1) = n(R_{\max} - 1)$.

Step 6: Per-unit: $|\Omega(n)/n + \Sigma(n)/n| \le R_{\max} - 1 \le 3$ (assuming $R_{\max} \le 4$).

Step 7: This is a valid bound but too loose to be useful. □

Example A.103 (Explain why μ is an infinite linear form in logs)**. Step 1:** $\mu = \sum_p \mathbb{E}[\log r_p]$.

Step 2: $\mathbb{E}[\log r_p] = p_{\text{both}} \cdot \log(r_p^{\text{both}}) + p_{\text{div}} \cdot \log(r_p^{\text{div}}) + p_{\text{neither}} \cdot \log(r_p^{\text{gen}})$.

Step 3: Each $r_p^{\cdots}$ is a rational number (ratio of products of rational local factors).

Step 4: So $\log(r_p^{\cdots})$ is the logarithm of a rational (algebraic) number.

Step 5: $\mu = \sum_p (c_{p1} \log \alpha_{p1} + c_{p2} \log \alpha_{p2} + c_{p3} \log \alpha_{p3})$.

Step 6: This is an infinite sum of rational multiples of logs of rationals.

Step 7: Baker's theorem applies to finite sums only. □

Example A.104 (Show the model overestimates K_Ω)**. Step 1:** Model predicts $K_\Omega^{\text{model}} \approx 0.5184$.

Step 2: Empirical $K_\Omega \approx 0.3640$.

Step 3: The model density $f_{\text{model}}(x)$ has the correct mean ($\mu = 0.2695$) and variance ($\sigma^2 = 0.6035$).

Step 4: But $f_{\text{model}}(x)$ has heavier tails than the actual distribution (empirically).

Step 5: Since $|e^x - 1|$ grows exponentially, tail weight matters enormously.

Step 6: The model's heavier tails produce a larger expectation.

Step 7: Therefore $K_\Omega^{\text{model}} > K_\Omega$. □

Example A.105 (State the minimal viable publishable result)**. Step 1:** Assume $\Omega(n)/n$ converges to K_Ω (open, but empirically verified).

Step 2: Then: $K_\Omega = 0.3640 \pm 0.0001$ (from $n \le 100{,}000$).

Step 3: $K_\Omega = \sum_B w_B \cdot K_\Omega(B)$ (proved in Chapter 2, conditional on convergence).

Step 4: $K_\Omega < 0.5$ (follows from branch decomposition and empirical bounds).

Step 5: The truncation error for $M = 100$ is < 0.012 (proved in Chapter 5, conditional on Conjecture 5.1).

Step 6: K_Ω is very likely irrational (numerical evidence, no proof).

Step 7: This is a conditional result with 1 open assumption (convergence). □

A.8 Chapter 8 Examples: Model and Data

Examples 8.1–8.10

Example A.106 (Model characteristic function)**.** $\varphi_{\text{model}}(0.1) = \prod_p \psi_p(0.1)$. Numerically: $\varphi_{\text{model}}(0.1) \approx 0.9973 + 0.0242i$.

Example A.107 (ODE initial condition)**.** $\varphi_{\text{model}}(0) = 1$, $\varphi'_{\text{model}}(0) = i\mu = 0.2695i$.

Example A.108 (Model density at $x = 0$)**.** $f_{\text{model}}(0) \approx 0.303$ (from Fourier inversion).

Example A.109 (Model peak)**.** $f_{\text{model}}(x)$ has a peak at $x \approx 0.21$ with height ≈ 1.19.

Example A.110 (Model vs normal)**.** Normal density $\mathcal{N}(0.2695, 0.6035)$ at $x = 0.21$: 0.506. Model: 1.19. Much higher peak (bimodality).

Example A.111 (Empirical $\Omega(10000)/10000$)**.** 0.363544 (Table 8.1).

Example A.112 (Branch constant for $(3,5)$)**.** $K_\Omega((3,5)) \approx 0.141$ (Table 8.2).

Example A.113 (Branch structure table)**.** For $m = 100$: $d \equiv 0 \pmod 3$ gives mean $|R - 1| = 0.5879$; $d \equiv 1 \pmod 3$ gives 0.2382 (Table 8.3).

Example A.114 (Convergence data)**.** $\Omega(1000)/1000 = 0.3628$, $\Omega(100000)/100000 = 0.3640$ (Table 8.5).

Example A.115 (Continued fraction of μ)**.** $\text{CF}(\mu) = [0; 3, 1, 2, 1, 1, 1, 12, 1, 1, 3, \dots]$.

Solved Examples 8.11–8.15

Example A.116 (Compute $\psi_3(t)$ for $t = 0.1$)**. Step 1:** $\psi_3(t) = p_{\text{both}} \cdot r_3^{\text{both}^{it}} + p_{\text{div}} \cdot r_3^{\text{div}^{it}} + p_{\text{neither}} \cdot r_3^{\text{gen}^{it}}$.

Step 2: $p_{\text{both}} = 1/9$, $p_{\text{div}} = 2/3 - 2/9 = 4/9$, $p_{\text{neither}} = 4/9$.

Step 3: $r_3^{\text{both}} = 1$, $r_3^{\text{div}} = 3$, $r_3^{\text{gen}} = 1$.

Step 4: $\psi_3(0.1) = (1/9) \cdot 1^{0.1i} + (4/9) \cdot 3^{0.1i} + (4/9) \cdot 1^{0.1i}$.

Step 5: $3^{0.1i} = \exp(0.1i \cdot \log 3) = \cos(0.1 \cdot 1.099) + i \cdot \sin(0.1 \cdot 1.099) = 0.9940 + 0.1096i$.

Step 6: $\psi_3(0.1) = (1/9) \cdot 1 + (4/9) \cdot (0.9940 + 0.1096i) + (4/9) \cdot 1 = 0.9984 + 0.0487i$.

Example A.117 (Verify $\mu = 0.2695420214$ from the ODE)**. Step 1:** $\mu = \varphi'(0)/i$ (from Lemma 8.2).

Step 2: $\varphi'(0) = \sum_p \psi'_p(0)$.

Step 3: $\psi'_p(0) = i \cdot p_{\text{div}} \cdot \log(r_p^{\text{div}})$.

Step 4: For $p = 3$: $\psi'_3(0) = i \cdot (4/9) \cdot \log(3) = i \cdot 0.4444 \cdot 1.099 = i \cdot 0.4883$.

Step 5: For $p = 5$: $\psi'_5(0) = i \cdot (8/25) \cdot \log(5/3) = i \cdot 0.32 \cdot 0.511 = i \cdot 0.1635$.

Step 6: For $p = 7$: $\psi'_7(0) = i \cdot (12/49) \cdot \log(7/5) = i \cdot 0.2449 \cdot 0.336 = i \cdot 0.0824$.

Step 7: Summing over all p: $\mu = 0.4883 + 0.1635 + 0.0824 + \cdots = 0.2695$. □

Example A.118 (Show the branch effect in data)**. Step 1:** From Table 8.1: $n = 30000$, $\Omega/n = 0.1407$ ($3 \mid m$, $5 \mid m$).

Step 2: $n = 40000$, $\Omega/n = 0.3640$ ($5 \mid m$, $3 \nmid m$).

Step 3: $n = 60000$, $\Omega/n = 0.1407$ ($3 \mid m$, $5 \mid m$).

Step 4: $n = 70000$, $\Omega/n = 0.4192$ ($5 \mid m$, $7 \mid m$, $3 \nmid m$).

Step 5: The branch effect is confirmed: $3 \mid m$ and $5 \mid m$ implies $\Omega/n \approx 0.14$.

Step 6: This matches Theorem 2.5 (branch decomposition). □

Example A.119 (Compute the CF of K_Ω to available precision)**. Step 1:** $K_\Omega \approx 0.3640$ (4 digits).

Step 2: $a_0 = \lfloor 0.3640 \rfloor = 0$. Remainder: 0.3640.

Step 3: $1/0.3640 = 2.7473$. $a_1 = \lfloor 2.7473 \rfloor = 2$. Remainder: 0.7473.

Step 4: $1/0.7473 = 1.3381$. $a_2 = 1$. Remainder: 0.3381.

Step 5: $1/0.3381 = 2.9577$. $a_3 = 2$. Remainder: 0.9577.

Step 6: $1/0.9577 = 1.0442$. $a_4 = 1$. Remainder: 0.0442.

Step 7: $1/0.0442 = 22.62$. $a_5 = 22$.

Step 8: CF $= [0; 2, 1, 2, 1, 22, \dots]$. The next term is huge (artifact).

Step 9: The CF is inconclusive with only 4 digits.

Example A.120 (Compare model prediction with empirical data)**. Step 1:** Model: $K_\Omega^{\text{model}} \approx 0.5184$.

Step 2: Empirical: $K_\Omega \approx 0.3640$.

Step 3: Relative error: $|0.5184 - 0.3640|/0.3640 = 0.1544/0.3640 = 42.4\%$.

Step 4: The model captures qualitative features (bimodality, light tails) but overestimates quantitatively.

Step 5: The discrepancy is due to tail behavior: the model has slightly heavier tails than reality.

Step 6: Since $|e^x - 1|$ amplifies tail weight exponentially, a small tail difference creates a large K_Ω difference.

Step 7: Conclusion: the model is useful for qualitative insight but not for precise computation. □

References

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